# Exploratory Verification

*in the age of*

# Probabilistic Code

**Mourjo Sen**

September 2026

$

```
$ mvn clean test
```

```
$ mvn clean test
```

```
…
tries = 187                    | # of calls to property
checks = 187                   | # of not rejected calls
generation = RANDOMIZED        | parameters are randomly generated
…

[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
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 Invariant 'no-overlap' failed after the following actions: [



 ]

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 Invariant 'no-overlap' failed after the following actions: [
     alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01

 ]

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 Invariant 'no-overlap' failed after the following actions: [
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     bob creates Meeting-2 in calendar default (UTC) from 10:00 to 10:01


 ]


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 Invariant 'no-overlap' failed after the following actions: [
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     bob creates Meeting-2 in calendar default (UTC) from 10:00 to 10:01
     bob invites alice to Meeting-2

 ]


…
tries = 187                   | # of calls to property
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# How did I miss that?

```
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```

# Namaste, Guten Tag!

- India → Netherlands
- 10 years in engineering
- Staff Engineer @ booking.com

# Everything's Changed

# LLMs Write Faster Than We Can Read

# LLMs Write Faster Than We Can Read

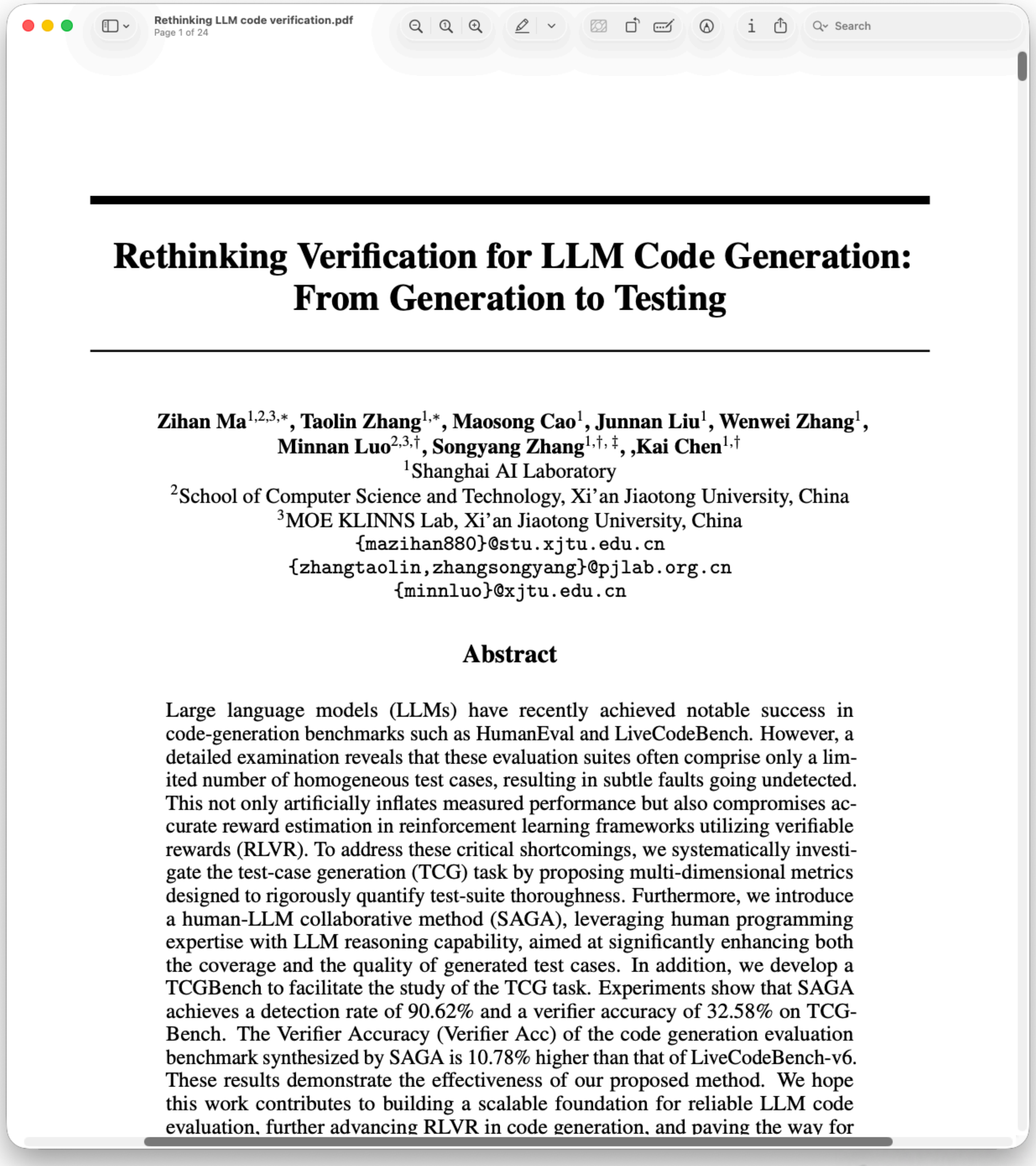
- Code generation is fast but probabilistic
  - Quality of software cannot be probabilistic

# LLMs Write Faster Than We Can Read

- Code generation is fast but probabilistic
  - Quality of software cannot be probabilistic
- Use LLMs for testing software?
  - Because we cannot read at that speed

# Are LLMs Good At Testing Software?

# Are LLMs Good At Testing Software?



Rethinking LLM code verification.pdf
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## Rethinking Verification for LLM Code Generation: From Generation to Testing


**Zihan Ma[1,2,3,*], Taolin Zhang[1,*], Maosong Cao[1], Junnan Liu[1], Wenwei Zhang[1], Minnan Luo[2,3,†], Songyang Zhang[1,†,‡], ,Kai Chen[1,†]**

[1]Shanghai AI Laboratory

[2]School of Computer Science and Technology, Xi'an Jiaotong University, China

[3]MOE KLINNS Lab, Xi'an Jiaotong University, China

{mazihan880}@stu.xjtu.edu.cn

{zhangtaolin,zhangsongyang}@pjlab.org.cn

{minnluo}@xjtu.edu.cn

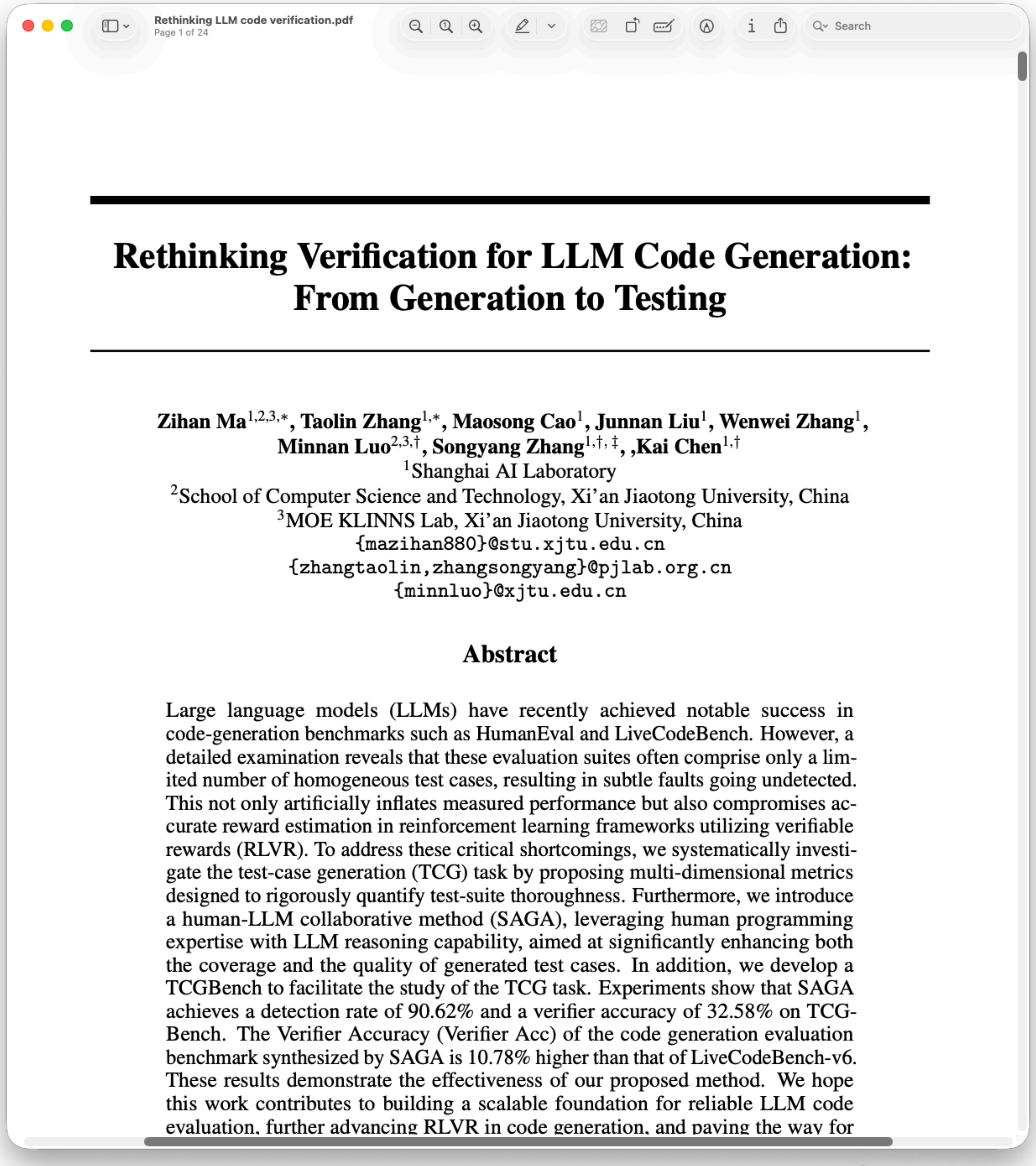

### Abstract


Large language models (LLMs) have recently achieved notable success in code-generation benchmarks such as HumanEval and LiveCodeBench. However, a detailed examination reveals that these evaluation suites often comprise only a limited number of homogeneous test cases, resulting in subtle faults going undetected. This not only artificially inflates measured performance but also compromises accurate reward estimation in reinforcement learning frameworks utilizing verifiable rewards (RLVR). To address these critical shortcomings, we systematically investigate the test-case generation (TCG) task by proposing multi-dimensional metrics designed to rigorously quantify test-suite thoroughness. Furthermore, we introduce a human-LLM collaborative method (SAGA), leveraging human programming expertise with LLM reasoning capability, aimed at significantly enhancing both the coverage and the quality of generated test cases. In addition, we develop a TCGBench to facilitate the study of the TCG task. Experiments show that SAGA achieves a detection rate of 90.62% and a verifier accuracy of 32.58% on TCGBench. The Verifier Accuracy (Verifier Acc) of the code generation evaluation benchmark synthesized by SAGA is 10.78% higher than that of LiveCodeBench-v6. These results demonstrate the effectiveness of our proposed method. We hope this work contributes to building a scalable foundation for reliable LLM code evaluation, further advancing RLVR in code generation, and paving the way for


Source: https://arxiv.org/abs/2507.06920, presented at NeurIPS 2025

# Are LLMs Good At Testing Software?

- LLMs struggle to catch bugs
  - ~60% chance of identifying a buggy solution

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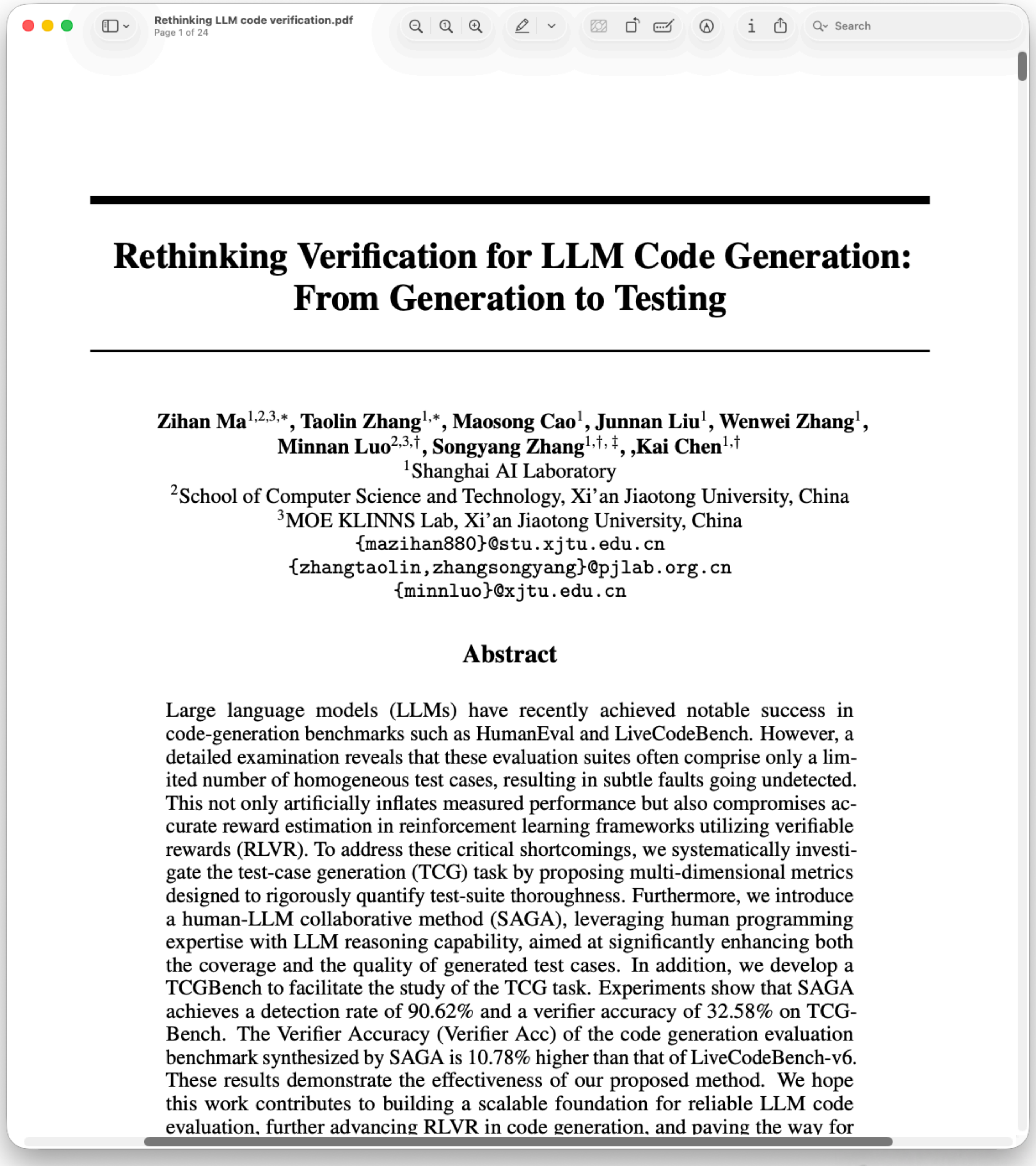
  - ~60% chance of identifying a buggy solution
- Inflated self-testing success
  - LLM-written tests have a high success rate against LLM-written source





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- Code generation is fast
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- But it is also probabilistic
    - If we can’t read it, we can’t verify its correctness
    - LLMs are not good at testing software
- Human comprehension
    - Our ability to understand and make meaning from information

# Playing To Our Strengths

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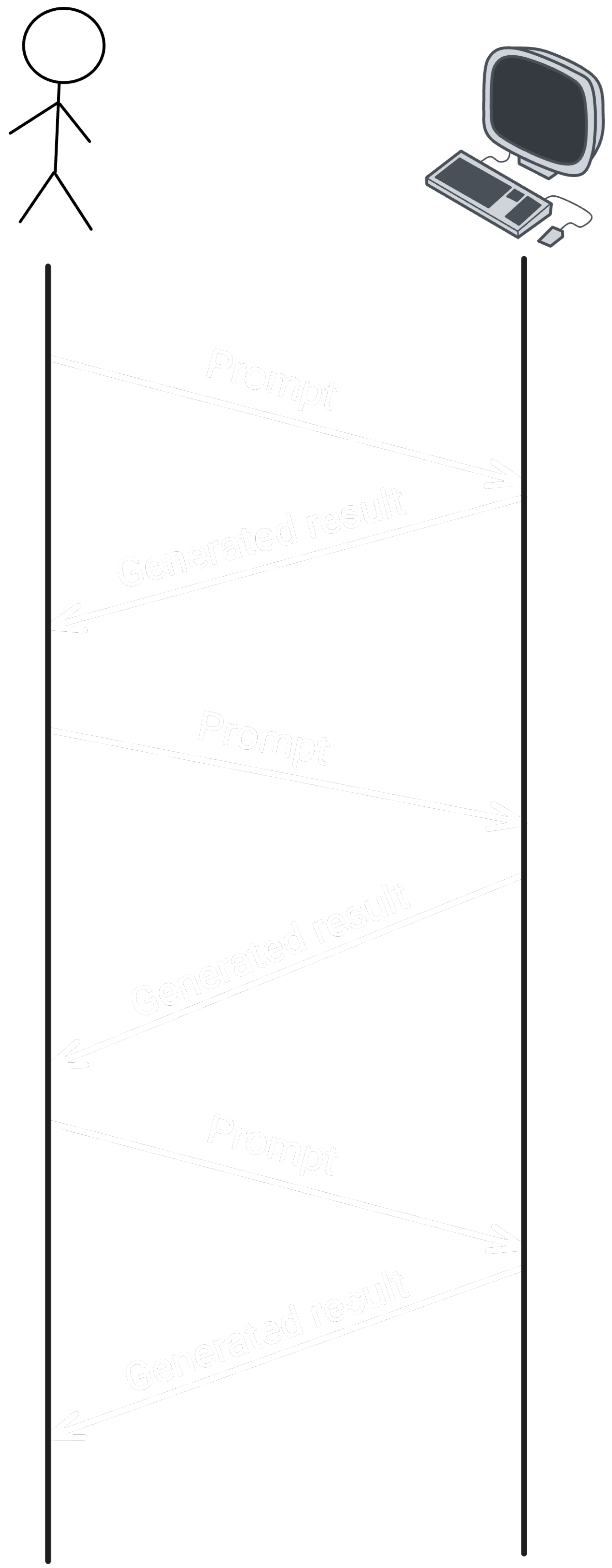

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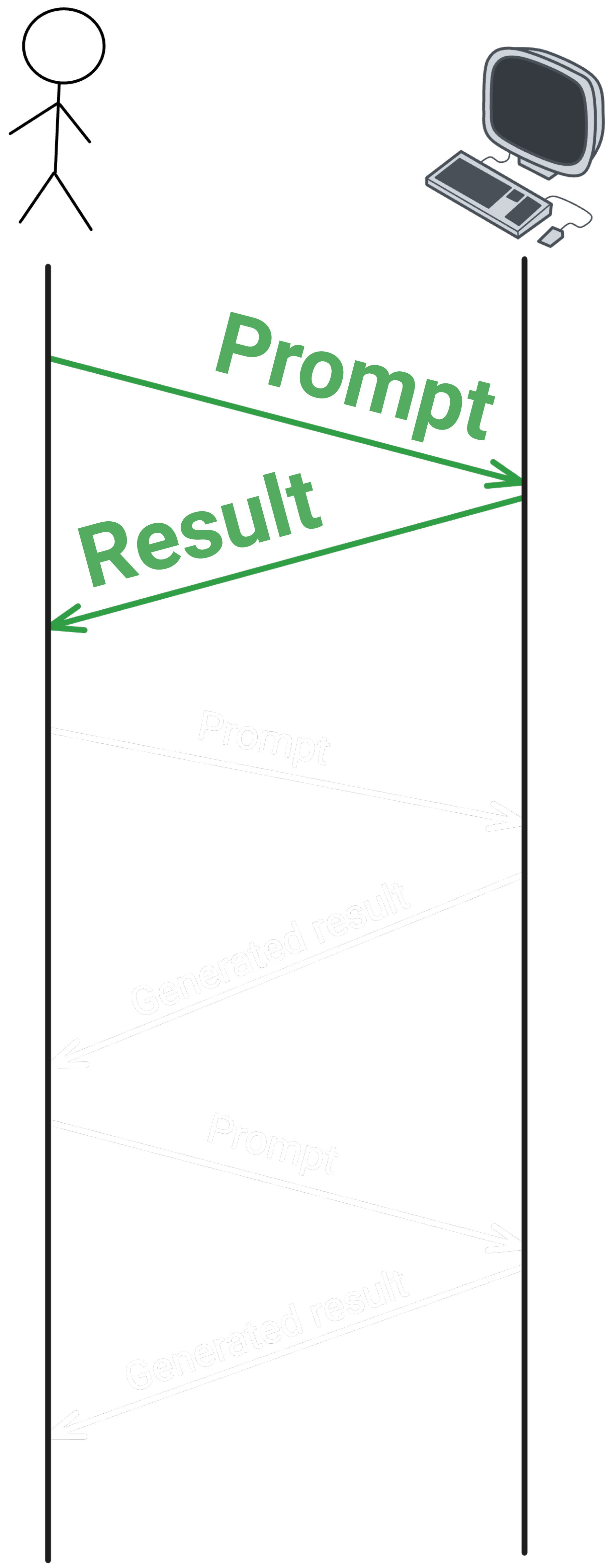

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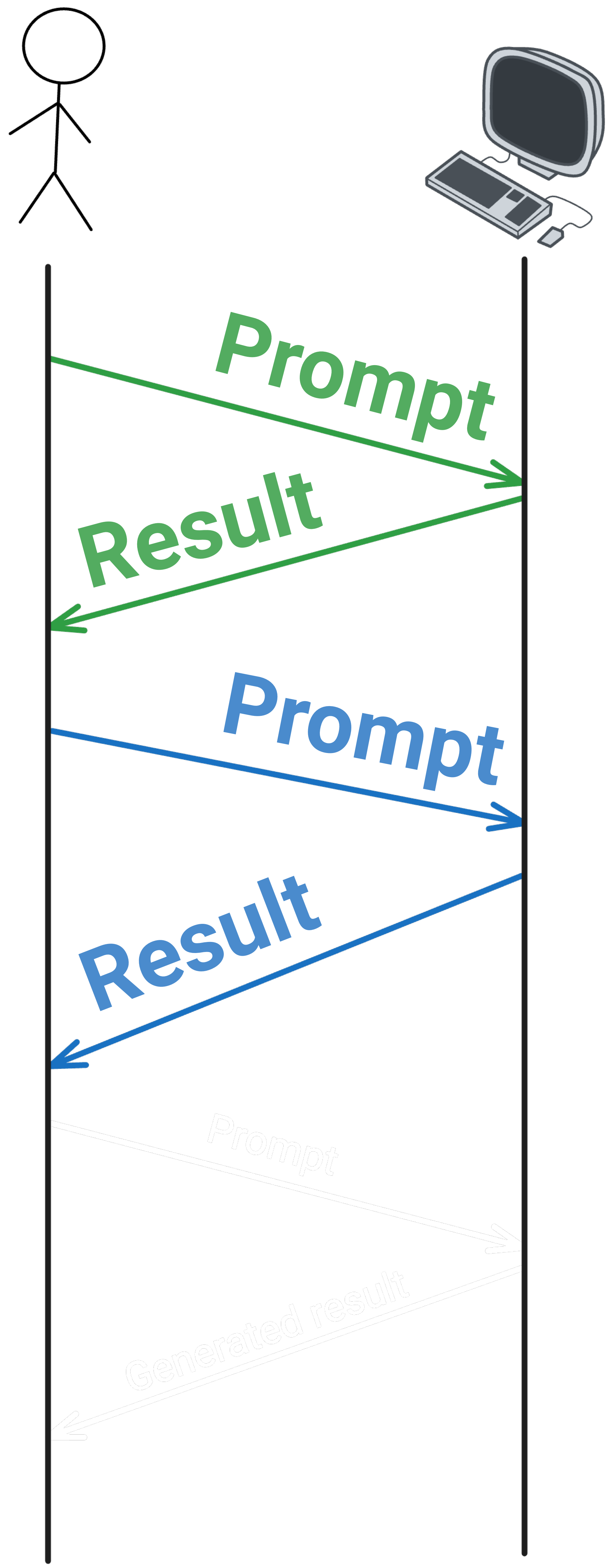

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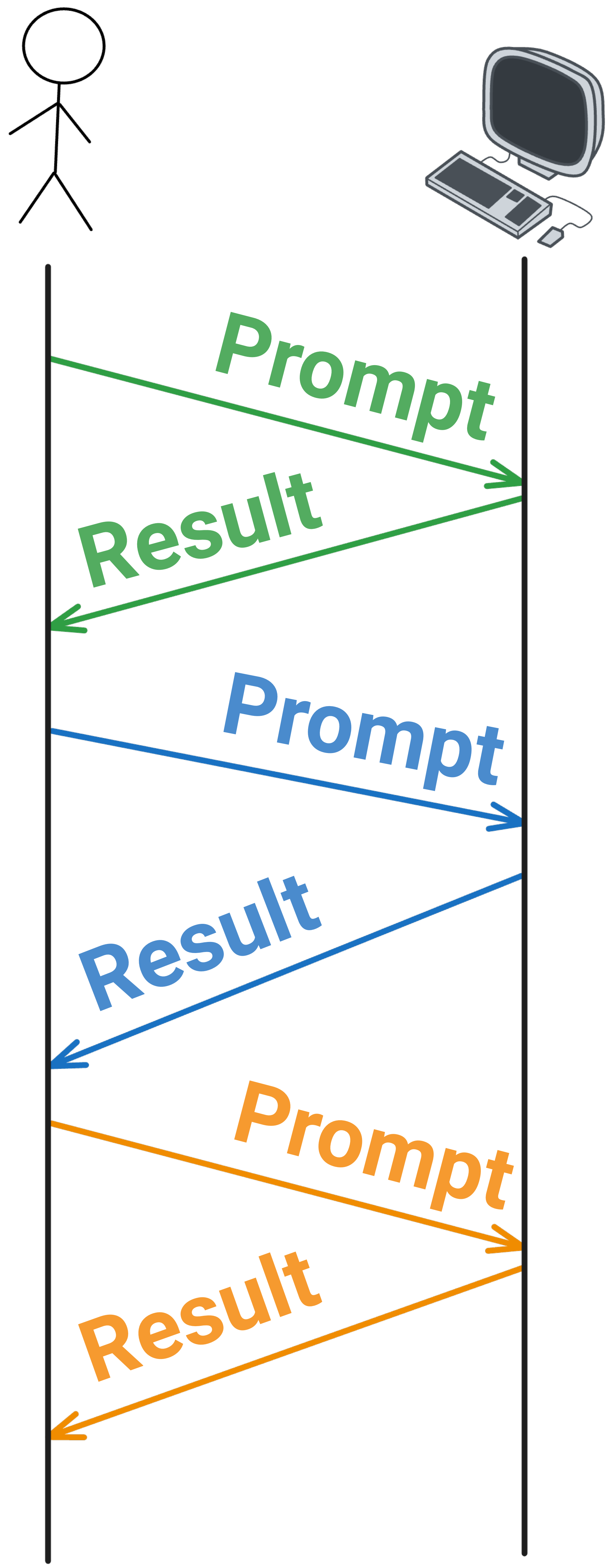

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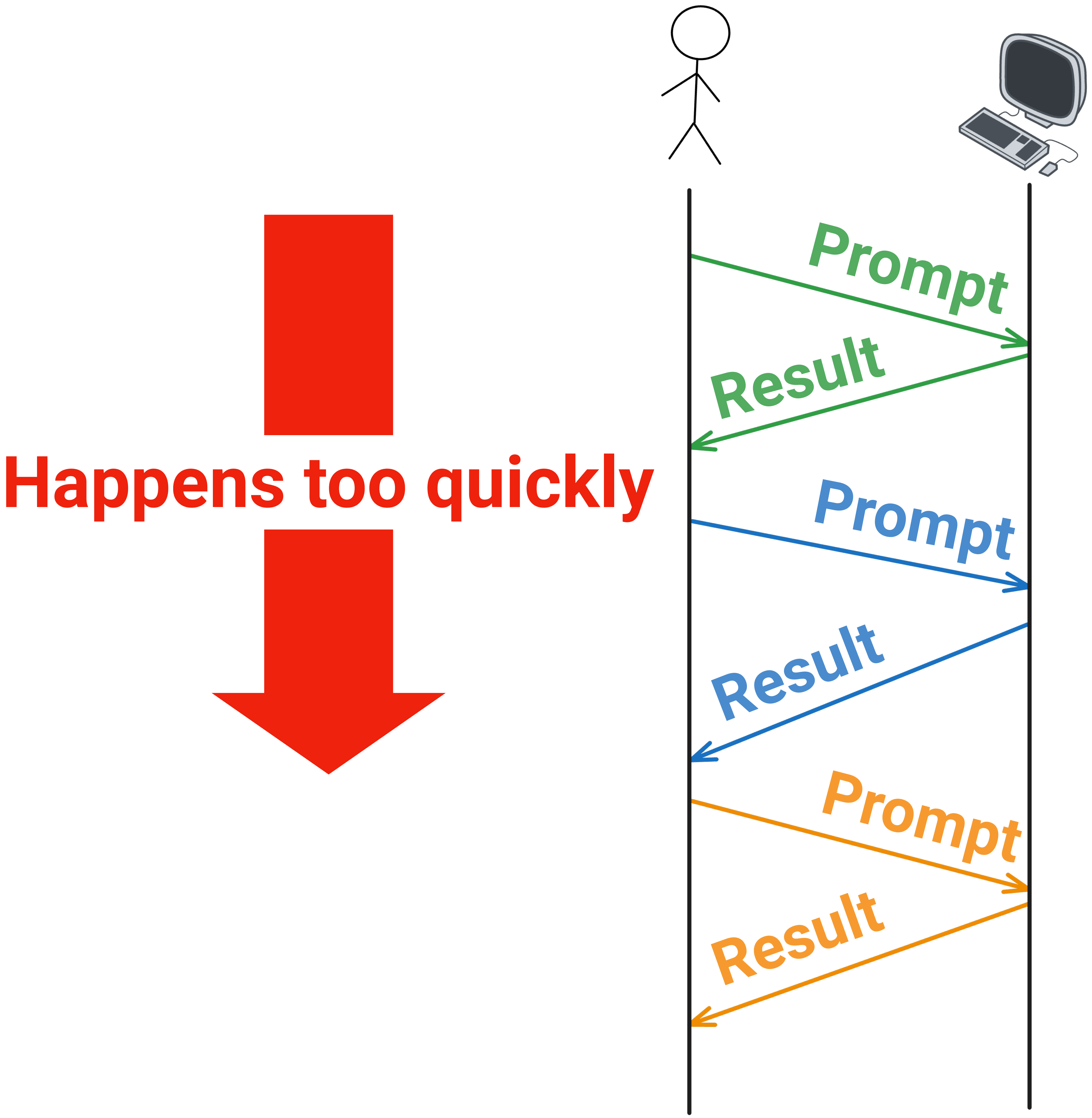

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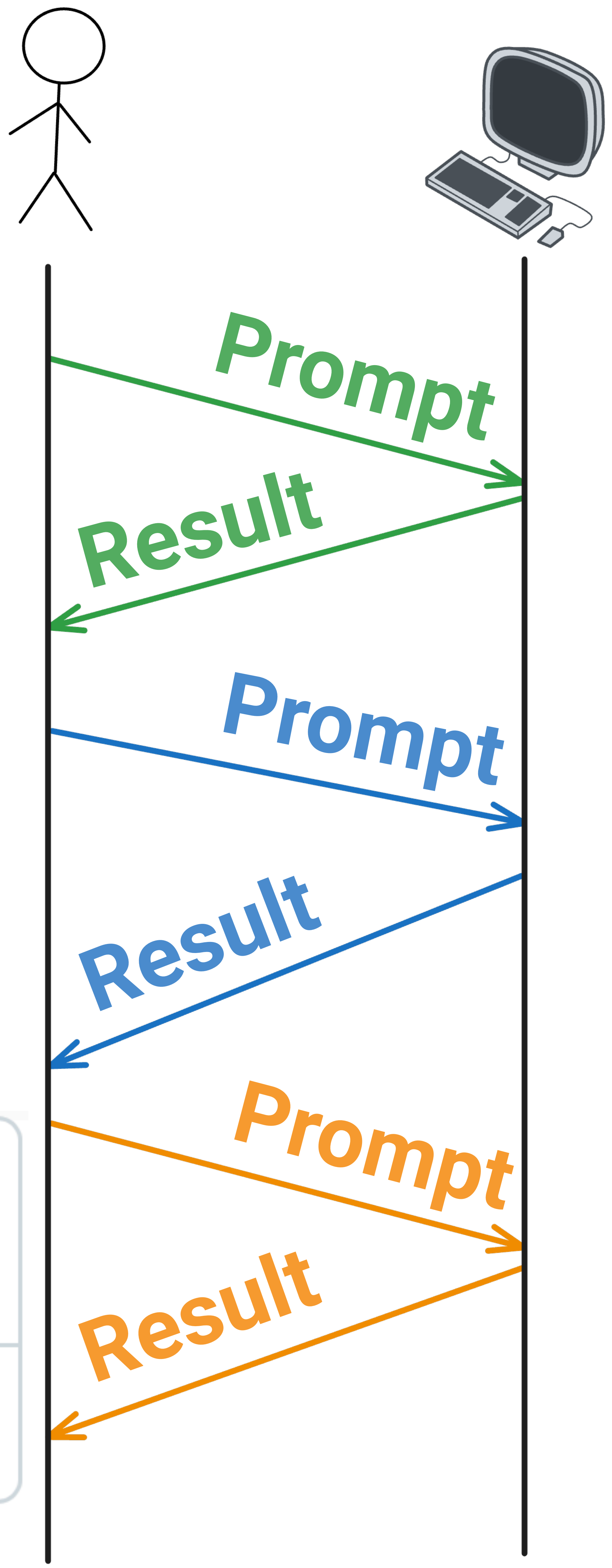



**Happens too quickly**

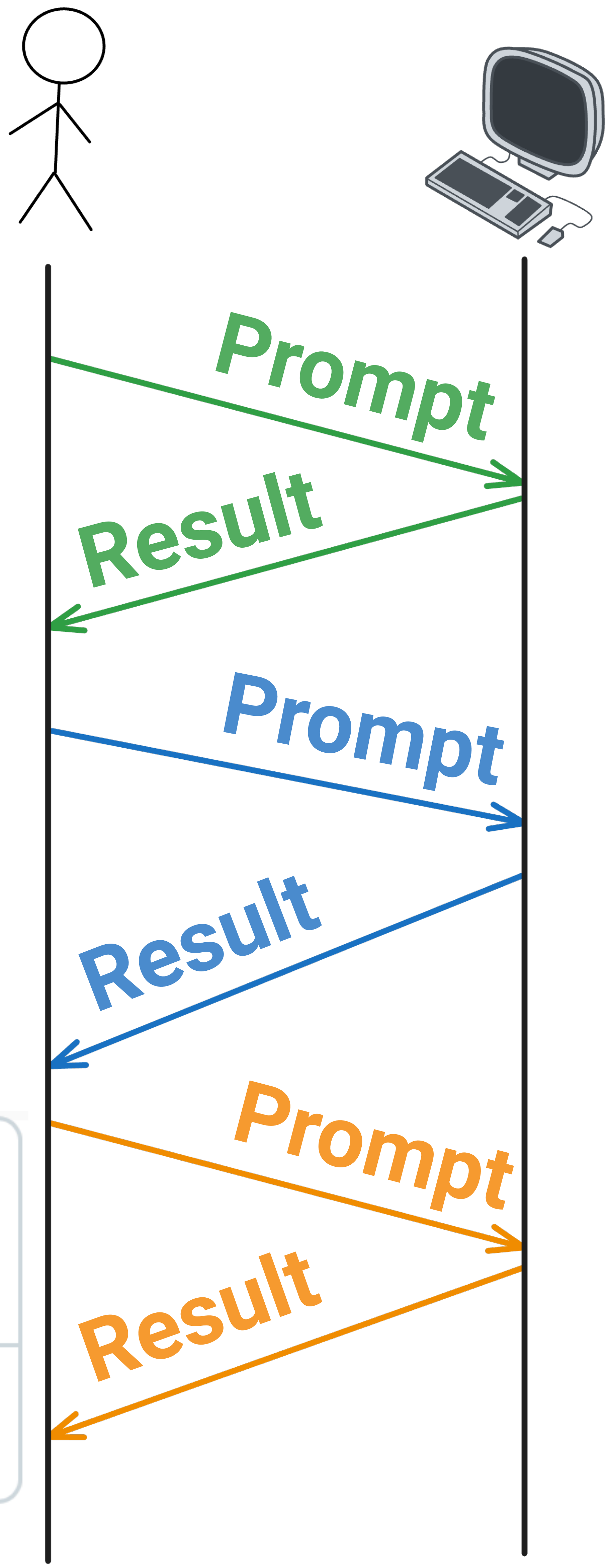
`feat: implement user, meeting, invitation endpoints`

**25 files changed** +1,030 -32

# Playing To Our Strengths

***Active*** **Participant**

***Superficial*** **Overseer**

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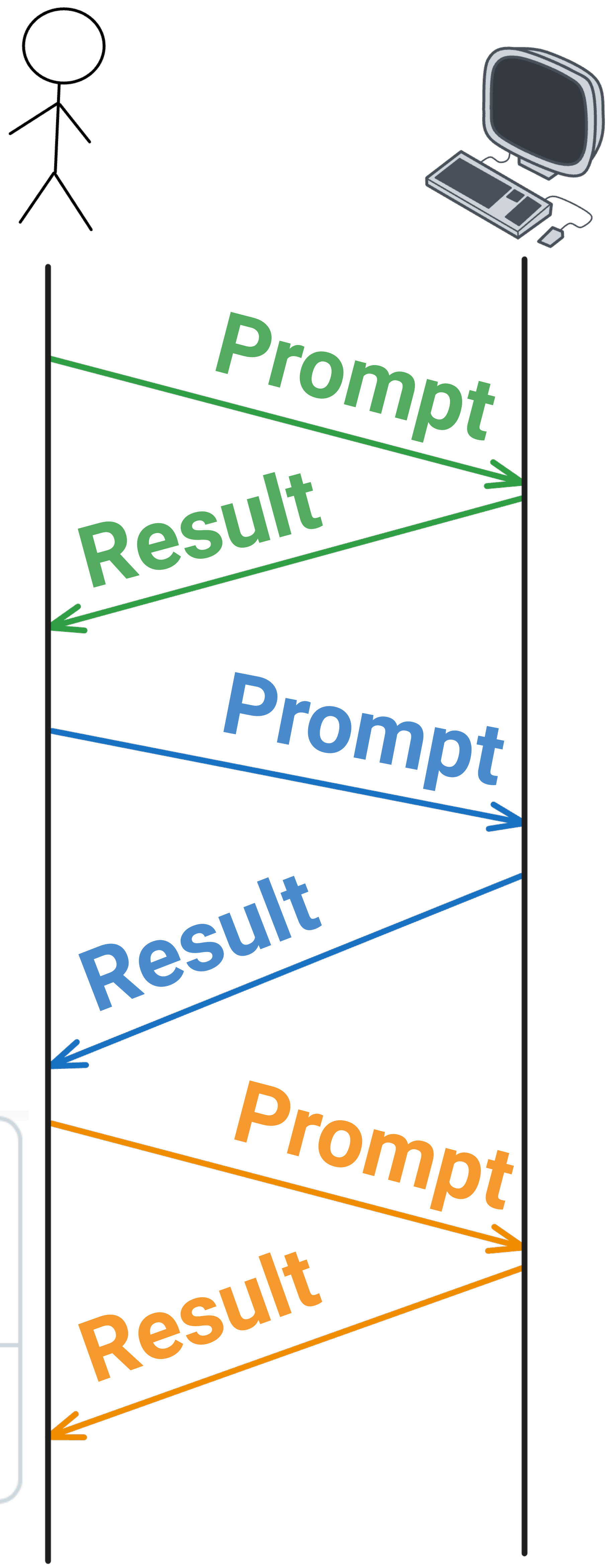


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Overseeing does not **engage** our **cognitive** processes

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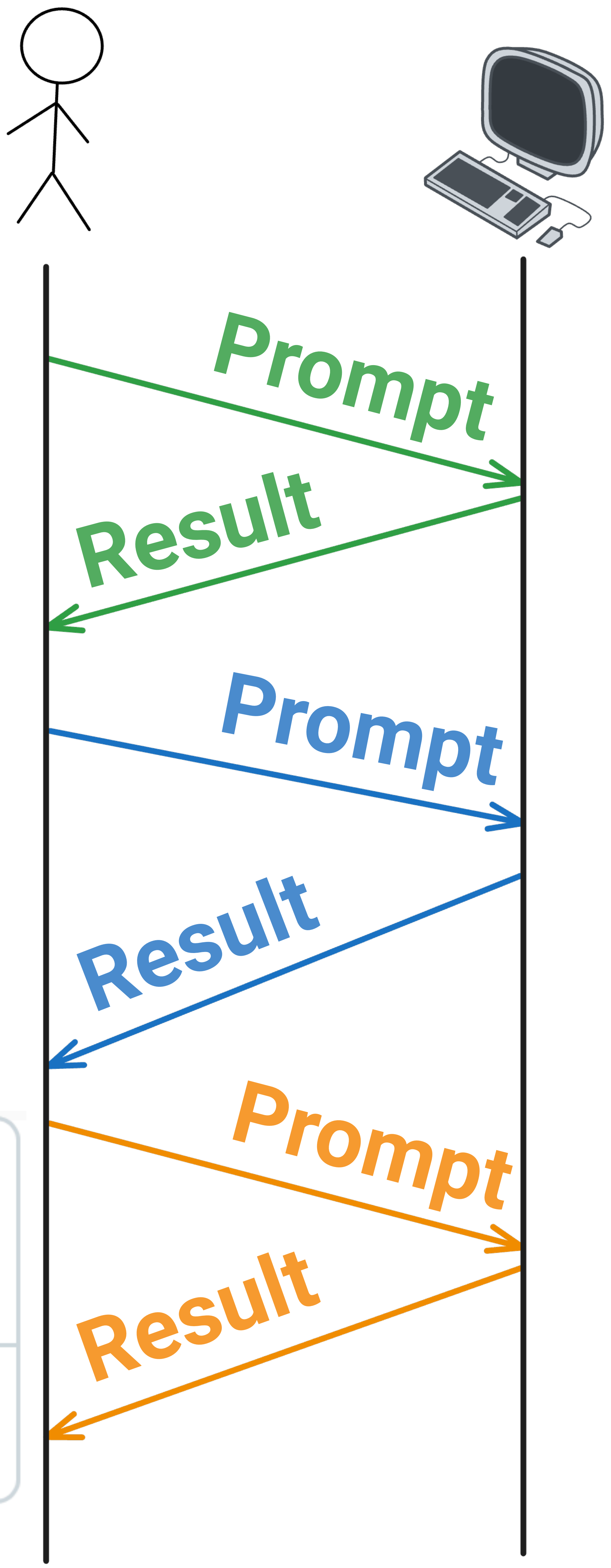


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*How did I miss that?*

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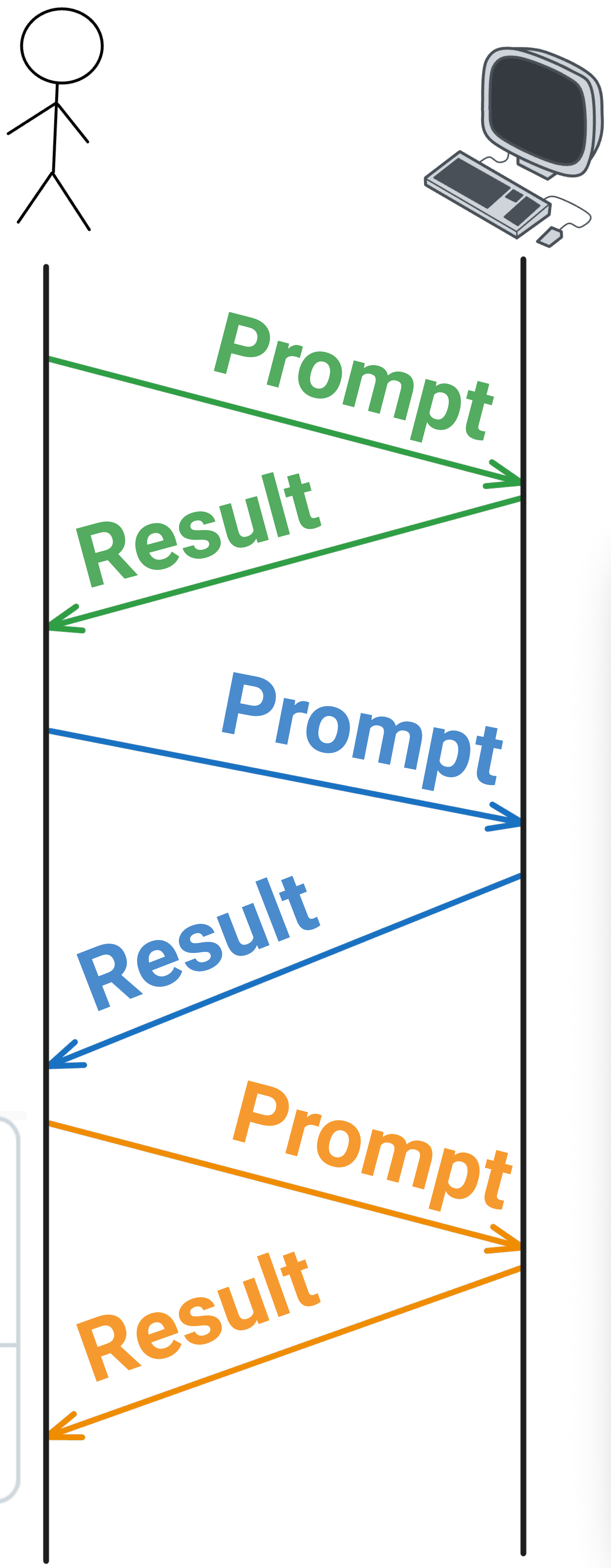


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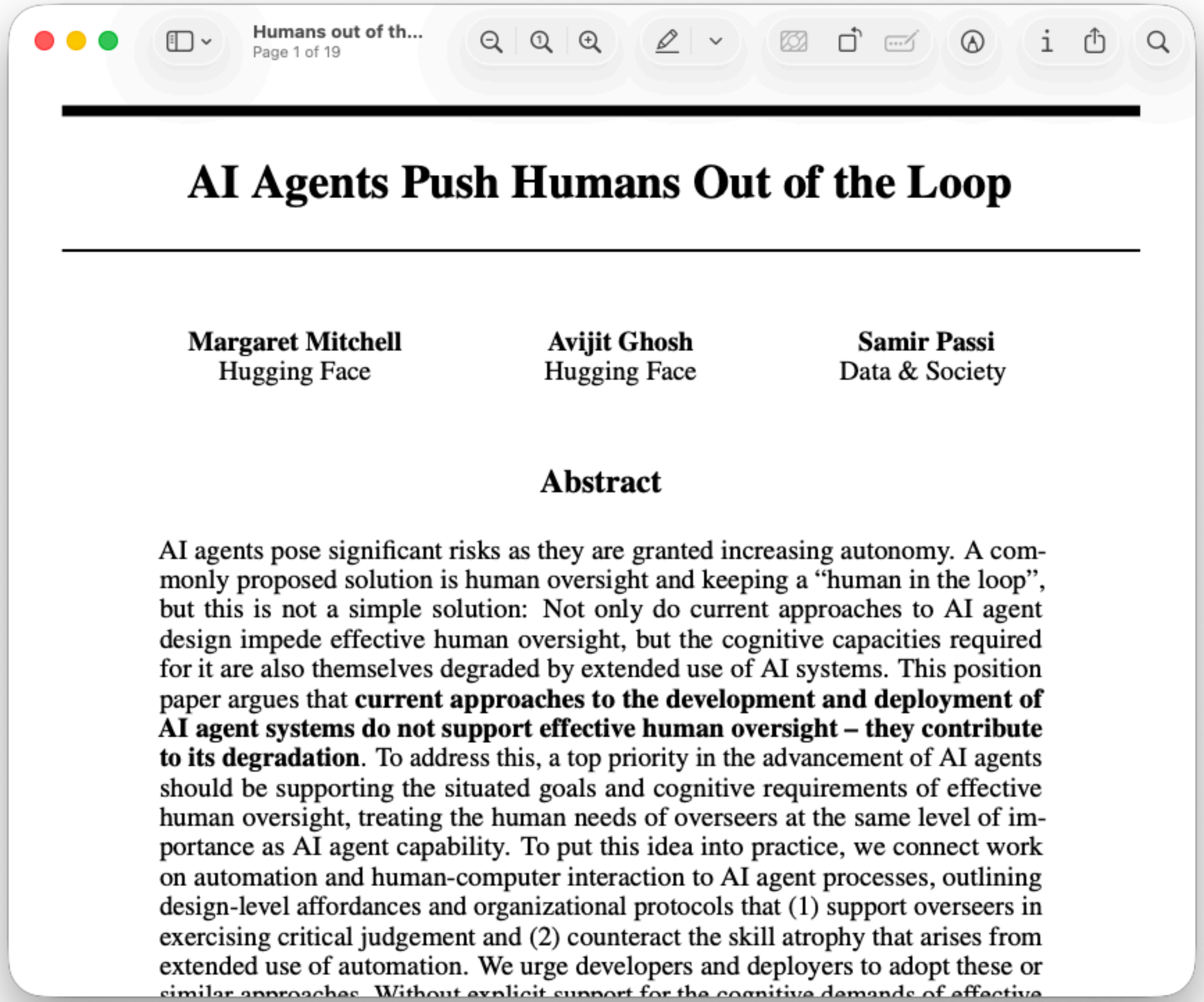


**AI Agents Push Humans Out of the Loop**


**Margaret Mitchell** Hugging Face

**Avijit Ghosh** Hugging Face

**Samir Passi** Data & Society



**Abstract**

AI agents pose significant risks as they are granted increasing autonomy. A commonly proposed solution is human oversight and keeping a "human in the loop", but this is not a simple solution: Not only do current approaches to AI agent design impede effective human oversight, but the cognitive capacities required for it are also themselves degraded by extended use of AI systems. This position paper argues that **current approaches to the development and deployment of AI agent systems do not support effective human oversight – they contribute to its degradation**. To address this, a top priority in the advancement of AI agents should be supporting the situated goals and cognitive requirements of effective human oversight, treating the human needs of overseers at the same level of importance as AI agent capability. To put this idea into practice, we connect work on automation and human-computer interaction to AI agent processes, outlining design-level affordances and organizational protocols that (1) support overseers in exercising critical judgement and (2) counteract the skill atrophy that arises from extended use of automation. We urge developers and deployers to adopt these or similar approaches. Without explicit support for the cognitive demands of effective

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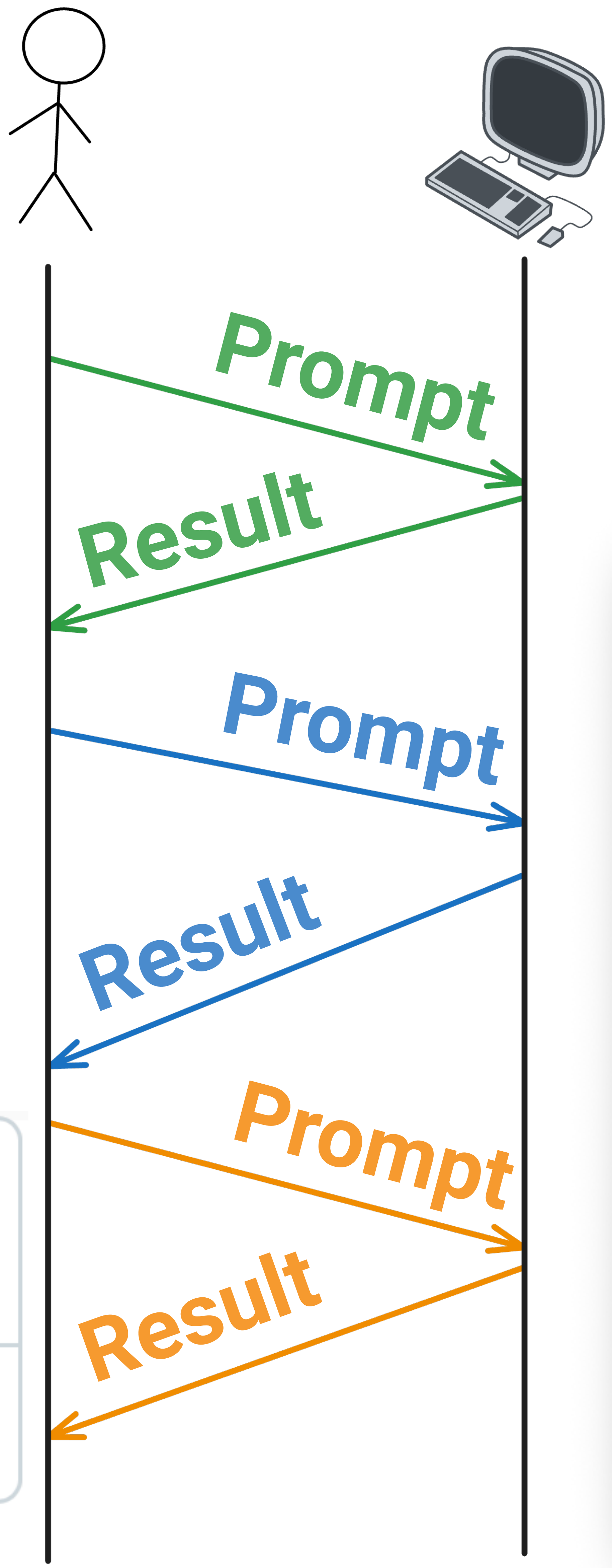


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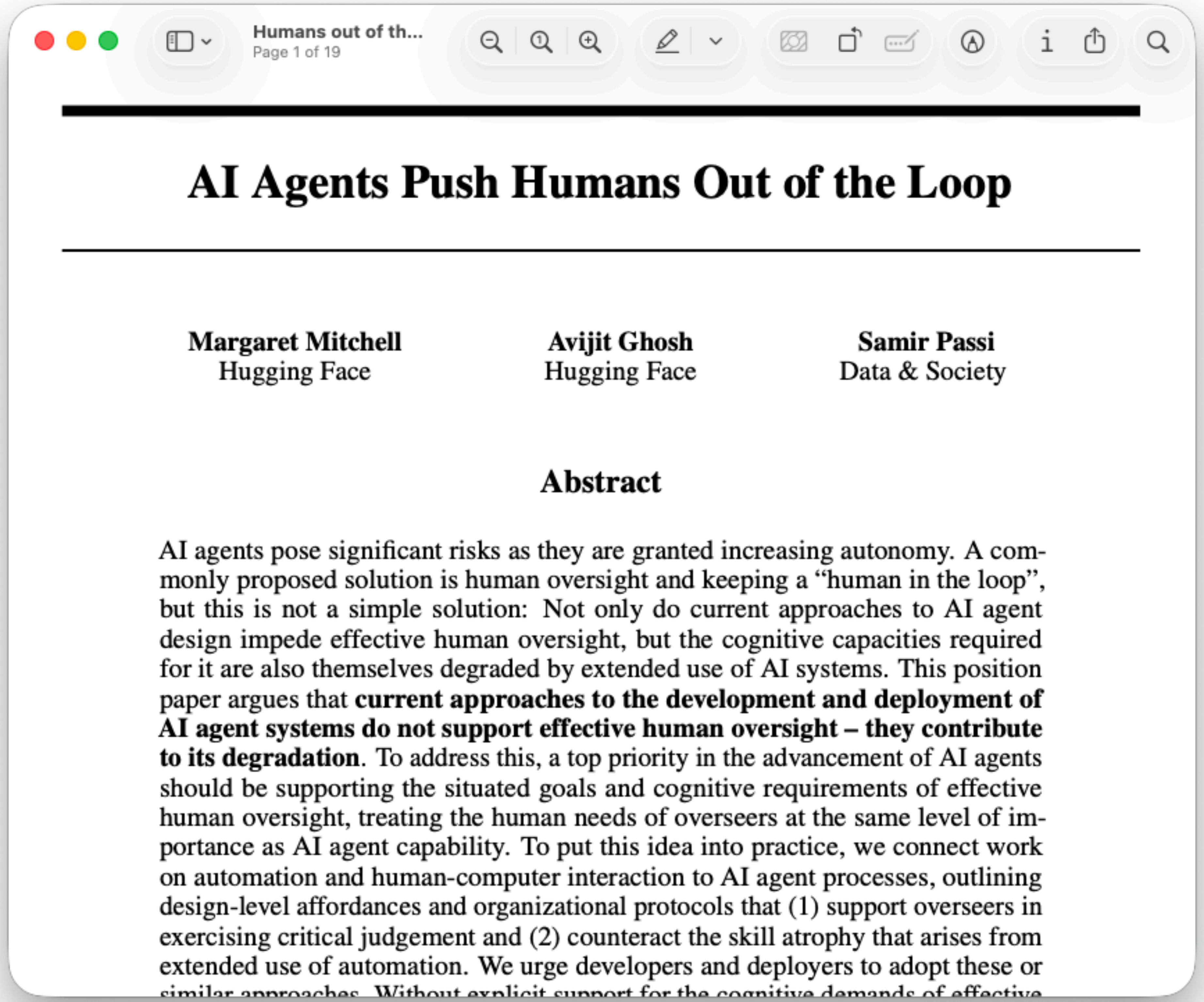


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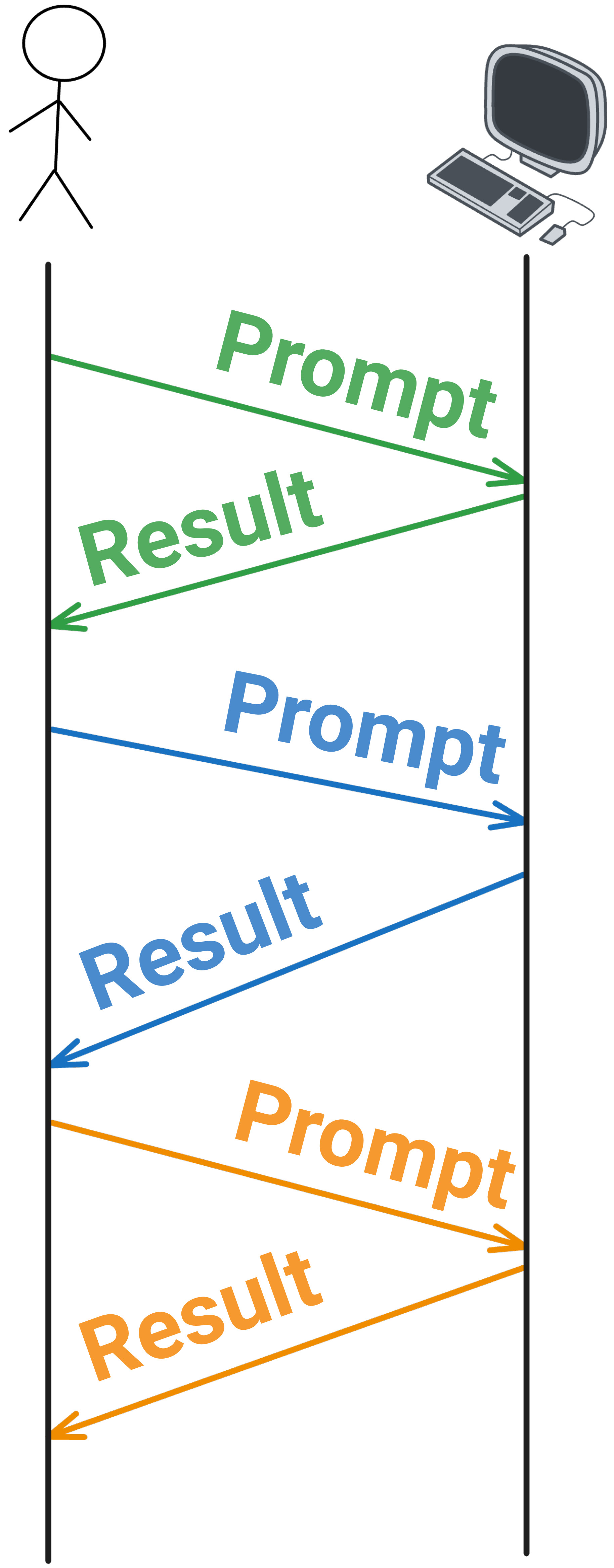

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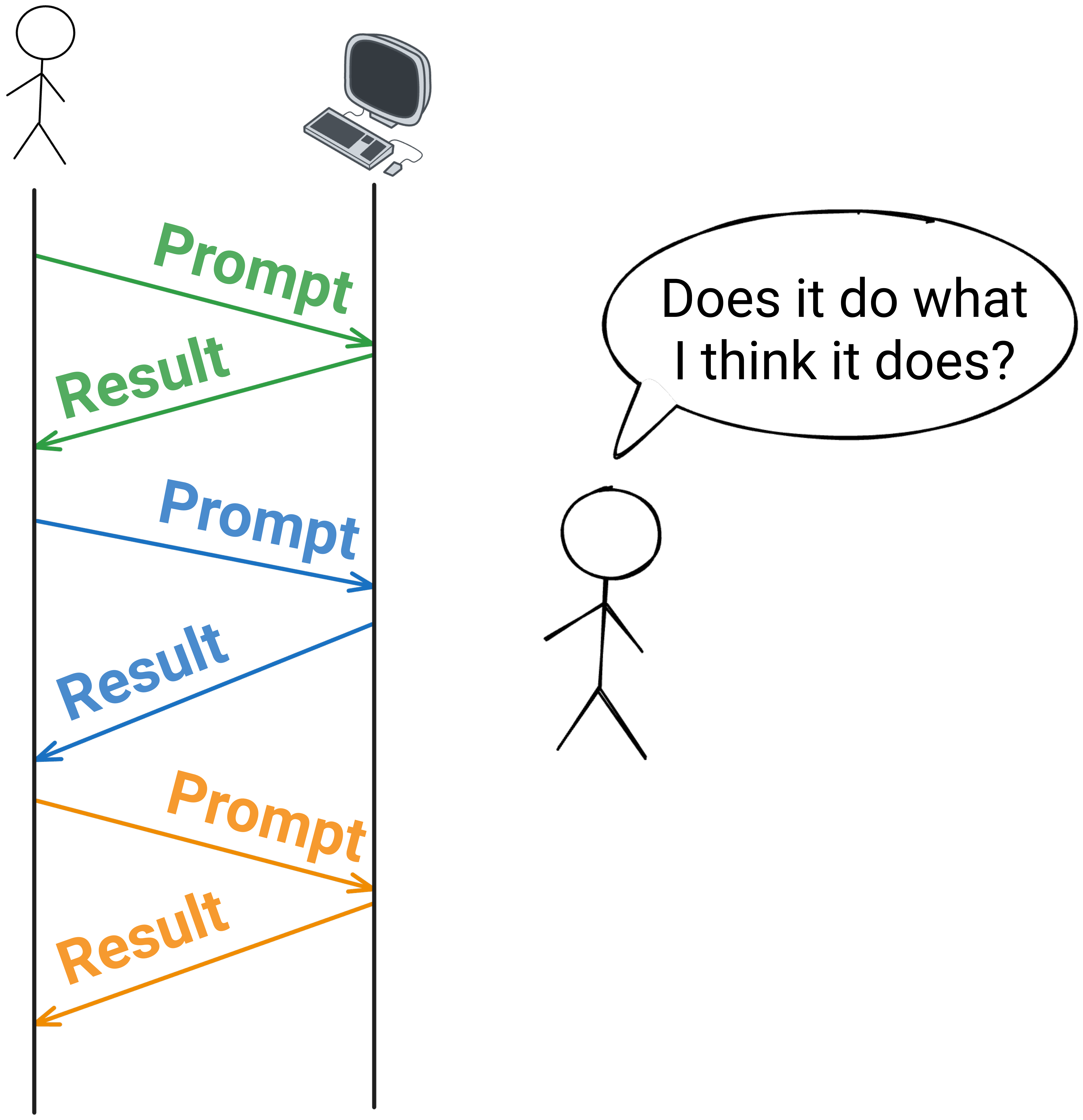

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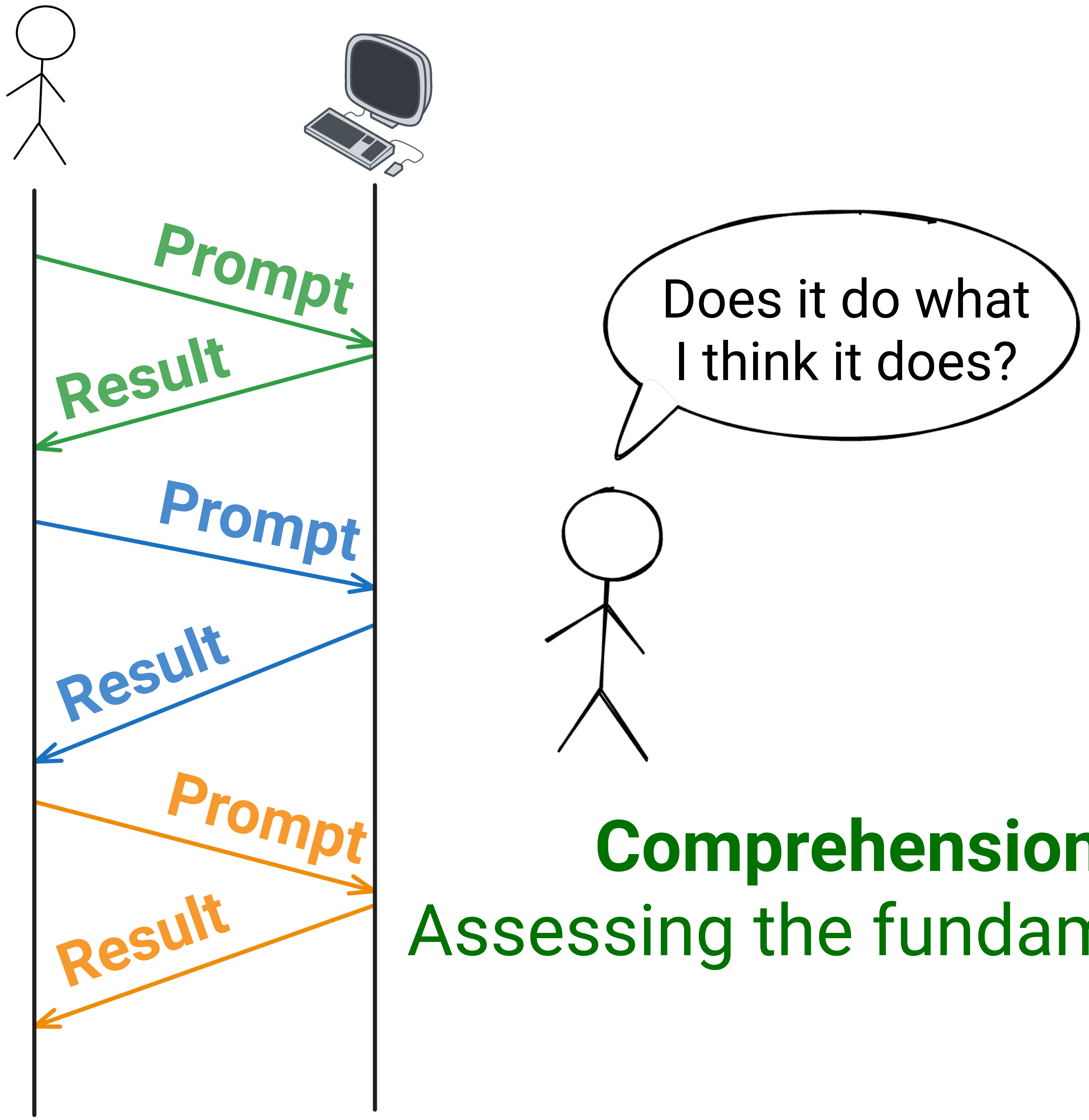


**Comprehension:**
Assessing the fundamentals

# Embedding Comprehension

into Software Engineering

# Fundamental Properties

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- A user never receives stack traces

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- Every element in a sorted list <= the next element
- One person cannot be in two meetings at the same time
- The amount of money remains the same after account transfers

# Fundamental Properties of Addition

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- Zero is the additive identity
  - $a + 0 = a$

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# Fundamental Properties of Addition

- Zero is the additive identity
  - $a + 0 = a$
- Adding something to its inverse gives back the identity
  - $a + (-a) = 0$
- Addition is commutative
  - $a + b = b + a$
- Addition is associative
  - $(a + b) + c = a + (b + c)$

# Testing the Properties of Addition

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```
assertEquals(a, add(a, 0), "Additive Identity");
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assertEquals(a, add(a, 0), "Additive Identity");
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assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
```

# Testing the Properties of Addition

```java
                    @ForAll double a, @ForAll double b
assertEquals(a, add(a, 0), "Additive Identity");
assertEquals(0, add(a, -a), "Additive Inverse");
assertEquals(add(a, b), add(b, a), "Commutativity");
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```java
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {
    assertEquals(a, add(a, 0), "Additive Identity");
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@Property
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {
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}

@Test
void exampleBasedAdditionTest() {
    assertEquals(1.5, add(1.5, 0));
    assertEquals(-101.8, add(-100, -1.8));
    assertEquals(99, add(-1, 100));
    assertEquals(230.5, add(130.5, 100));
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The problem space is explored

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```

Verifies for multiple values of ***a*** and ***b*** at run time

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    assertEquals(0, add(a, -a), "Additive Inverse");
    assertEquals(add(a, b), add(b, a), "Commutativity");
    assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
}

@Test
void exampleBasedAdditionTest() {
    assertEquals(1.5, add(1.5, 0));
    assertEquals(-101.8, add(-100, -1.8
    assertEquals(99, add(-1, 100));
    assertEquals(230.5, add(130.5, 100));
}
```


# Testing the Properties of Addition

```
@Property
void proper
    assertEq
    assertEq
    assertEq
    assertEq
}

@Test
void exampl
    assertEq
    assertEq
    assertEq
    assertEq
}
```

```
                                |-----------jqwik-----------
tries = 285                     | # of calls to property
checks = 285                    | # of not rejected calls
generation = RANDOMIZED         | parameters are randomly generated
...

Original Sample
---------------
  a: 15.6
  b: -1.49

  Original Error
  --------------
  org.opentest4j.AssertionFailedError:
    Associativity ==> expected: <15.11> but was: <15.110000000000001>
```

# Testing the Properties of Addition

```
@Property
void proper
    assertEq
    assertEq
    assertEq
    assertEq
}

@Test
void exampl
    assertEq
    assertEq
    assertEq
    assertEq
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Original Sample
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  a: 15.6
  b: -1.49

  Original Error
  --------------
  org.opentest4j.AssertionFailedError:
    Associativity ==> expected: <15.11> but was: <15.110000000000001>
```

```
jshell>
```

# Testing the Properties of Addition

```
@Property
void proper
    assertEq
    assertEq
    assertEq
    assertEq
}

@Test
void exampl
    assertEq
    assertEq
    assertEq
    assertEq
}
```

```
                              |-----------jqwik-----------
tries = 285                   | # of calls to property
checks = 285                  | # of not rejected calls
generation = RANDOMIZED       | parameters are randomly generated
...

Original Sample
---------------
  a: 15.6
  b: -1.49

  Original Error
  --------------
  org.opentest4j.AssertionFailedError:
    Associativity ==> expected: <15.11> but was: <15.110000000000001>
```

```
jshell> 1 + (15.6 + -1.49)
$1 ==> 15.11
```

# Testing the Properties of Addition

```
@Property
void proper
    assertEq
    assertEq
    assertEq
    assertEq
}

@Test
void exampl
    assertEq
    assertEq
    assertEq
    assertEq
}
```

```
                                |-----------jqwik-----------
tries = 285                     | # of calls to property
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---------------
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  Original Error
  --------------
  org.opentest4j.AssertionFailedError:
    Associativity ==> expected: <15.11> but was: <15.110000000000001>
```

```
jshell> 1 + (15.6 + -1.49)
$1 ==> 15.11

jshell> (1 + 15.6) + -1.49
$2 ==> 15.110000000000001
```

# Testing the Properties of Addition

```
@Property
void proper
    assertEq
    assertEq
    assertEq
    assertEq
}

@Test
void exampl
    assertEq
    assertEq
    assertEq
    assertEq
}
```

```
                              |-----------jqwik-----------
tries = 285                   | # of calls to property
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generation = RANDOMIZED       | parameters are randomly generated
...

Original Sample
---------------
  a: 15.6
  b: -1.49

  Original Error
  --------------
  org.opentest4j.AssertionFailedError:
    Associativity ==> expected: <15.11> but was: <15.110000000000001>
```

```
jshell> 1 + (15.6 + -1.49)
$1 ==> 15.11

jshell> (1 + 15.6) + -1.49
$2 ==> 15.110000000000001

jshell> 15.6d == 15.6f
$3 ==> false
```

# Testing the Properties of Addition

```java
@Property
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {
    assertEquals(a, add(a, 0), "Additive Identity");
    assertEquals(0, add(a, -a), "Additive Inverse");
    assertEquals(add(a, b), add(b, a), "Commutativity");
    assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
}
```


```java
@Test
void exampleBasedAdditionTest() {
    assertEquals(1.5, add(1.5, 0));
    assertEquals(-101.8, add(-100, -1.8));
    assertEquals(99, add(-1, 100));
    assertEquals(230.5, add(130.5, 100));
}
```

# Examples → Exploration

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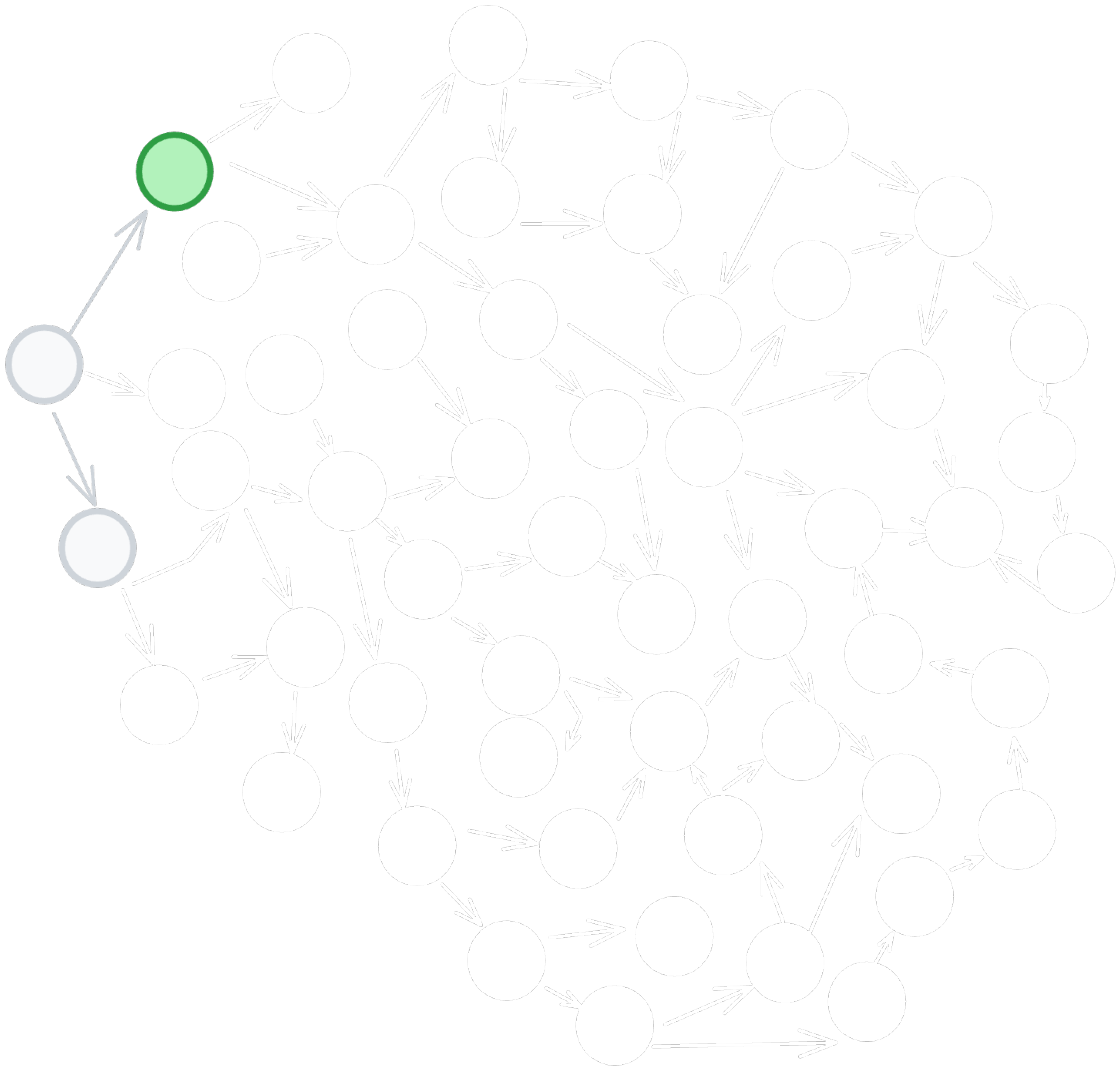
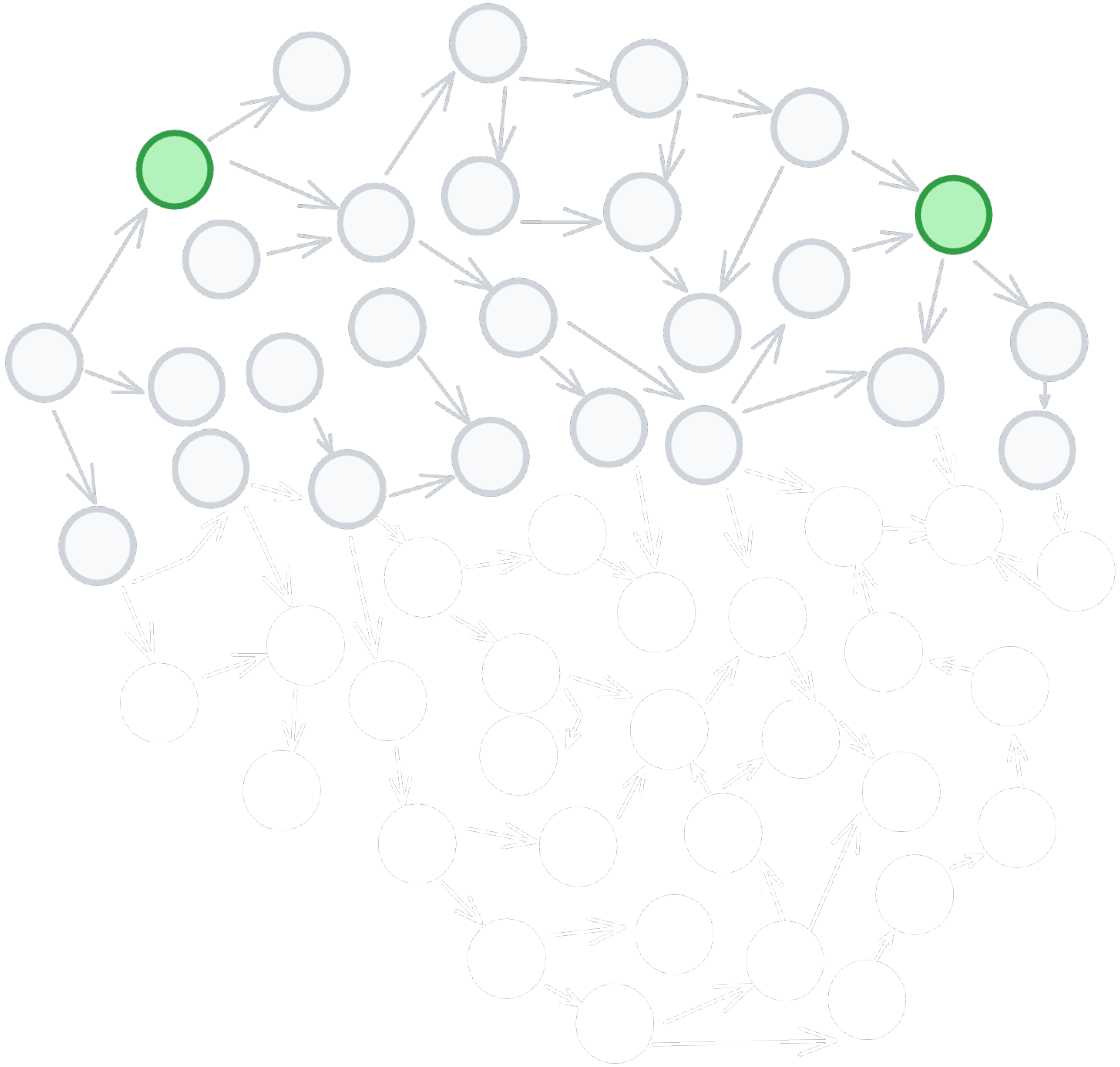
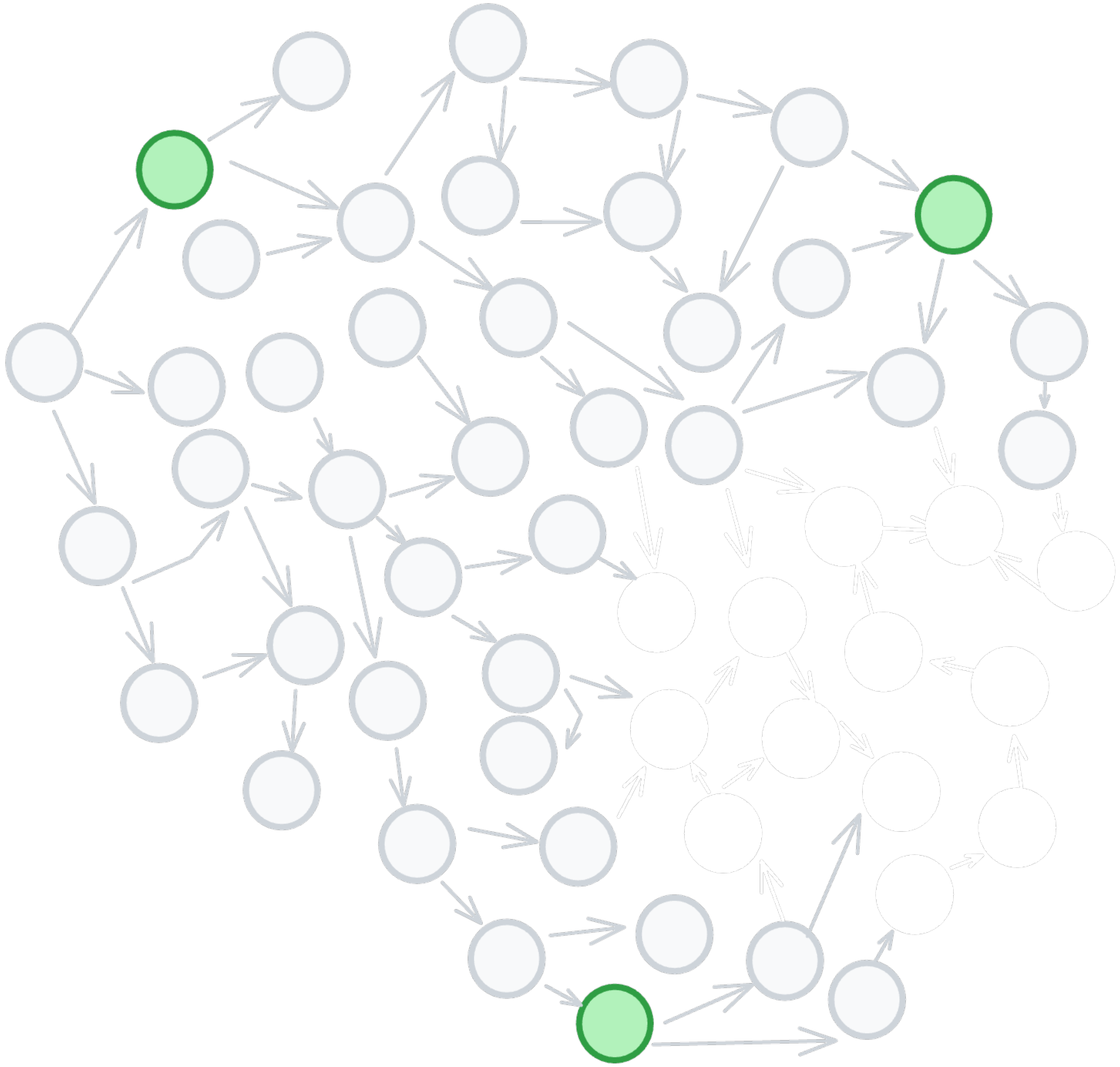


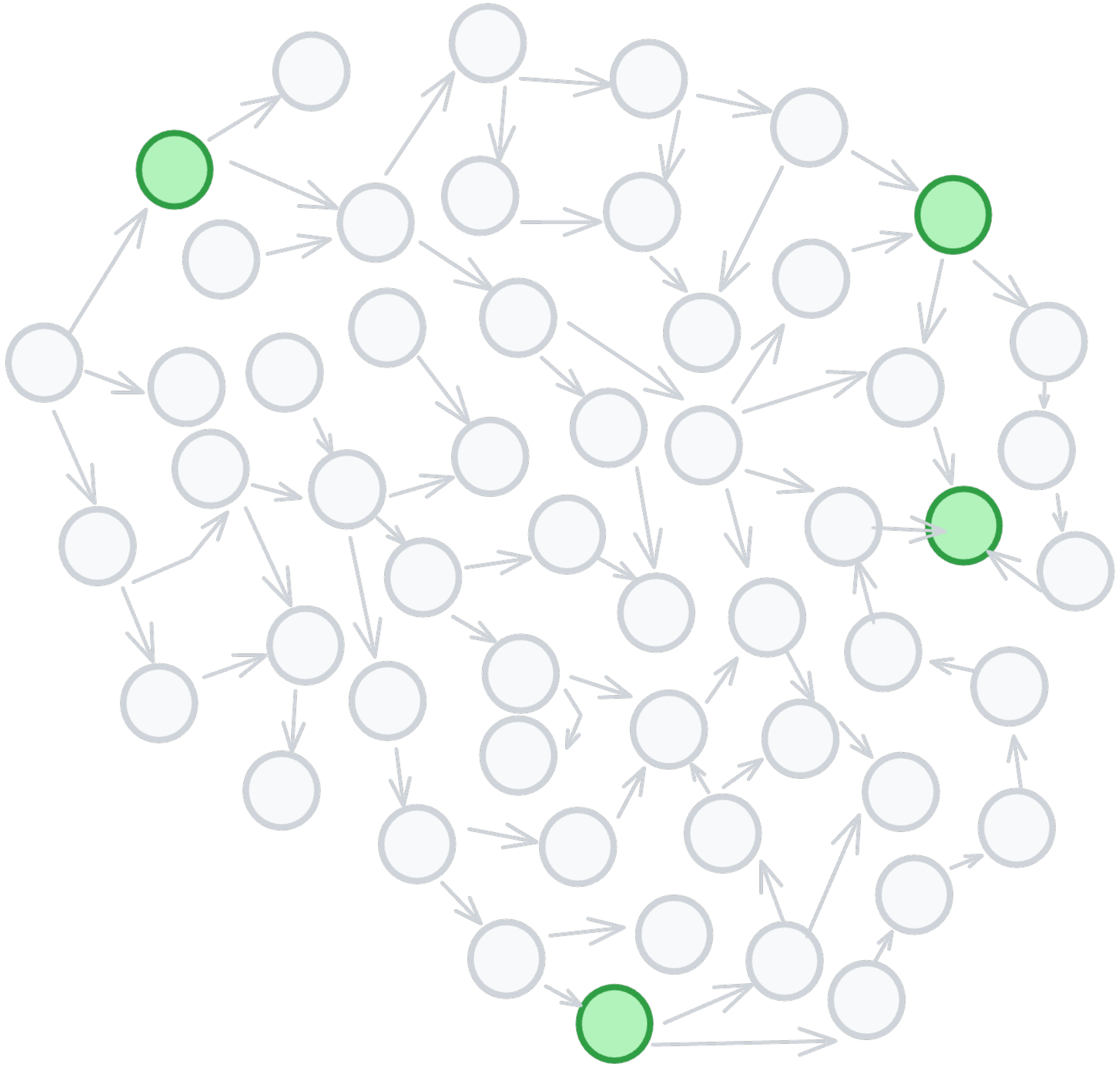
# Examples → Exploration



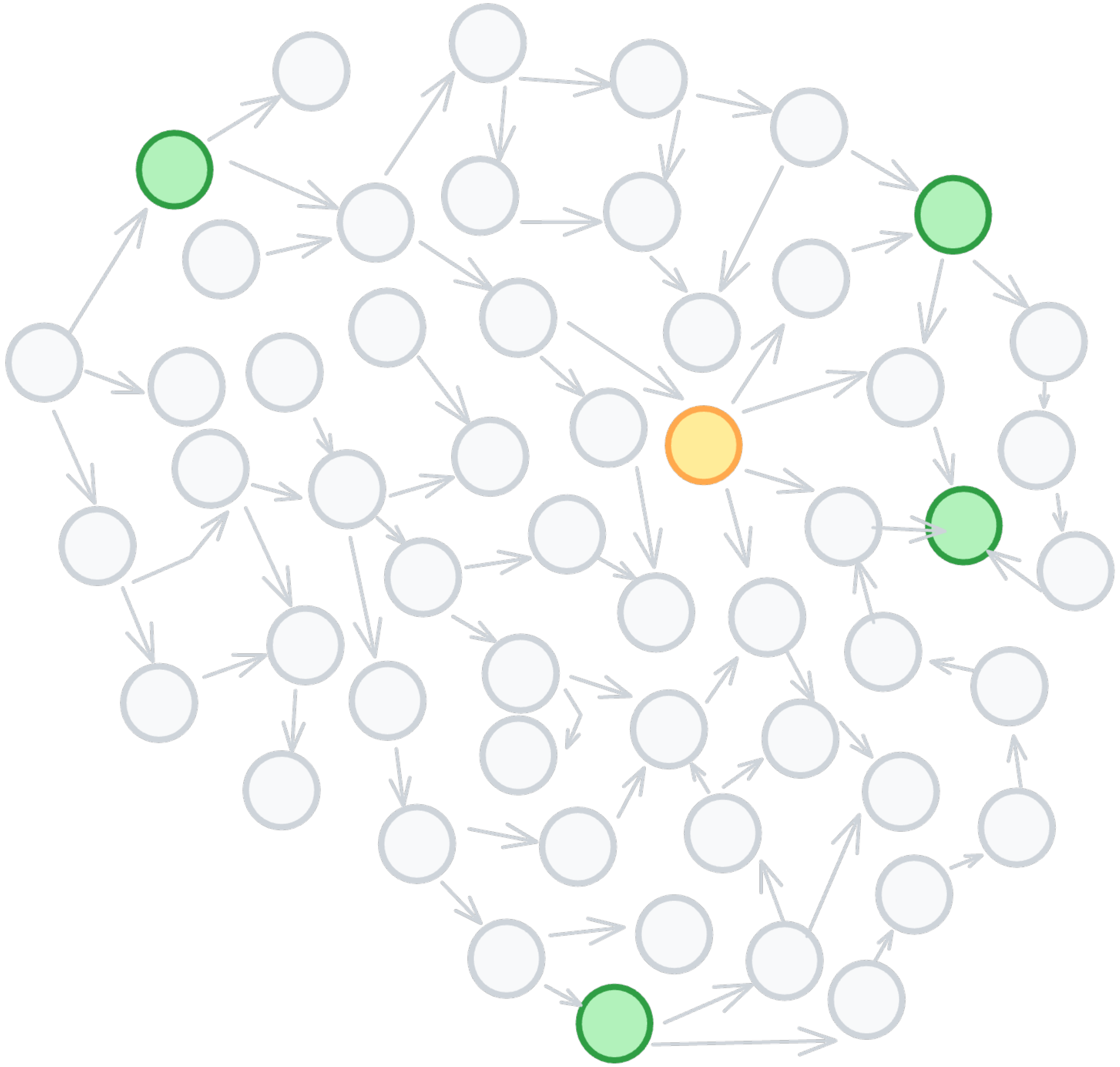
# Examples → Exploration



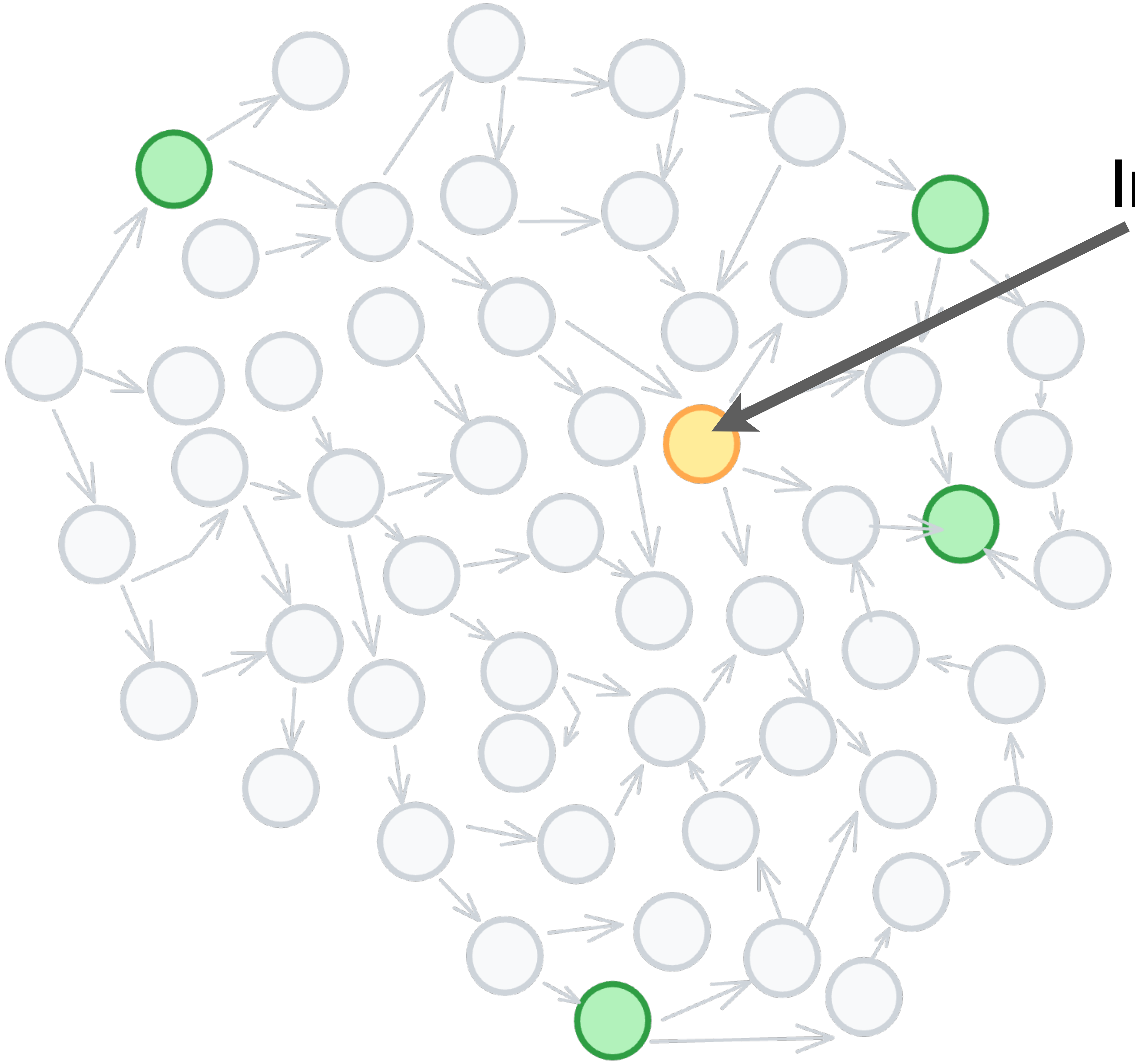
# Examples → Exploration



# Examples → Exploration

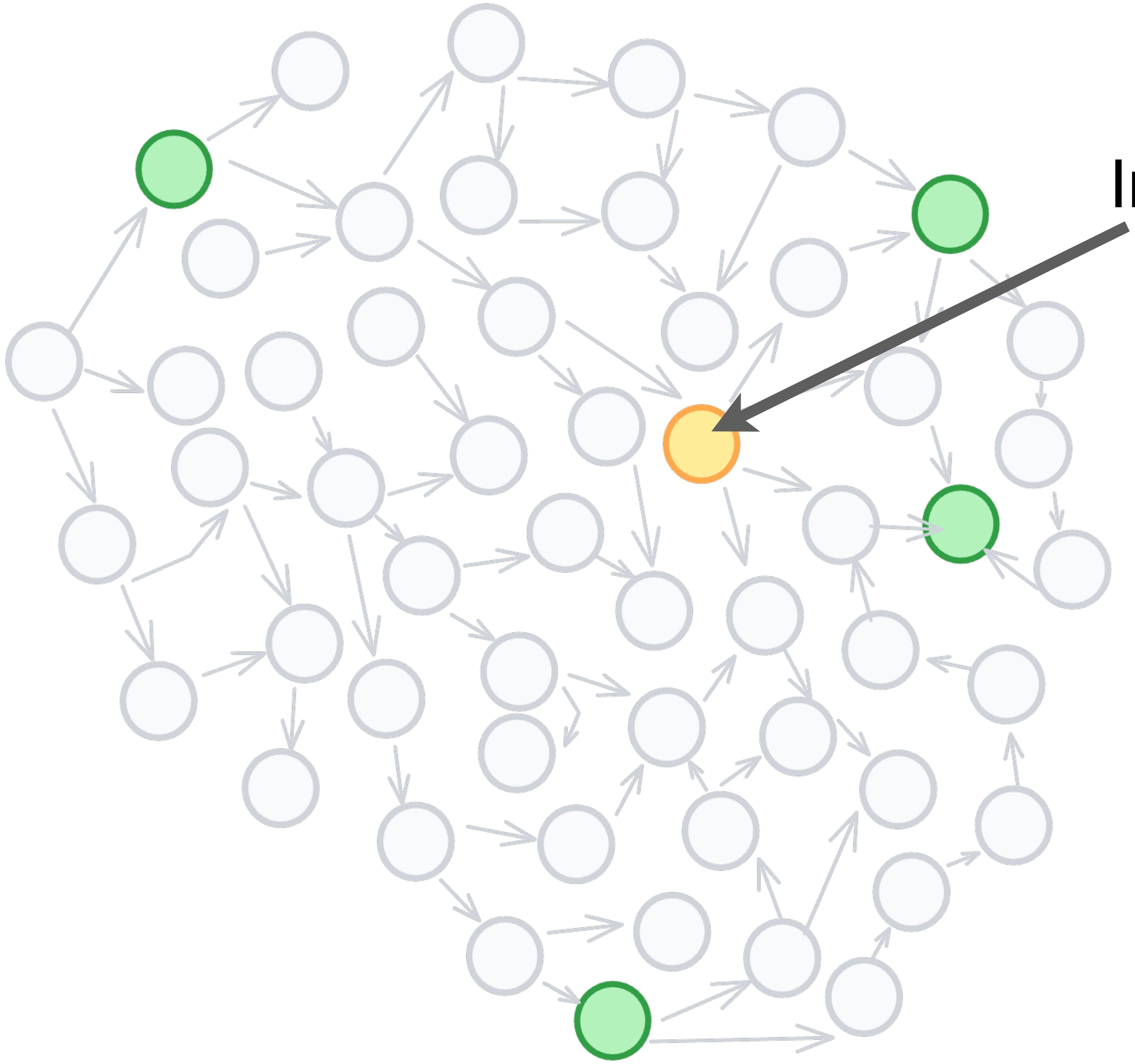


# Examples → Exploration



Intuitively we might have added this test case

# Examples → Exploration



Intuitively we might have added this test case

**We can't be sure that the LLM will**

# Examples → Exploration

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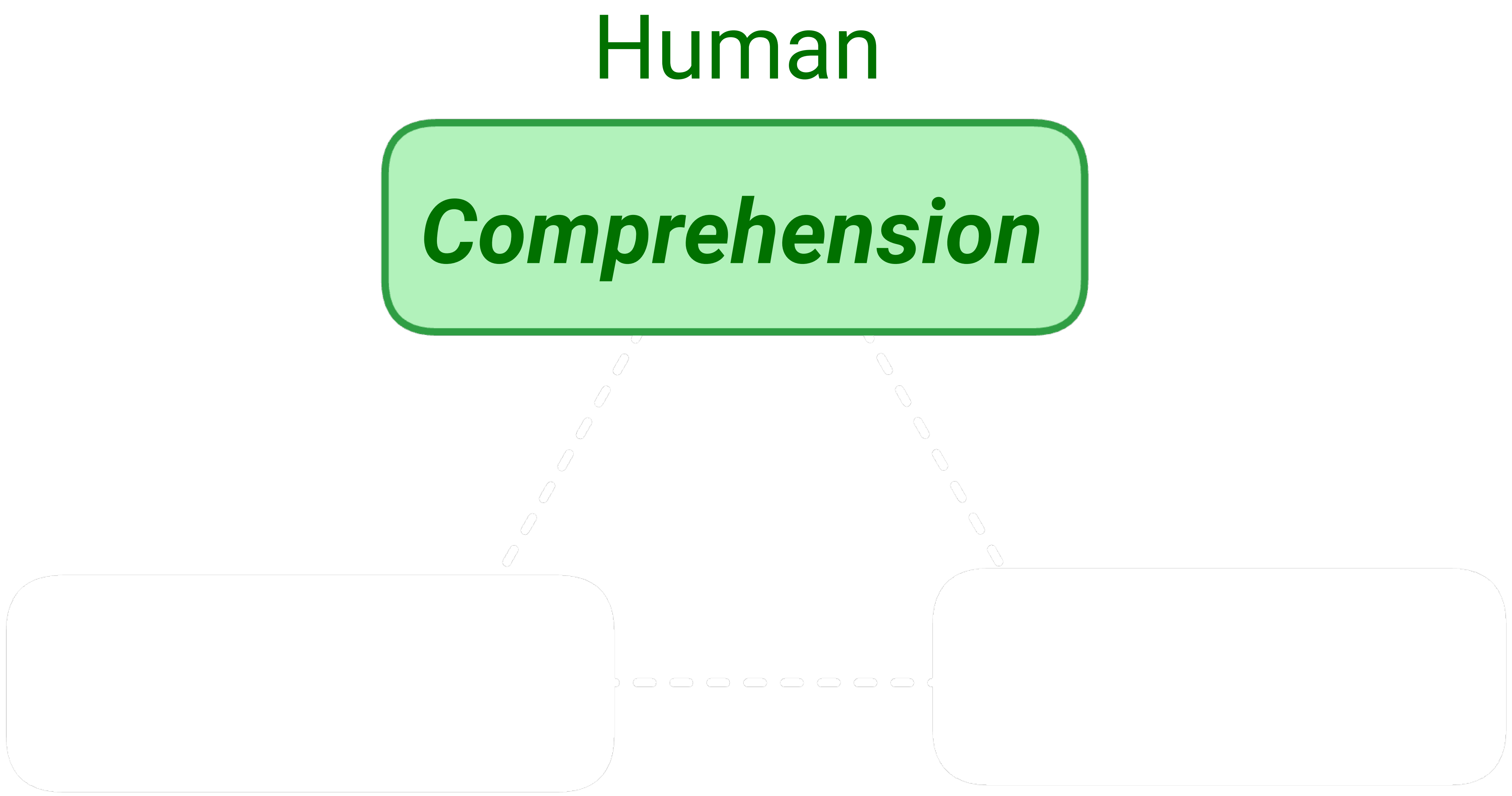
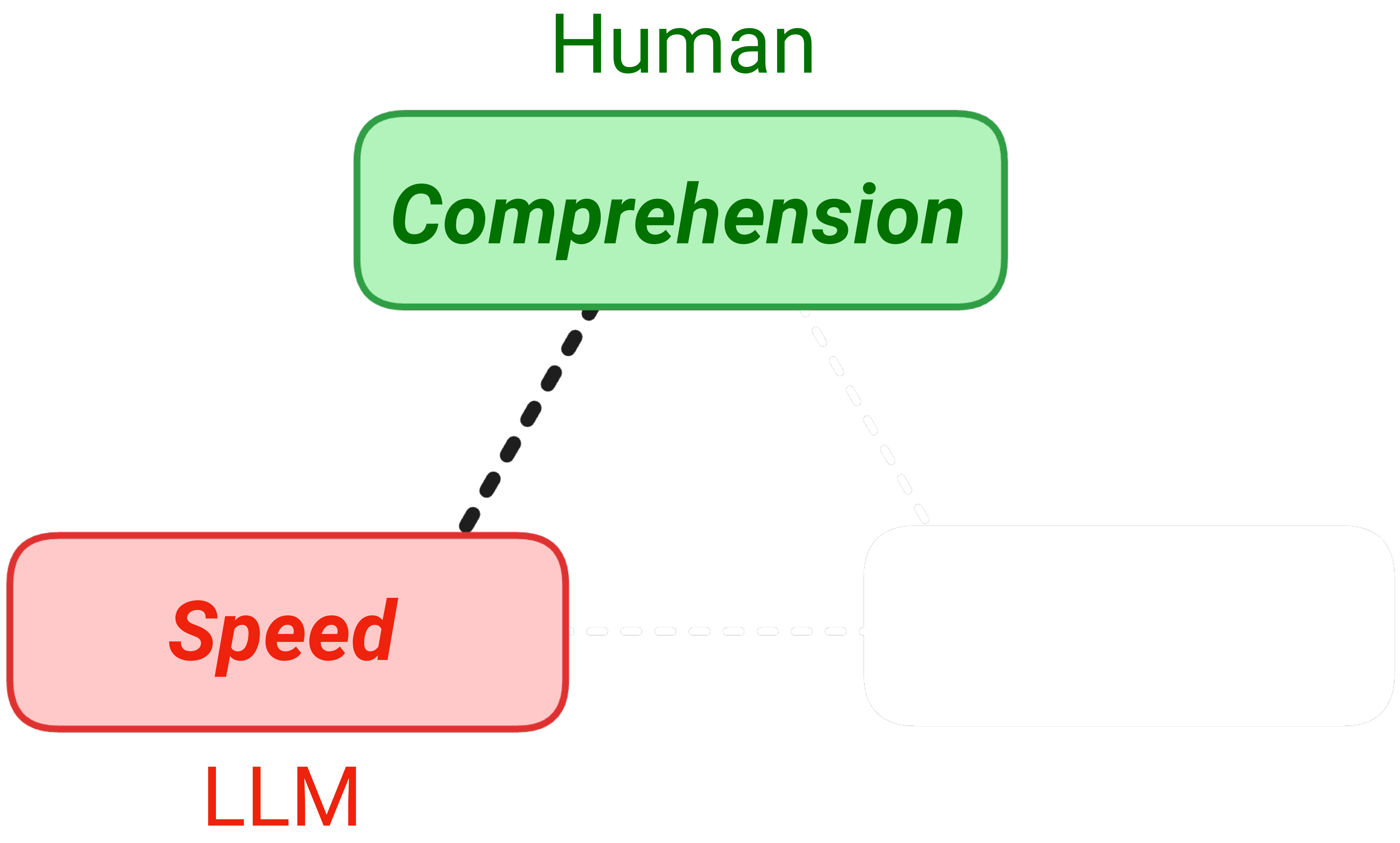


# Examples → Exploration



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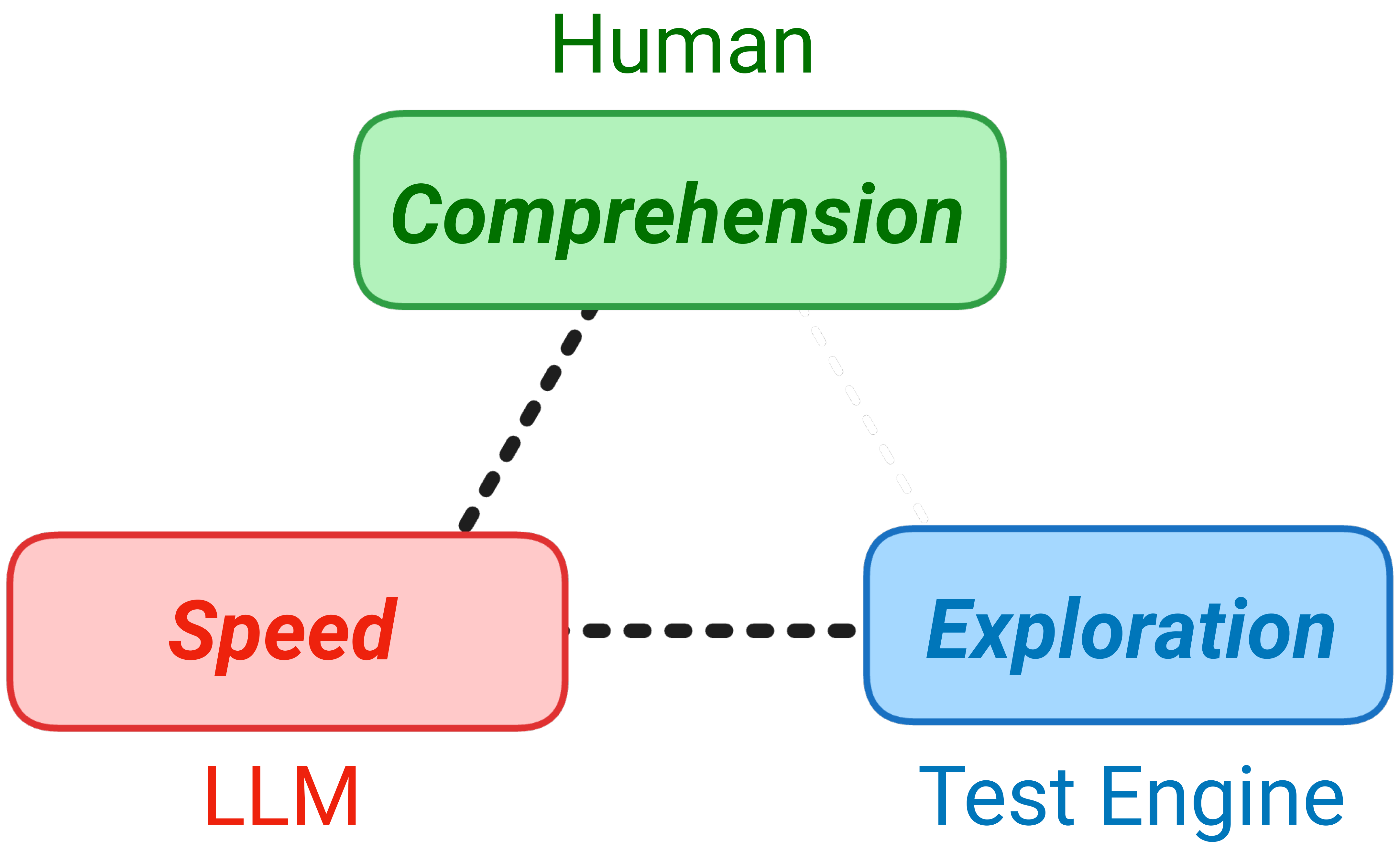
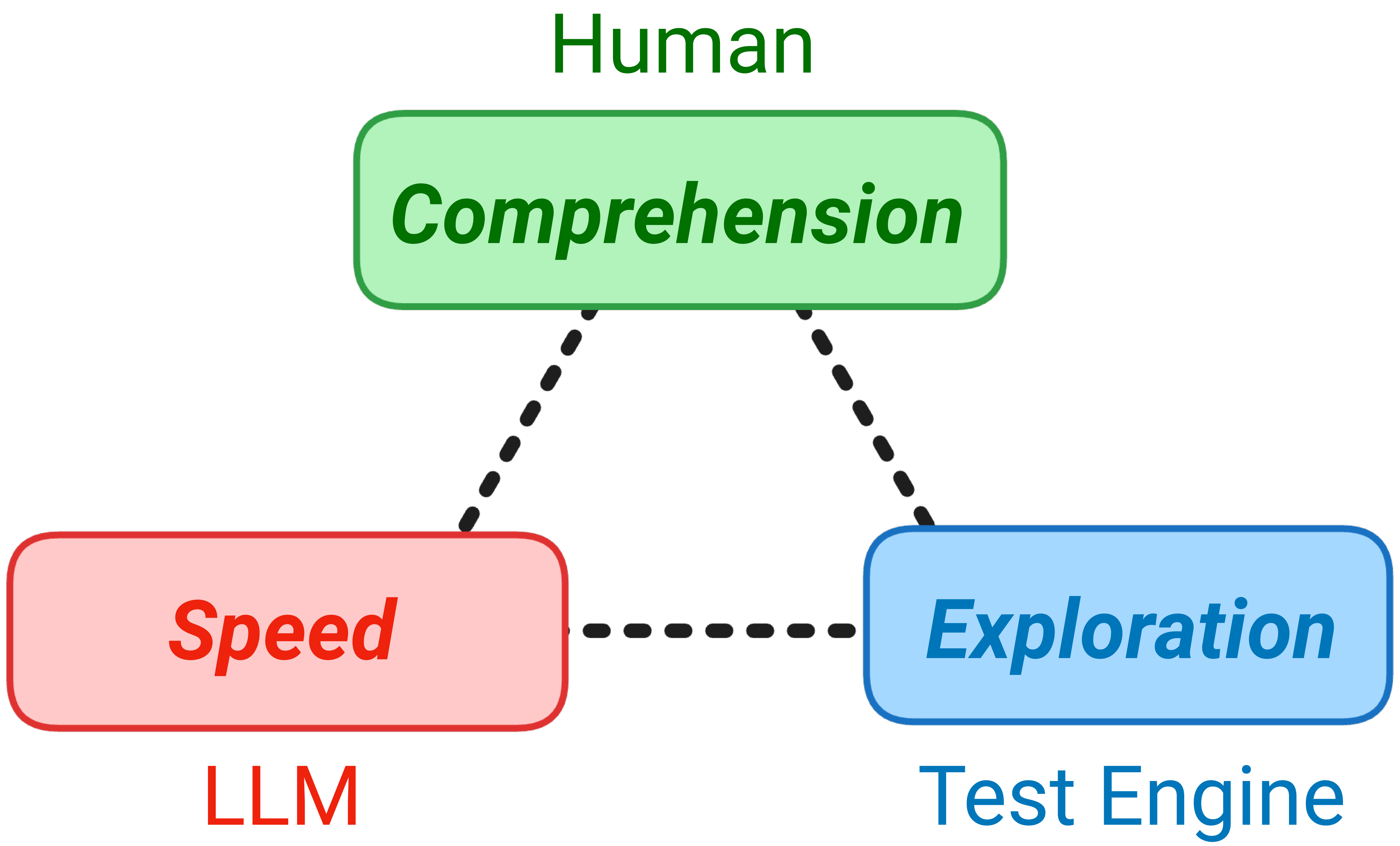


# Examples → Exploration

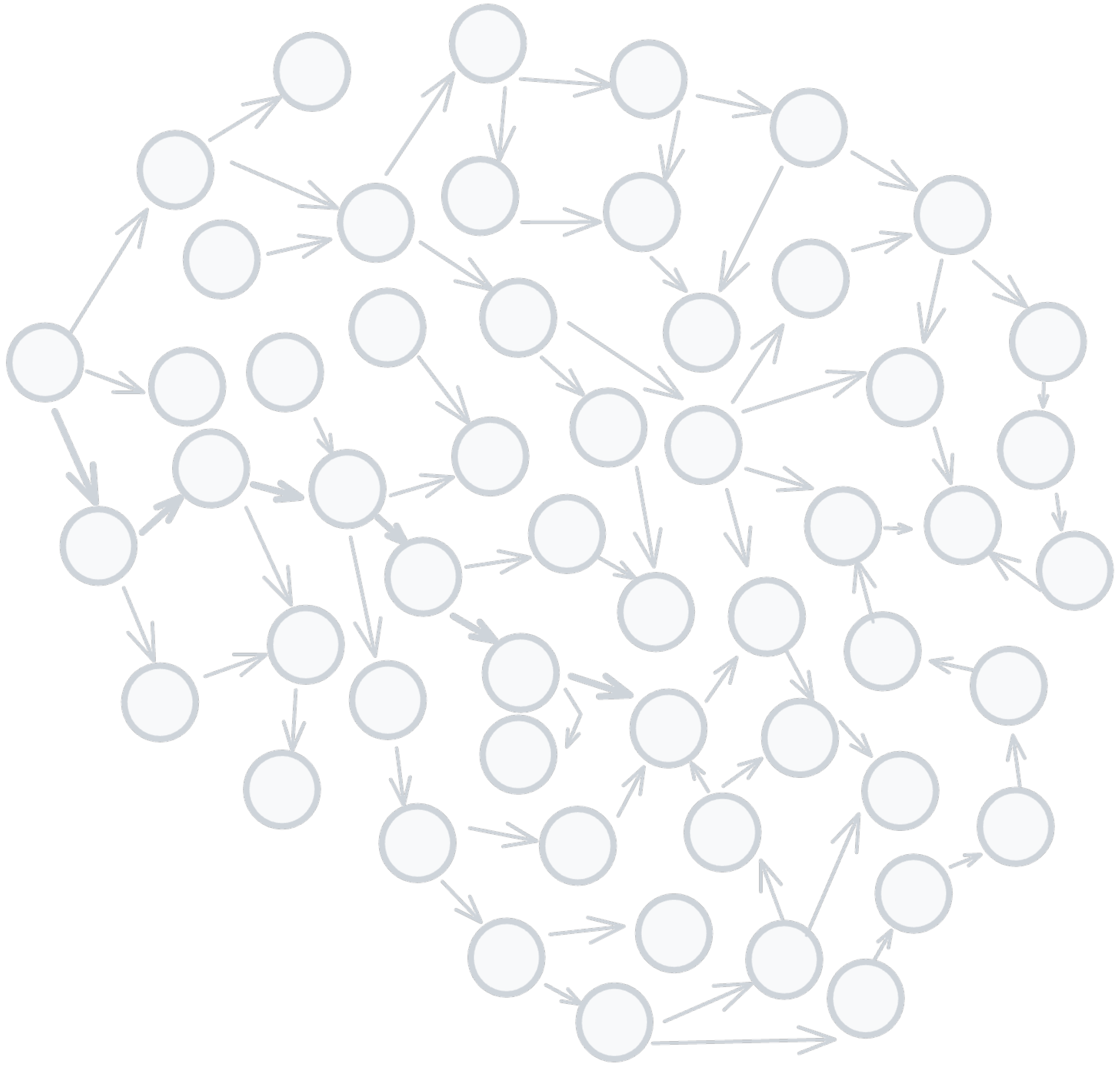


# Examples → Exploration

LLMs generate a solution

# Examples → Exploration

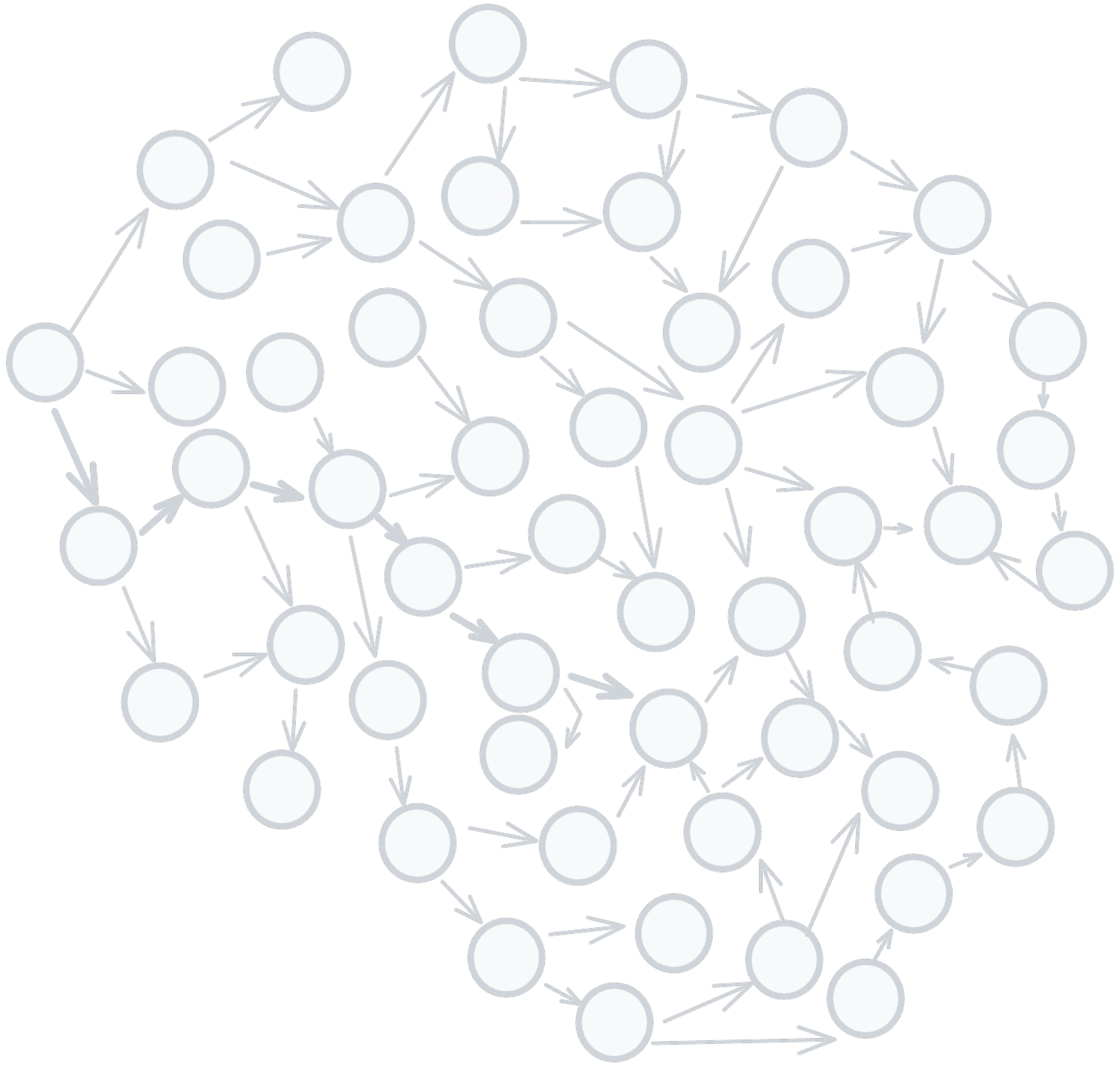
LLMs generate a solution



# Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true



# Examples → Exploration

LLMs generate a solution

Humans define what
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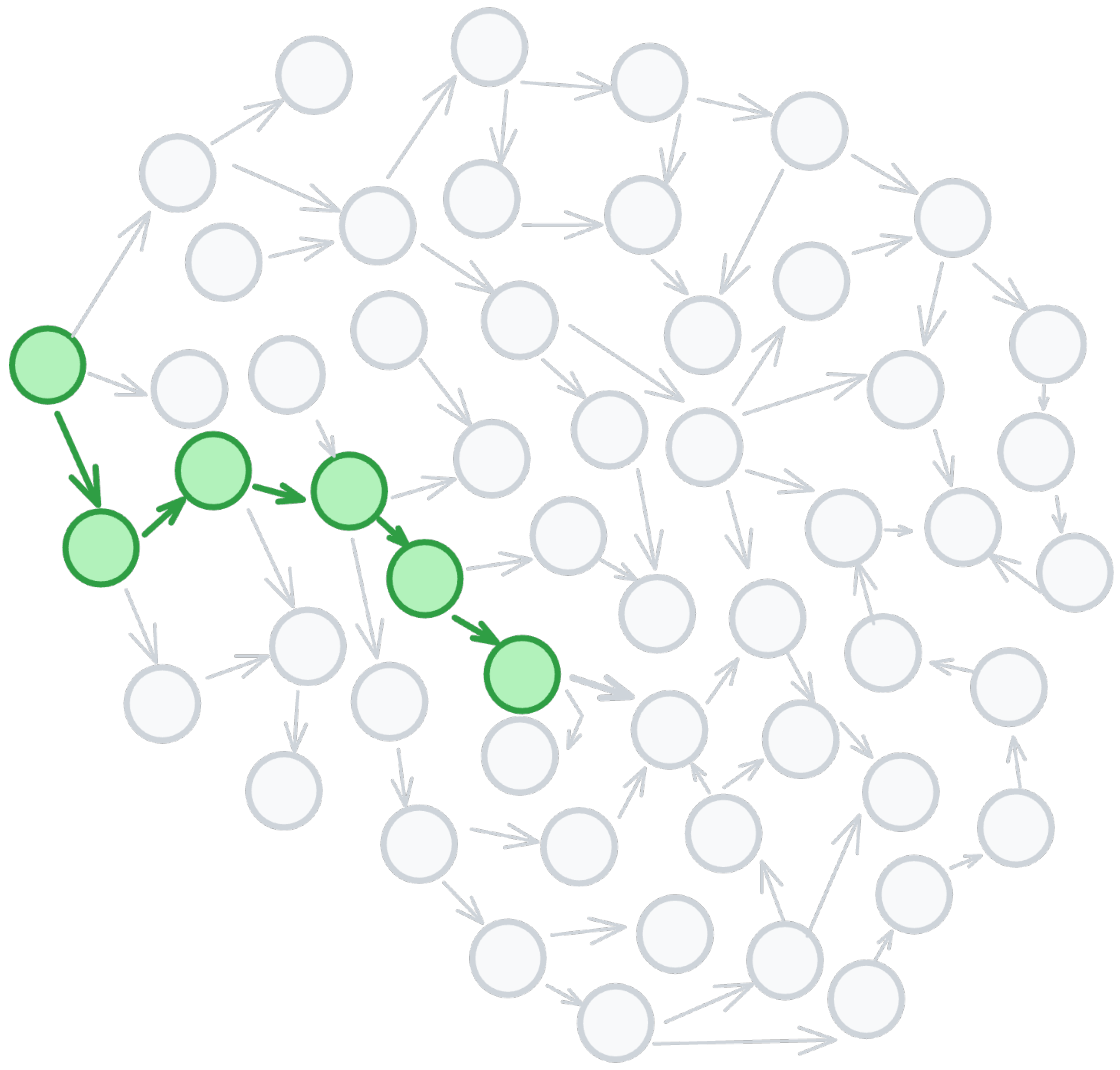
The test engine explores
the problem space

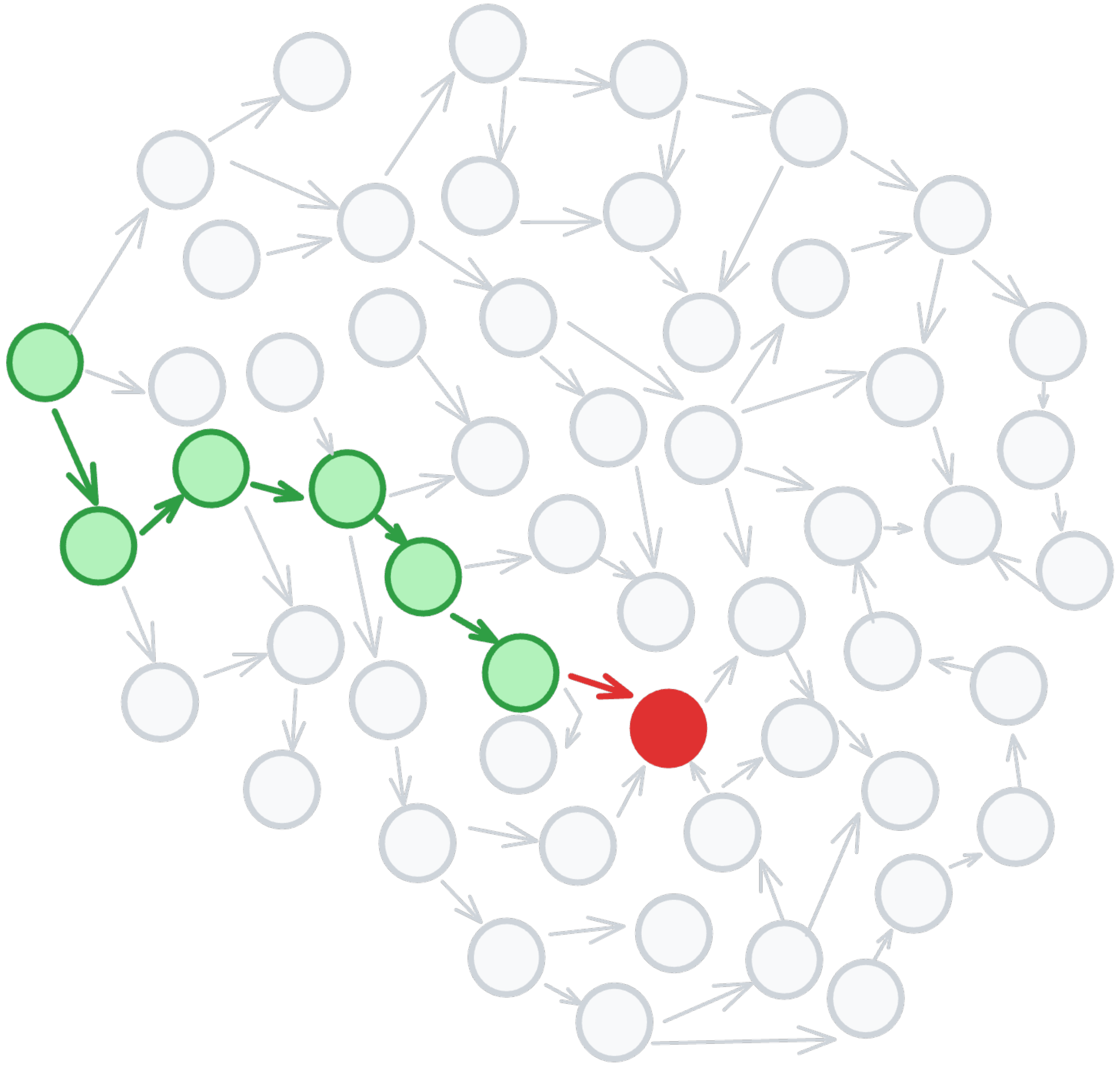


# Examples → Exploration

LLMs generate a solution

Humans define what must always hold true

The test engine explores the problem space



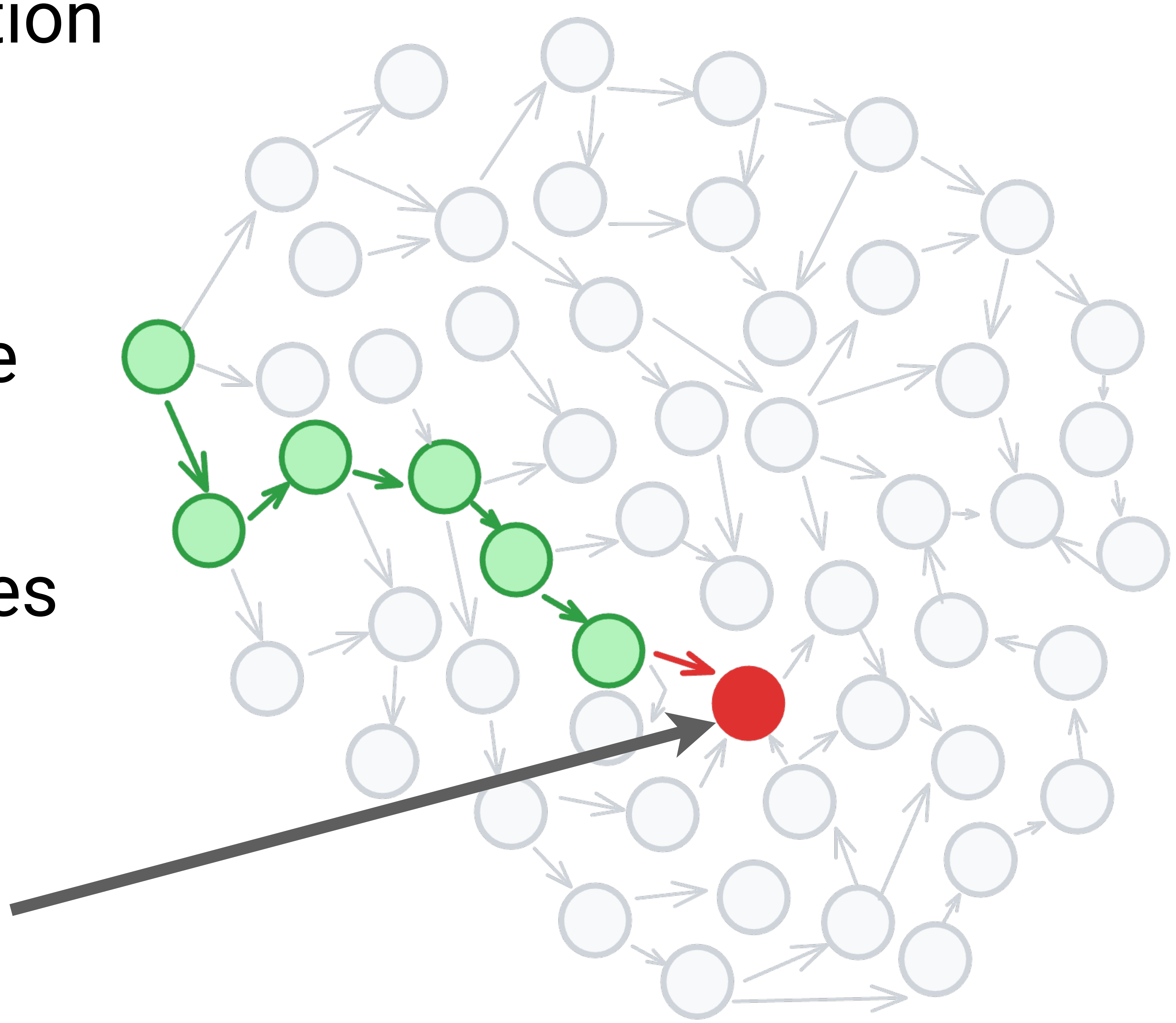
# Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

The test engine explores
the problem space

The test engine ***finds***
a counter example



# Failure

# Failure → Shrinking

# Failure → Shrinking → Understanding

# Failure → Shrinking → Understanding

```
Original Sample
---------------
  integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
             249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,
             8045, -373, 147465, -3, -58, 514, -177, 97, 1386116950, 2953, 3804]
  Expecting actual:
    1386116950
  to be less than or equal to:
    3804
```

# Failure → Shrinking → Understanding

```
Original Sample
---------------
  integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
             249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,
             8045, -373, 147465, -3, -58, 514, -177, 97, 1386116950, 2953, 3804]
  Expecting actual:
    1386116950
  to be less than or equal to:
    3804
```

# Failure → Shrinking → Understanding

```
Shrunk Sample (11 steps)
------------------------
  integers: [0, -1]
  Expecting actual:
    0
  to be less than or equal to:
    -1

Original Sample
---------------
  integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
             249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,
             8045, -373, 147465, -3, -58, 514, -177, 97, 1386116950, 2953, 3804]
  Expecting actual:
    1386116950
  to be less than or equal to:
    3804
```

# Failure → Shrinking → Understanding

```
Shrunk Sample (11 steps)
------------------------
  integers: [0, -1]
  Expecting actual:
    0
  to be less than or equal to:
    -1
```

```
for (int i = 0; i < n - 1; i++) {
   for (int j = 0; j < n - i - 2; j++) {
       if (arr.get(j) > arr.get(j + 1)) {
           int temp = arr.get(j);
           arr.set(j, arr.get(j + 1));
           arr.set(j + 1, temp);
       }
   }
}
```

# Failure → Shrinking → Understanding

```
Shrunk Sample (11 steps)
------------------------
  integers: [0, -1]
  Expecting actual:
    0
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for (int i = 0; i < n - 1; i++) {
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```

Last item never gets sorted

# Failure → Shrinking → Understanding

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         int temp = arr.get(j);
         arr.set(j, arr.get(j + 1));
         arr.set(j + 1, temp);
      }
   }
}
```

Last item never gets sorted

Shrunk samples are easier to **understand**

# How Did I Miss That?

# Meeting Scheduling

# Meeting Scheduling

Existing Meeting

# Meeting Scheduling

# Meeting Scheduling

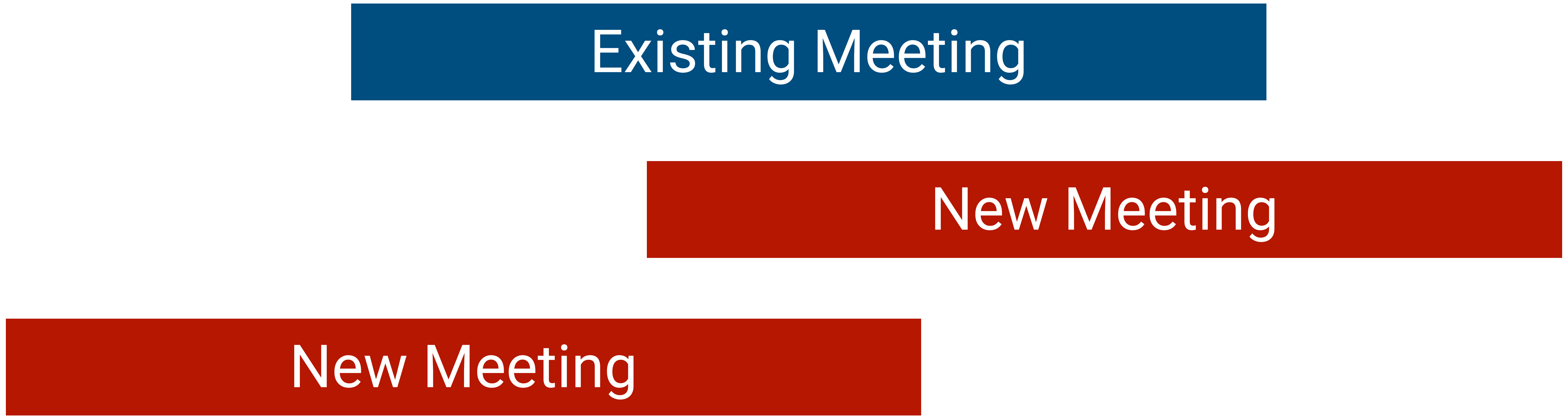
Existing Meeting

New Meeting

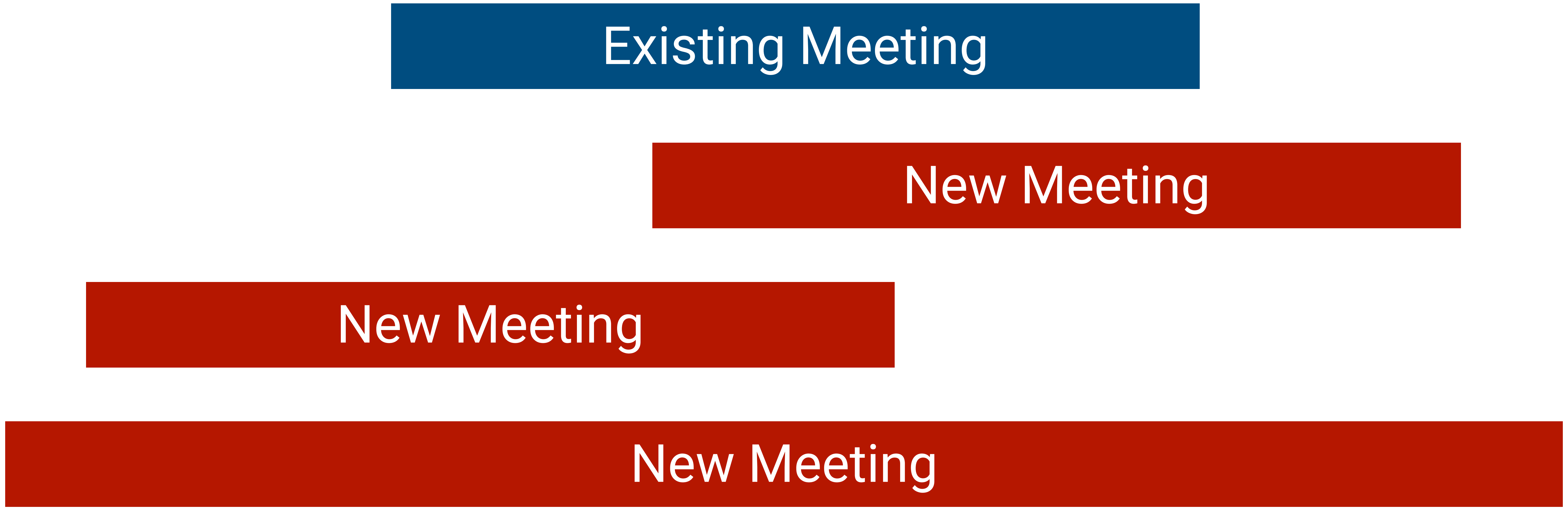
New Meeting

# Meeting Scheduling

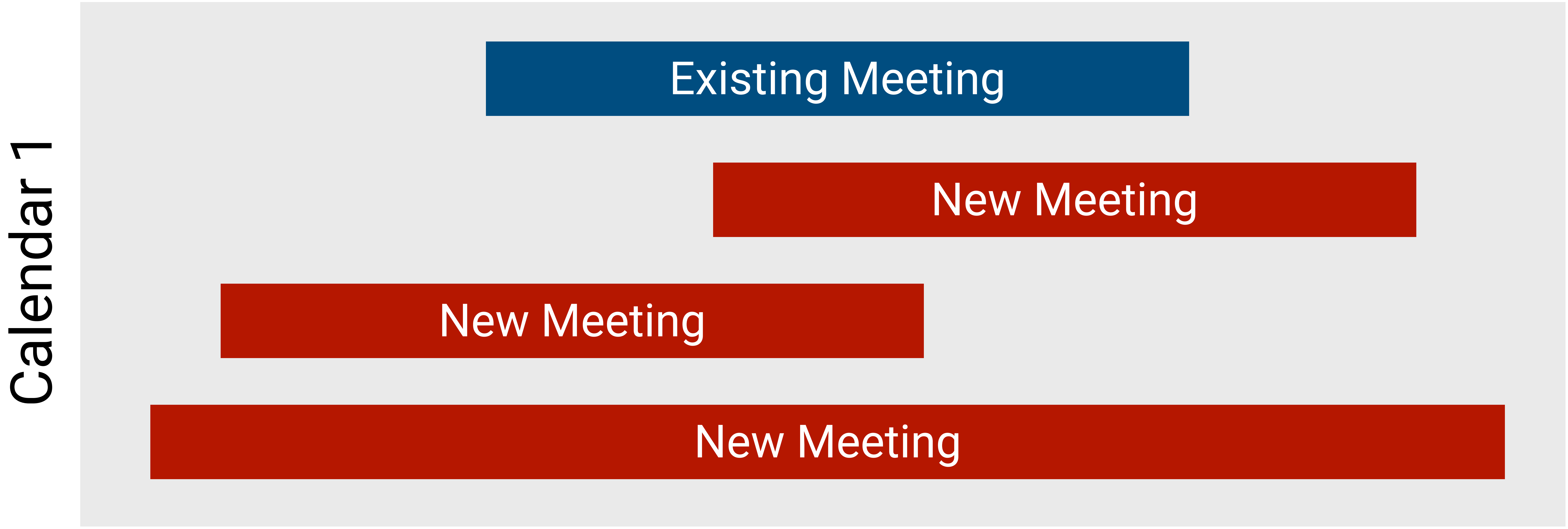
# Meeting Scheduling



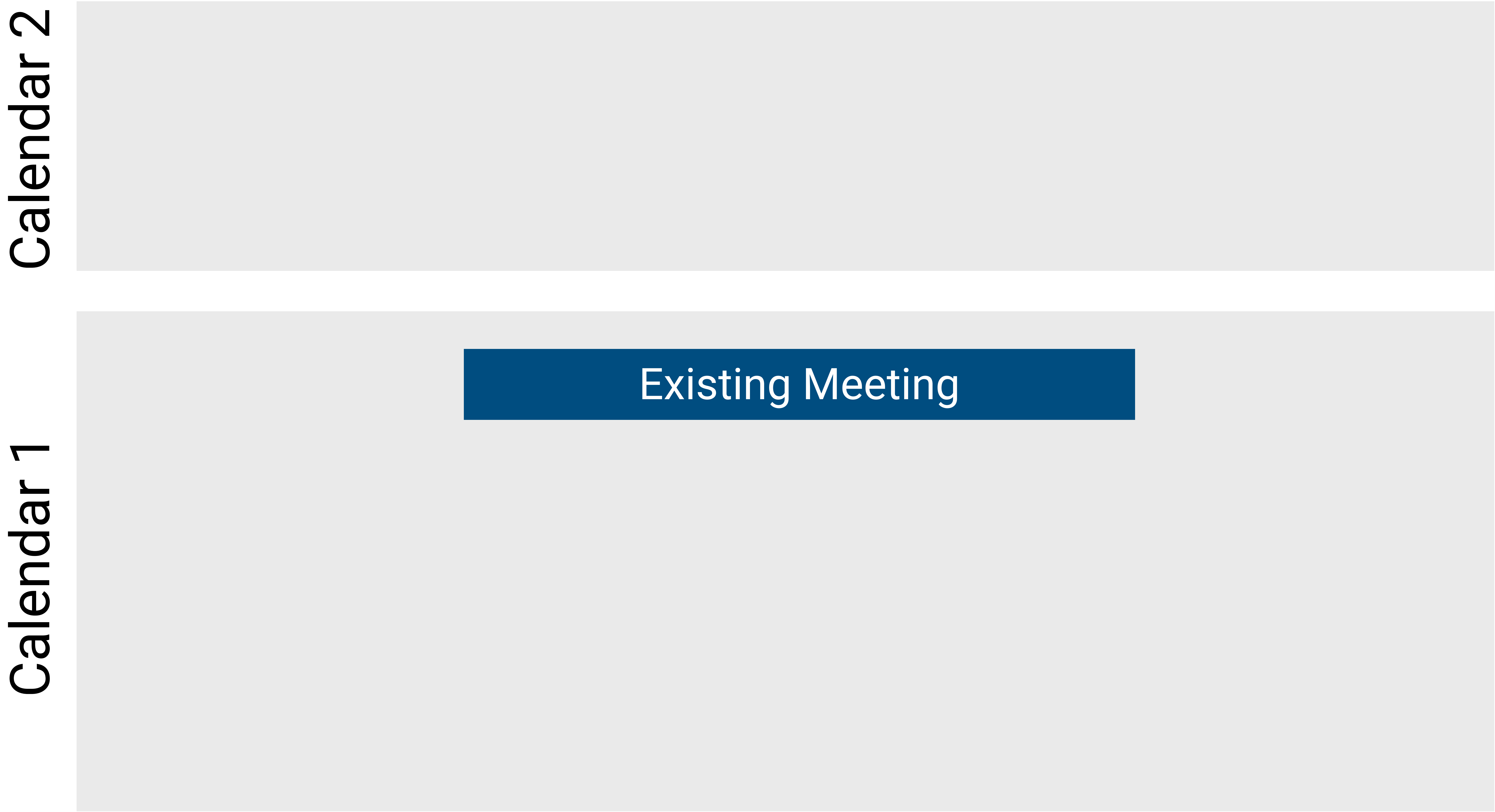
# Meeting Scheduling



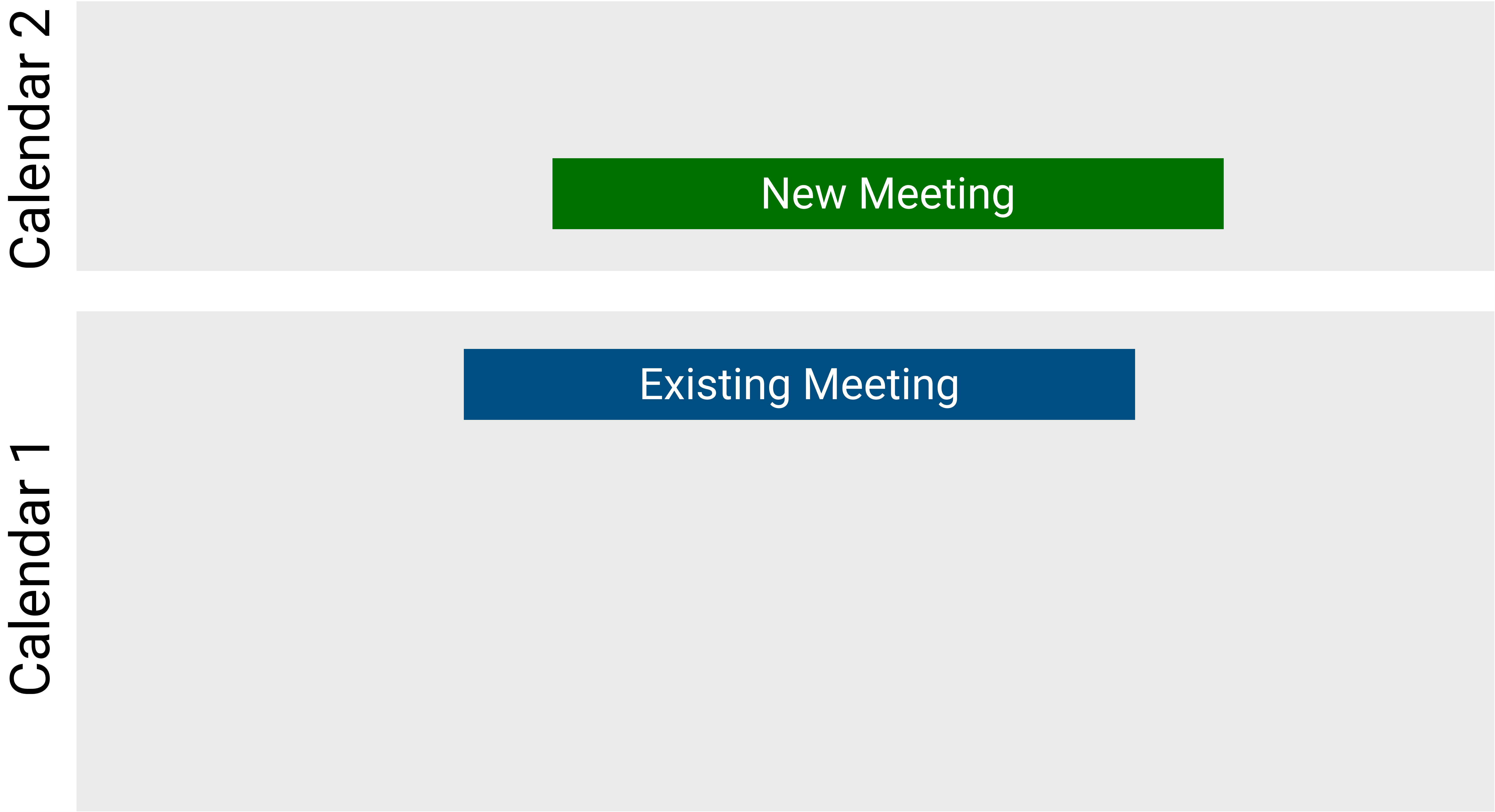
# Meeting Scheduling



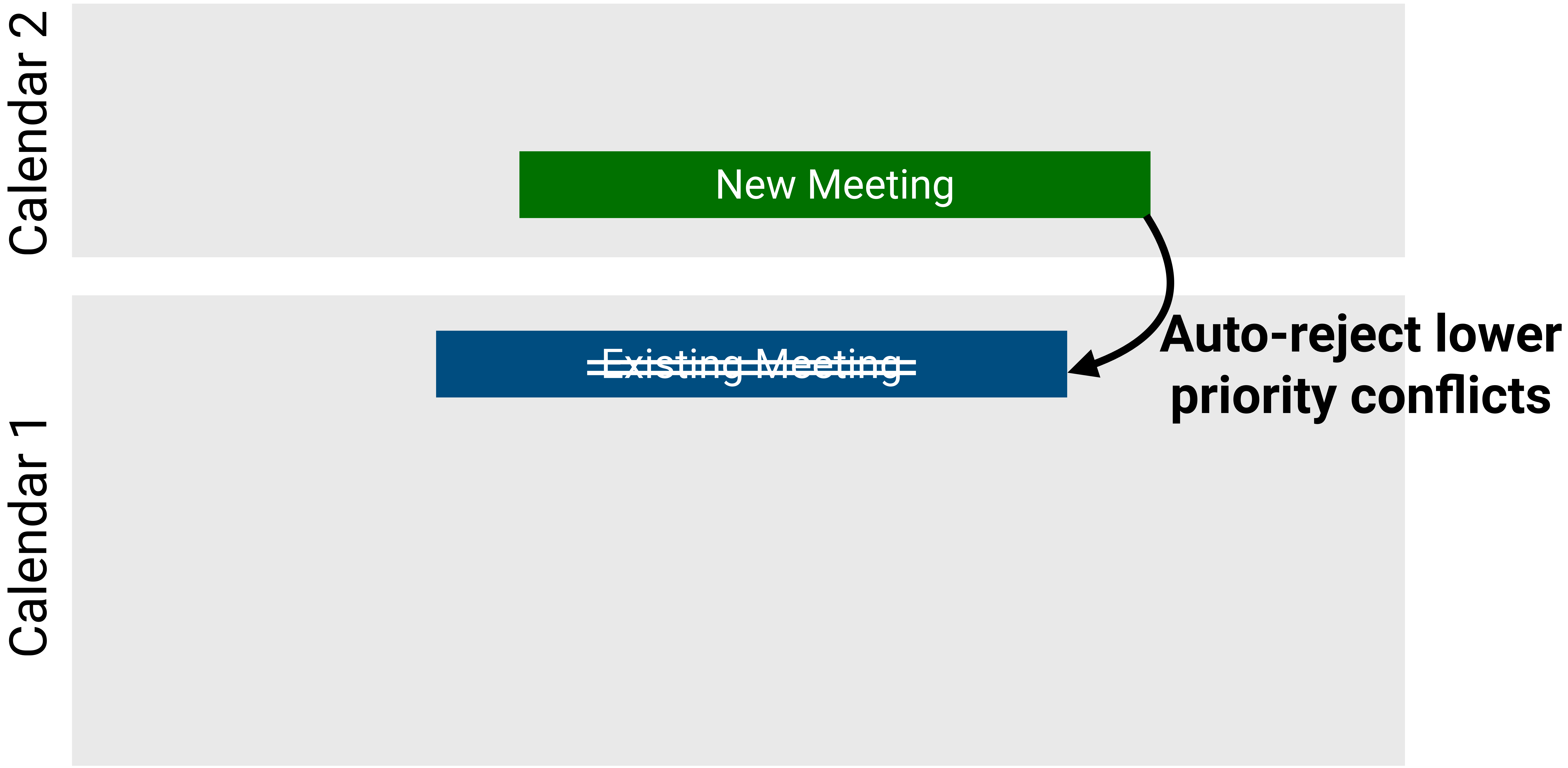
# Meeting Scheduling



# Meeting Scheduling



# Meeting Scheduling



# Meeting Scheduling, Vibe Coded

```
agy

Antigravity CLI 1.1.27
[redacted]@gmail.com (Google AI Pro)
Claude Opus 4.6 (Thinking)
~/repos/prompt-meetings

> This application called "prompt-meetings" is designed to be
  used for creating/managing meetings. Create the following
  endpoints:
  - create a user with a unique username
  - view all users in the system
  - create a meeting with a start and end time
  - invite other users by user name to a meeting
  - accept an invite
  - reject and invite
  - view pending invitations for a user
  - view all meetings
  - view all pending invitations received by a user

                                    Claude Opus 4.6 (Thinking)
```

# Meeting Scheduling, Vibe Coded

GitHub - mourjo/prompt-mee

github.com/mourjo/prompt-meetings

README Apache-2.0 license

## Prompt Meetings

This is an online API for creating and managing meetings. It allows users to register, create and manage meetings, send invitations, track responses, and coordinate calendar schedules within an organization.

Unlike traditional testing, which relies on manually writing individual test cases, this project uses property-based testing to uncover subtle bugs. Vibe-coded, this project highlights the need for property-based tests more than ever.

### Capabilities

- **User Management**: Support for creating new users with unique usernames and retrieving the directory of all registered users.
- **Calendar Management**: Support for creating new calendars with a unique name and priority by any existing user.

Source: github.com/mourjo/prompt-meetings

# Meeting Scheduling, Vibe Coded

# Meeting Scheduling, Vibe Coded

> *This application called "prompt-meetings" is designed to be used for creating/managing meetings.*
> *create the following endpoints:*
> *- create a user with a unique username*
> *- view all users in the system*
> *- create a meeting with a start and end time*
> *...*

# Meeting Scheduling, Vibe Coded

> *I want to introduce now a concept of calendars. A calendar contains meetings. Every meeting must be part of a calendar. Calendars have priorities denoted by a floating point number. Migrate the existing meetings to a default calendar. Update the database schema. Update the create-meeting endpoints...*

# Meeting Scheduling, Vibe Coded

> *I want to introduce now a concept of calendars. A calendar contains meetings. Every meeting must be part of a calendar. Calendars have priorities denoted by a floating point number. Migrate the existing meetings to a default calendar. Update the database schema. Update the create-meeting endpoints...*

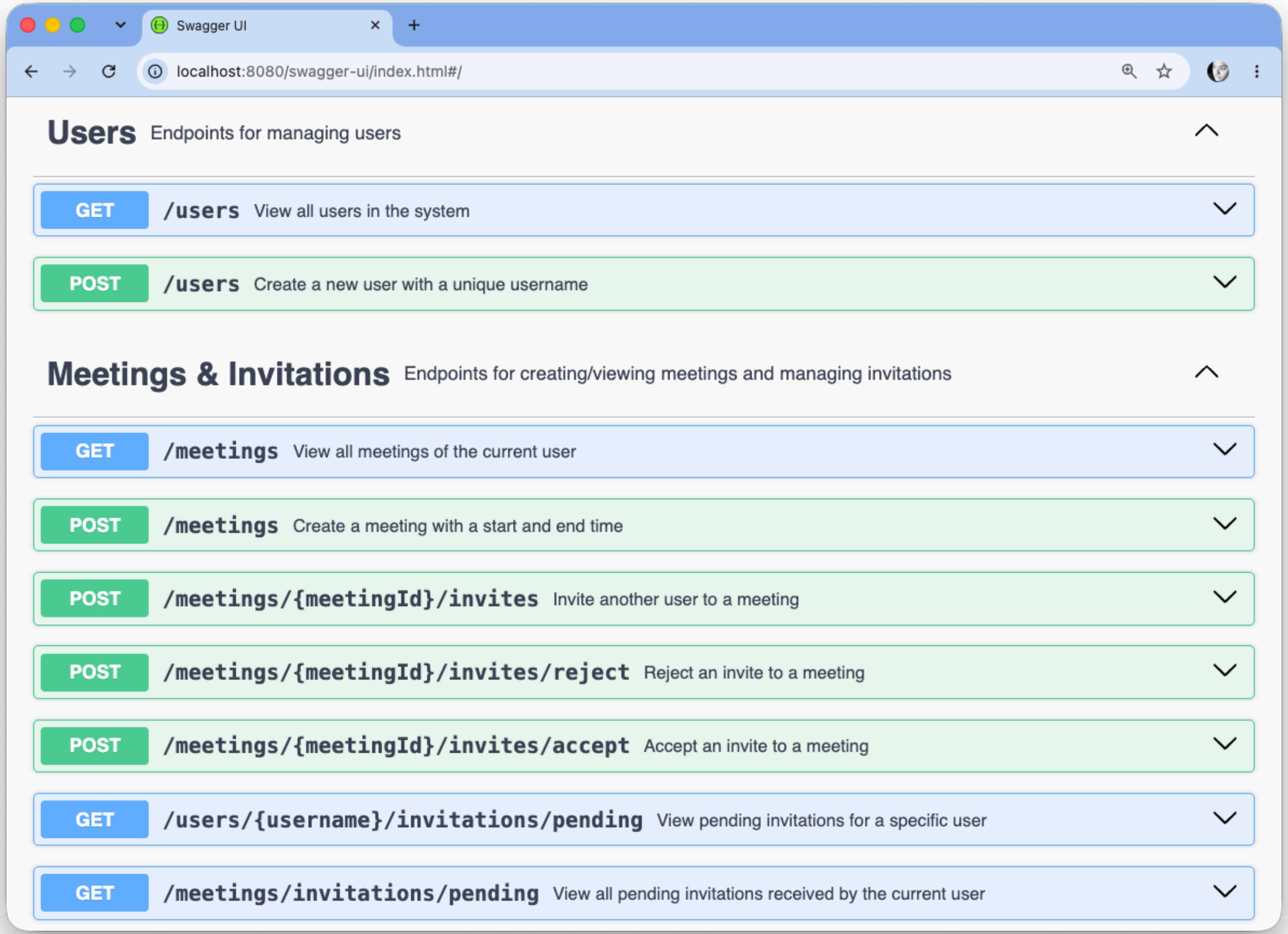
# Meeting Scheduling, Vibe Coded

> *Now I want to use the priority of calendars – when a meeting is being created, check if any of my meetings are conflicting with this. If the conflicting meeting is originating from a calendar of a lower priority, reject that meeting. If the conflict is coming from a calendar of the same priority or higher, do not allow its creation...*

# Meeting Scheduling, Vibe Coded

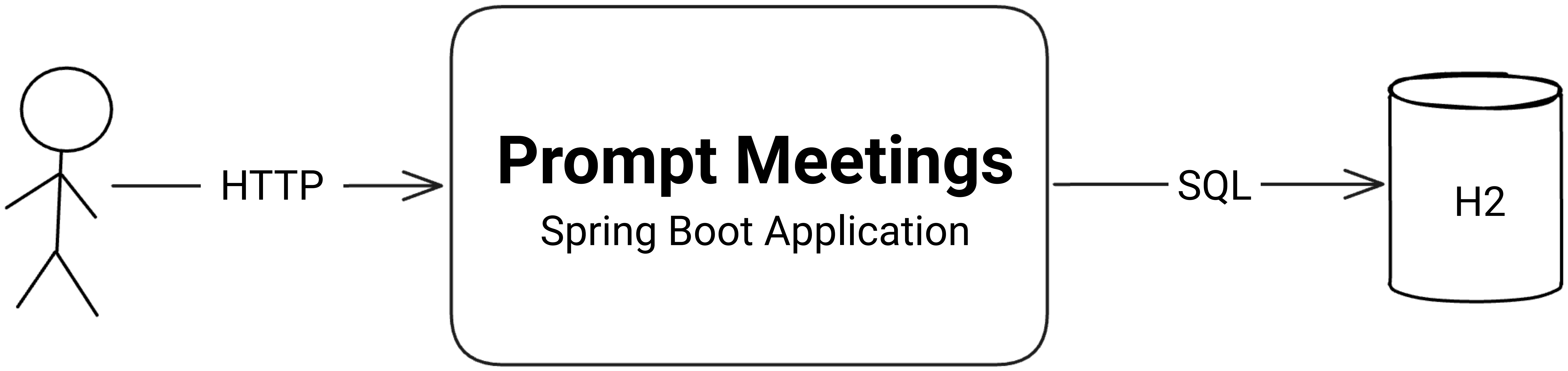
```
$ mvn clean test
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
...
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.349 s
[INFO] Finished at: 2026-08-16T17:54:28+02:00
[INFO] ------------------------------------------------------------------------
```

# Meeting Scheduling, Vibe Coded




Source: github.com/mourjo/prompt-meetings

# Meeting Scheduling, Vibe Coded




Source: github.com/mourjo/prompt-meetings

# Meeting Scheduling, Vibe Coded

> *I want to add property based tests using jqwik. Use version 1.9.3. Use the spring boot extension.*
> *- There should be 4 users in the system (this should be a constant, do this in the setup)*
> *- There should be 3 calendars of priorities default, medium, high, this should be a constant, do this in the setup...*

# Meeting Scheduling, Vibe Coded

> *...the property based test should perform sequences of actions on behalf of users*
> *- create-meeting*
> *- invite someone else to a meeting*
> *- accept an invitation*
> *- reject an invitation*

# Meeting Scheduling, Vibe Coded

> *...the invariant should always hold true: no user can ever part of two meetings at the same time – no sequence of actions should ever allow this.*

# Meeting Scheduling, Vibe Coded

```
# ~/repos/prompt-meetings → mvn clean test
[INFO] Scanning for projects...
...
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   MeetingSchedulerPropertyTests.noOverlappingMeetingsForAnyUser:90
>lambda$noOverlappingMeetingsForAnyUser$0:80 Invariant 'no-overlap' failed after the following actions: [
    alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    alice accepts invitation to meeting Meeting-0
]
final state: me.mourjo.prompt.meetings.MeetingSchedulerPropertyTests$CalendarMeetingsState@7fdfe501
User charlie is in overlapping meetings: Meeting-0 [10:00 - 10:01] and Meeting-1 [10:00 - 10:01]
[INFO]
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
```

# Meeting Scheduling, Vibe Coded

```
Invariant 'no-overlap' failed after the following actions: [
    alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    alice accepts invitation to meeting Meeting-0
]
```

# Meeting Scheduling, Vibe Coded

```
Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
    alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
    bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04
    charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28
    dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
    bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37
    alice rejects invitation to Meeting-2
    dan accepts invitation to meeting Meeting-2
    dan invites alice to Meeting-2
    alice invites charlie to Meeting-1
    alice rejects invitation to Meeting-0
    charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
    dan rejects invitation to Meeting-3
    charlie accepts invitation to meeting Meeting-6
    dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
    alice invites dan to Meeting-4
    charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
    dan invites bob to Meeting-9
    charlie invites alice to Meeting-4
    bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
    dan accepts invitation to meeting Meeting-4
]
```

# Meeting Scheduling, Vibe Coded

```
Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
    alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
    bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04
    charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28
    dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
    bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37
    alice rejects invitation to Meeting-2
    dan accepts invitation to meeting Meeting-2
    dan invites alice to Meeting-2
    alice invites charlie to Meeting-1
    alice rejects invitation to Meeting-0
    charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
    dan rejects invitation to Meeting-3
    charlie accepts invitation to meeting Meeting-6
    dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
    alice invites dan to Meeting-4
    charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
    dan invites bob to Meeting-9
    charlie invites alice to Meeting-4
    bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
    dan accepts invitation to meeting Meeting-4
]
```

# Meeting Scheduling, Vibe Coded

```
Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
    alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
    bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04
    charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28
    dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
    bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37
    alice rejects invitation to Meeting-2
    dan accepts invitation to meeting Meeting-2
    dan invites alice to Meeting-2
    alice invites charlie to Meeting-1
    alice rejects invitation to Meeting-0
    charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
    dan rejects invitation to Meeting-3
    charlie accepts invitation to meeting Meeting-6
    dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
    alice invites dan to Meeting-4
    charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
    dan invites bob to Meeting-9
    charlie invites alice to Meeting-4
    bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
    dan accepts invitation to meeting Meeting-4
]
```

# Meeting Scheduling, Vibe Coded

```
Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
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    alice rejects invitation to Meeting-2
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    dan invites alice to Meeting-2
    alice invites charlie to Meeting-1
    alice rejects invitation to Meeting-0
    charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
    dan rejects invitation to Meeting-3
    charlie accepts invitation to meeting Meeting-6
    dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
    alice invites dan to Meeting-4
    charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
    dan invites bob to Meeting-9
    charlie invites alice to Meeting-4
    bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
    dan accepts invitation to meeting Meeting-4
]
```

# Meeting Scheduling, Vibe Coded

@Property

# Meeting Scheduling, Vibe Coded

```
@Property
                                        @ForAll("actions") ActionChain<CalendarState> chain
```

# Meeting Scheduling, Vibe Coded

```
@Property
                                    @ForAll("actions") ActionChain<CalendarState> chain
```

```
@Provide
Arbitrary<ActionChain<CalendarState>> actions() {
    return ActionChain.startWith(CalendarMeetingsState::new)
        .withAction(new CreateMeetingAction())
        .withAction(new InviteUserAction())
        .withAction(new AcceptInviteAction())
        .withAction(new RejectMeetingAction());
}
```

# Meeting Scheduling, Vibe Coded

```
@Property
                                        @ForAll("actions") ActionChain<CalendarState> chain
```

# Meeting Scheduling, Vibe Coded

```java
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
```

# Meeting Scheduling, Vibe Coded

```
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut -> {









    }).run();
```

# Meeting Scheduling, Vibe Coded

```
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut -> {

        for (String user : USERS) {







        }
    }).run();
```

# Meeting Scheduling, Vibe Coded

```java
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut -> {

        for (String user : USERS) {
            var meetings = meetingService.getMyMeetings(user).stream()
                .filter(m -> "ACCEPTED".equalsIgnoreCase(m.userStatus()))
                .toList();








        }
    }).run();
```

# Meeting Scheduling, Vibe Coded

```java
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut -> {

        for (String user : USERS) {
            var meetings = meetingService.getMyMeetings(user).stream()
                .filter(m -> "ACCEPTED".equalsIgnoreCase(m.userStatus()))
                .toList();

            for (int i = 0; i < meetings.size(); i++) {
                for (int j = i + 1; j < meetings.size(); j++) {




            }
        }
    }).run();
```

# Meeting Scheduling, Vibe Coded

```java
@Property
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut -> {

        for (String user : USERS) {
            var meetings = meetingService.getMyMeetings(user).stream()
                .filter(m -> "ACCEPTED".equalsIgnoreCase(m.userStatus()))
                .toList();

            for (int i = 0; i < meetings.size(); i++) {
                for (int j = i + 1; j < meetings.size(); j++) {
                    if (overlaps(meetings.get(i), meetings.get(j))) {
                        throw new AssertionError(...);
                    }
                }
            }
        }
    }).run();
```

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

# Meeting Scheduling, Vibe Coded

```
[ERROR]  Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

*Charlie creates Meeting-0*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

*Charlie creates Meeting-0*

*Charlie creates an overlapping meeting in a higher priority calendar*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

*Charlie creates Meeting-0*
*Charlie creates an overlapping meeting in a higher priority calendar*
*Charlie accepts Meeting-0*

# Meeting Scheduling, Vibe Coded

> *When accepting a meeting ensure that the invitation is in pending state - ie, do not allow acceptance of a meeting which is already accepted or rejected.*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

*Charlie creates a Meeting-0*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

*Charlie creates a Meeting-0*

*Alice creates an overlapping meeting*

# Meeting Scheduling, Vibe Coded

```
[ERROR]  Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

*Charlie creates a Meeting-0*
*Alice creates an overlapping meeting*
*Alice invites Charlie to their meeting*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

*Charlie creates a Meeting-0*
*Alice creates an overlapping meeting*
*Alice invites Charlie to their meeting*
*Charlie accepts the invitation*

# Meeting Scheduling, Vibe Coded

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

```
[ERROR]   Invariant 'no-overlap' failed after the following actions: [
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

# Meeting Scheduling, Vibe Coded

[

```
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
```

] Four actions

Three actions

[

```
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
```

]

# Meeting Scheduling, Vibe Coded

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
  alice invites charlie to Meeting-1
  charlie accepts invitation to meeting Meeting-1
]
```

Two users

One user

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
  charlie accepts invitation to meeting Meeting-0
]
```

# Meeting Scheduling, Vibe Coded

```
[
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
    alice invites charlie to Meeting-1
    charlie accepts invitation to meeting Meeting-1
]
```

One calendar

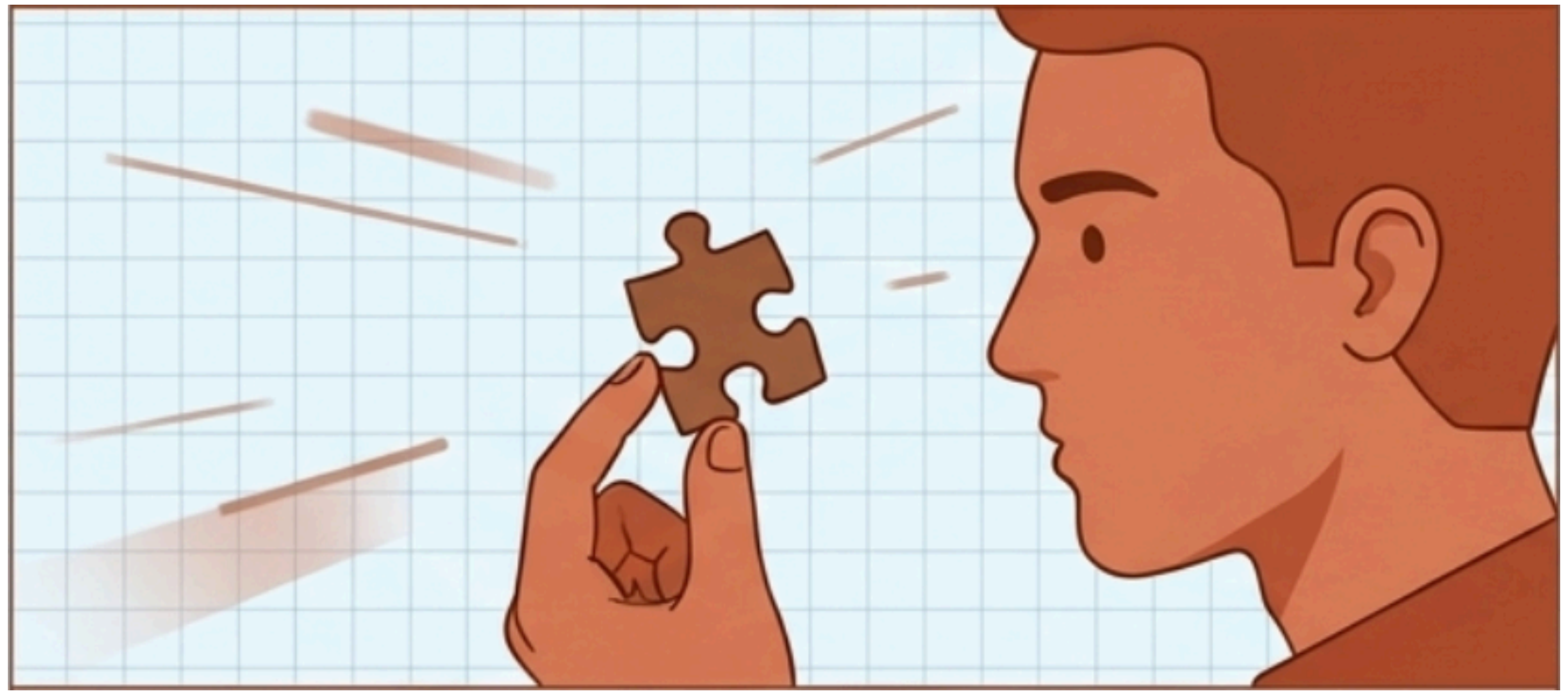
Two calendars

```
[
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    charlie accepts invitation to meeting Meeting-0
]
```

# More Than Just Testing

# “For This” → “For All”

# “For This” → “For All”



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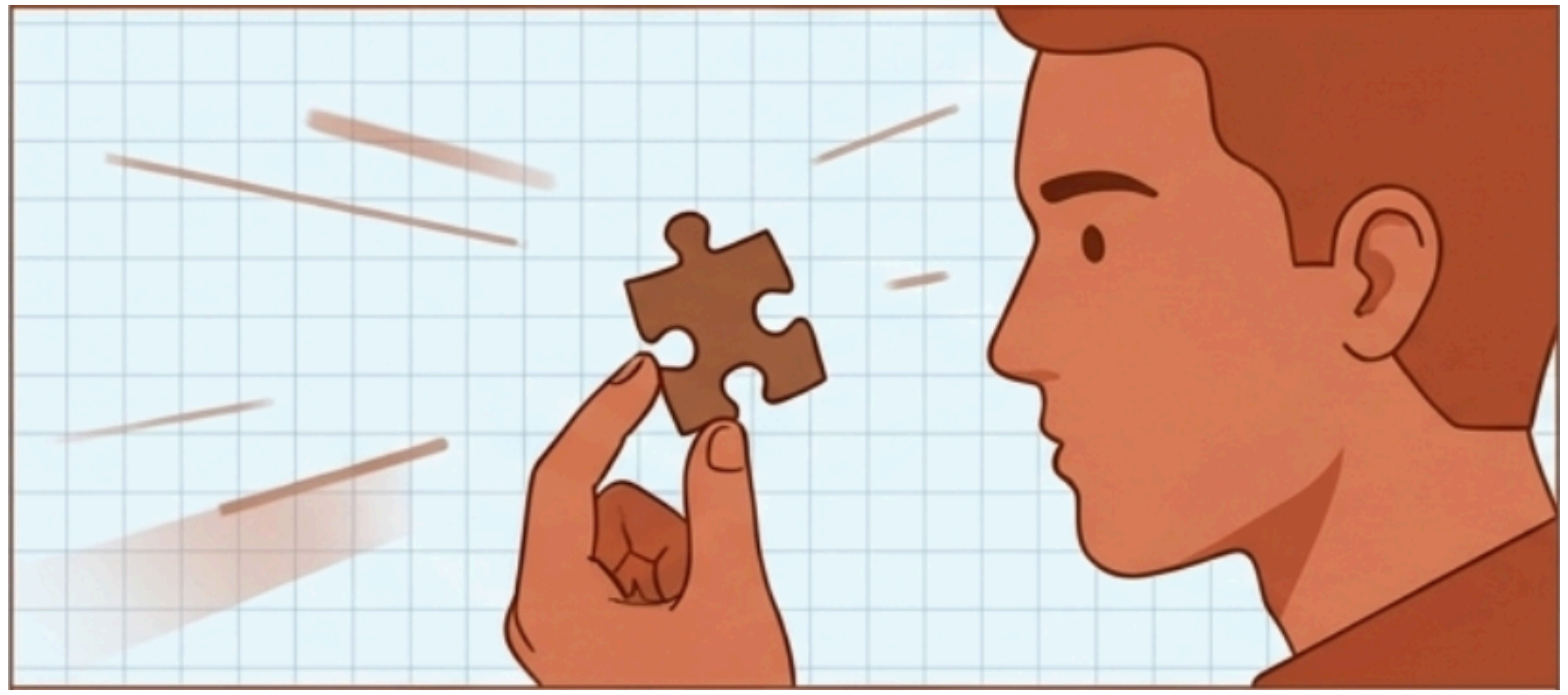
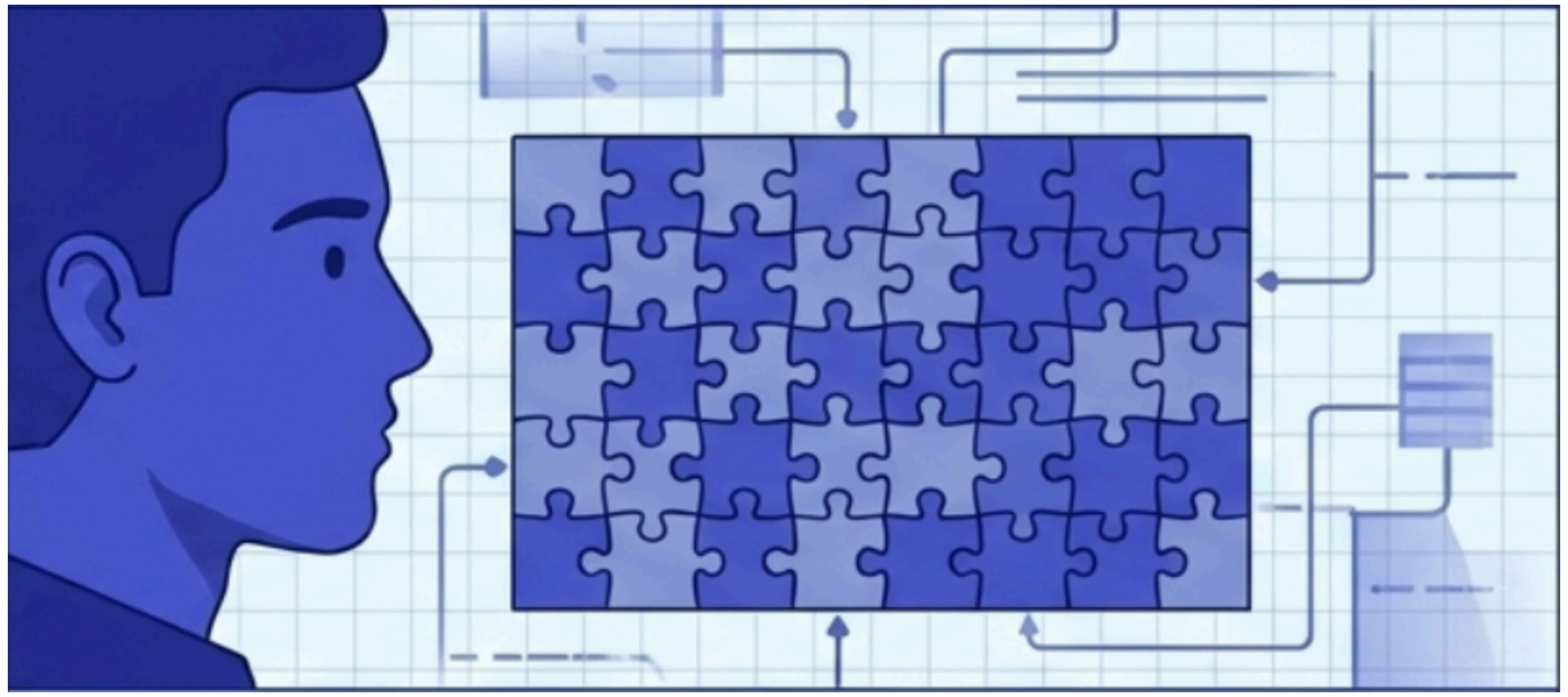
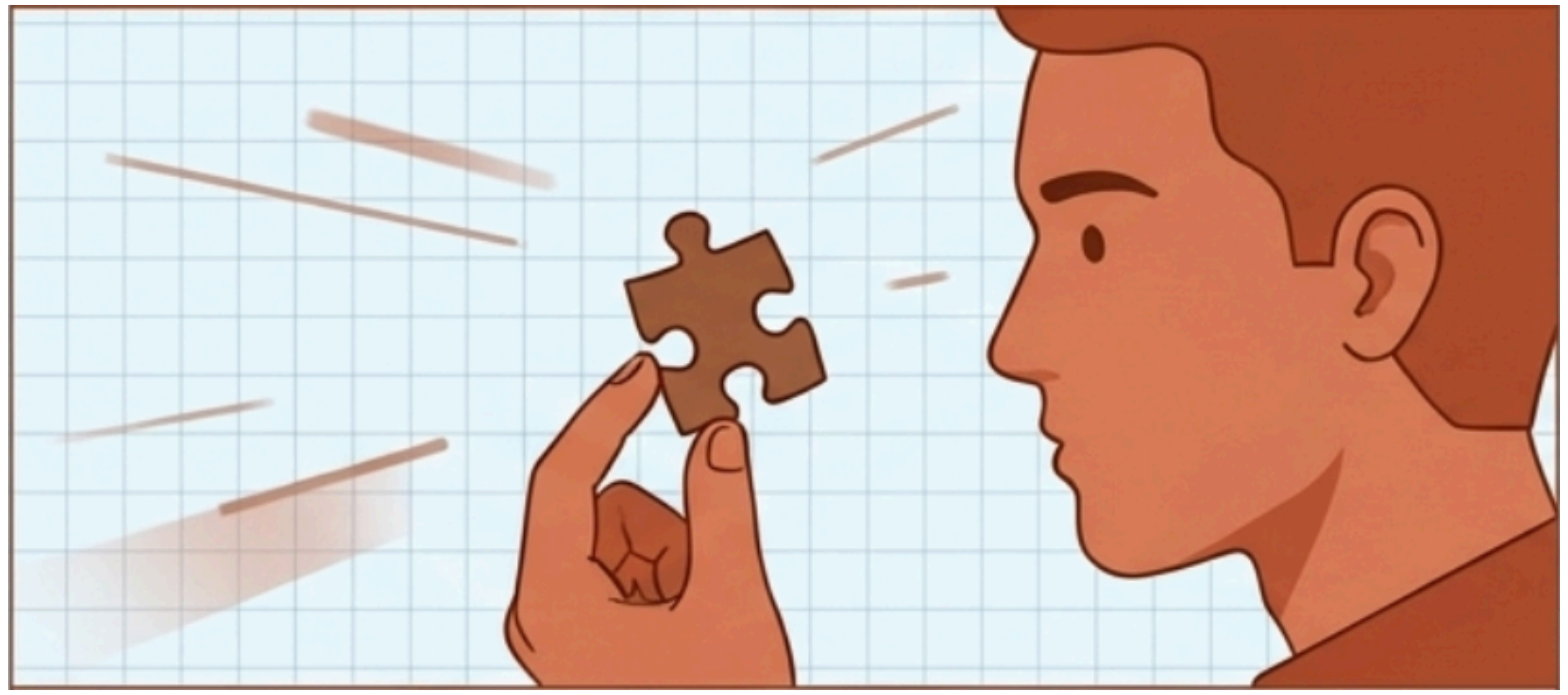
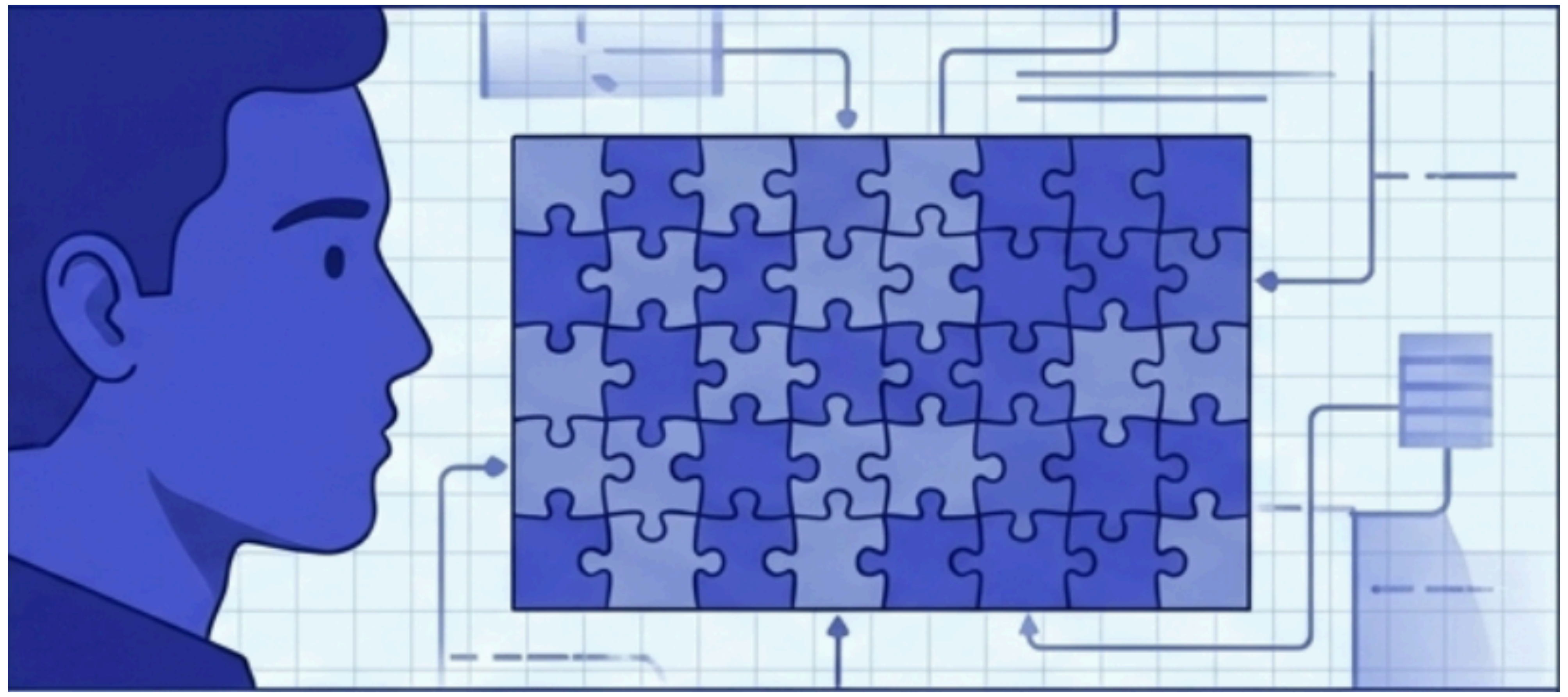




# “For This” → “For All”

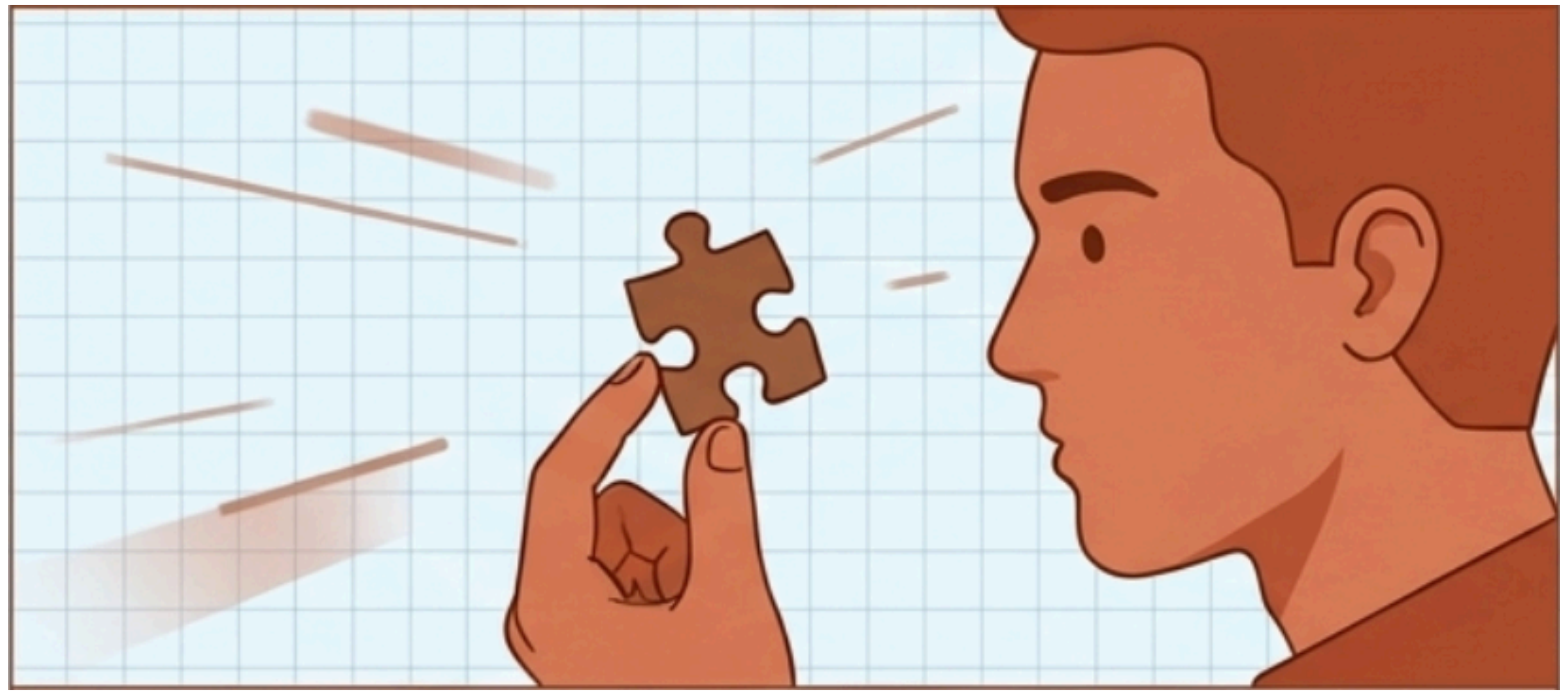
- Defining what must always hold true

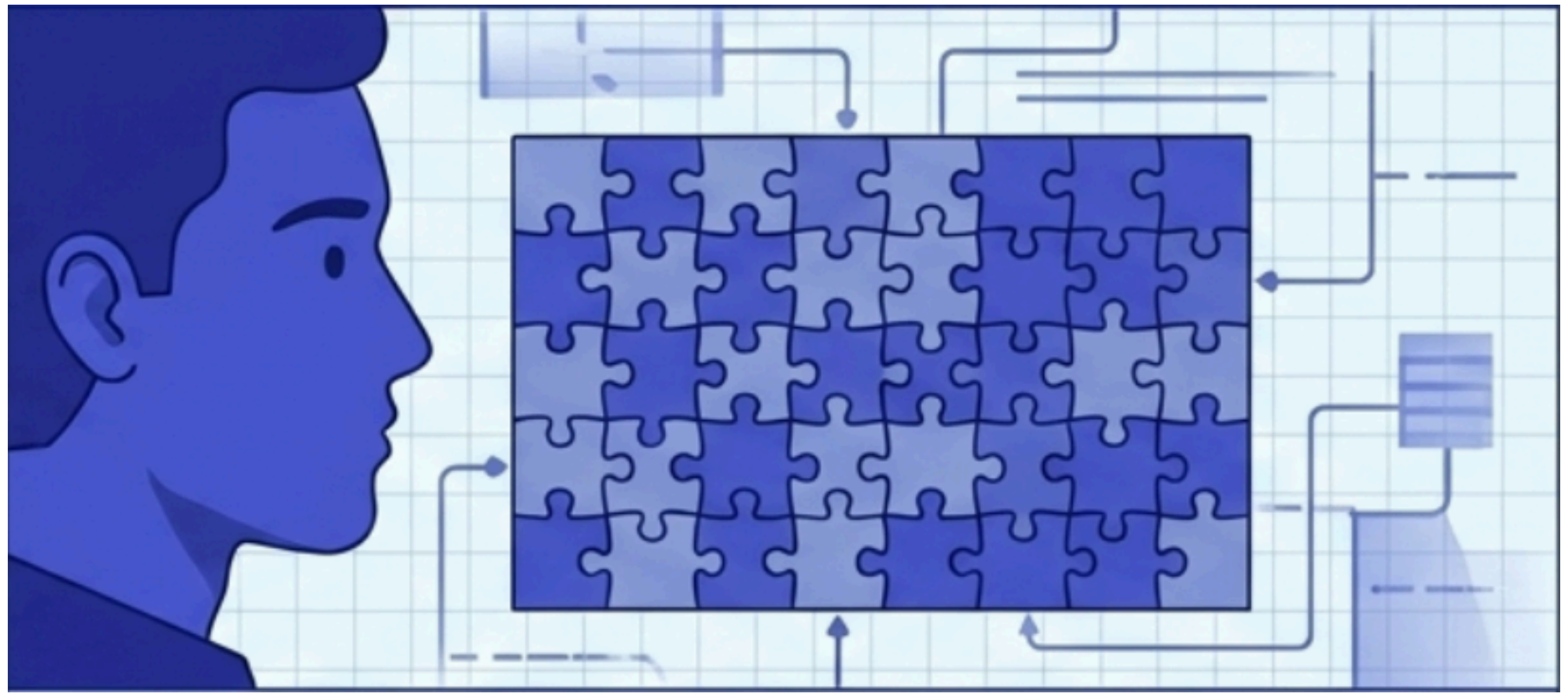




# “For This” → “For All”

- Defining what must always hold true
- Understanding counter examples





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> *Add a new property based test which ensures this invariant:* ***for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted*** *by the current user.*

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> “*Add a new property based test which ensures this invariant:* ***for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted*** *by the current user.*”

Meeting 1

# “For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

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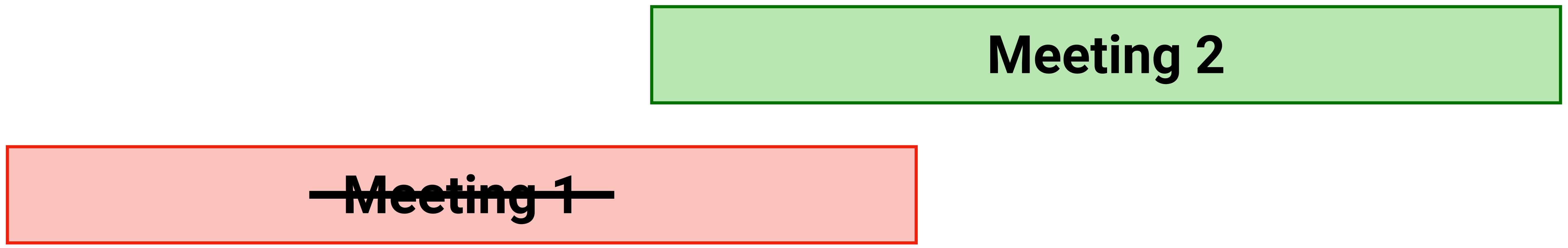

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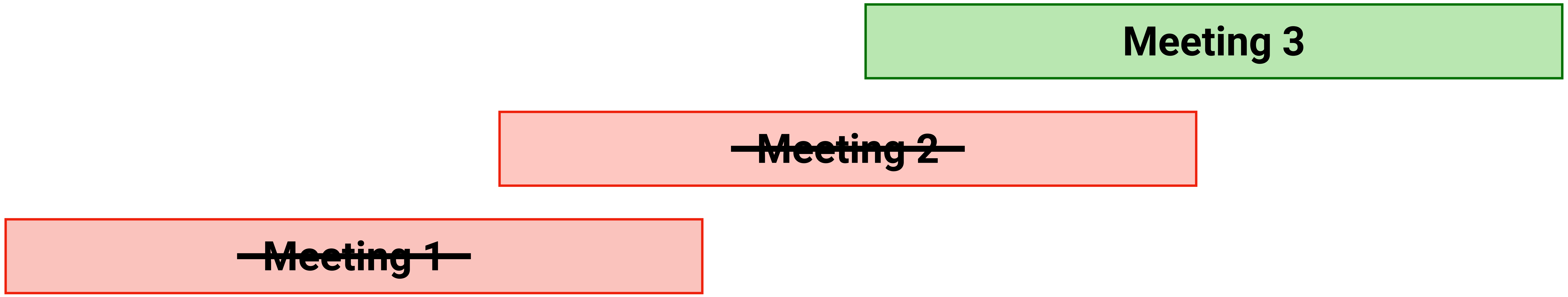

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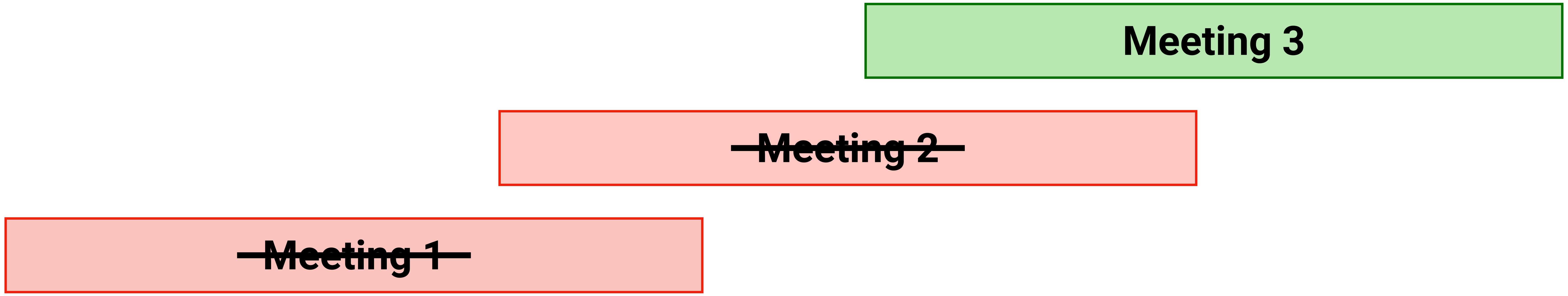



No overlapping accepted meeting

# Counter Examples are Transformative

# Counter Examples are Transformative

Mental Model

# Counter Examples are Transformative

What must ***always*** hold true

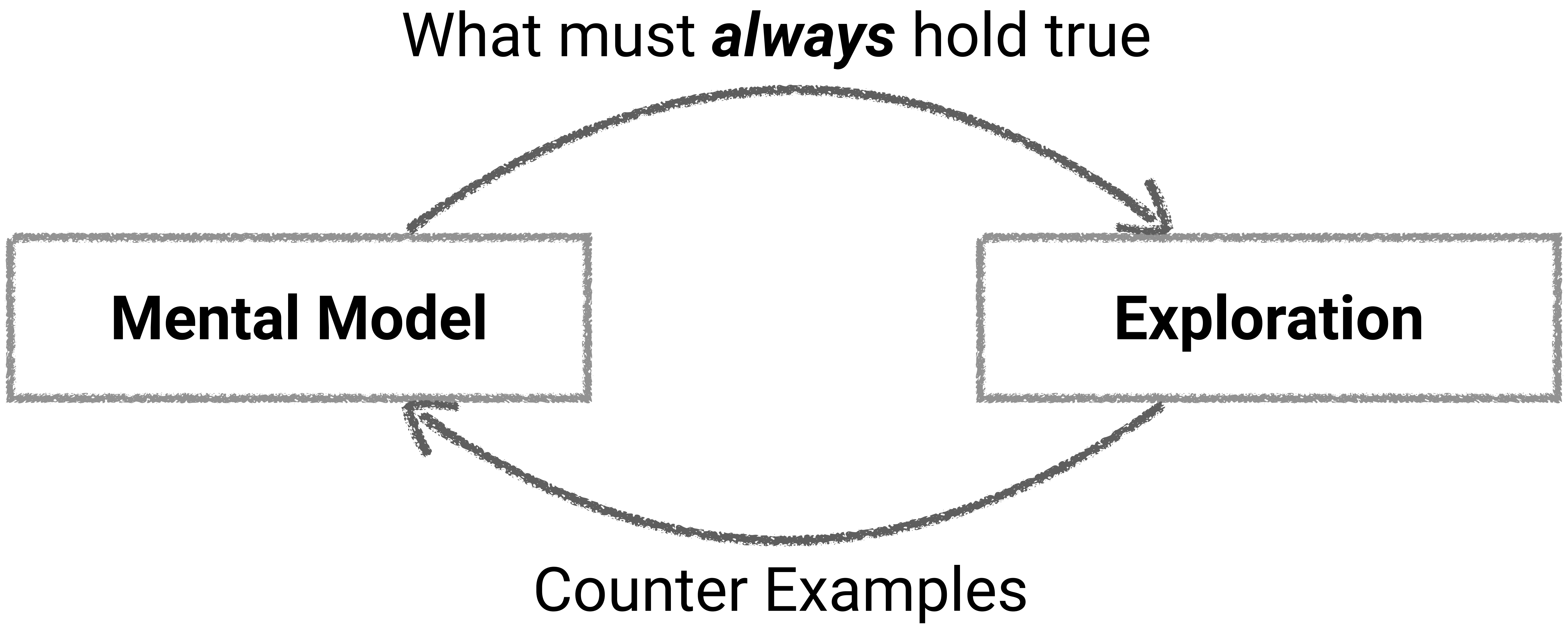
**Mental Model**

# Counter Examples are Transformative

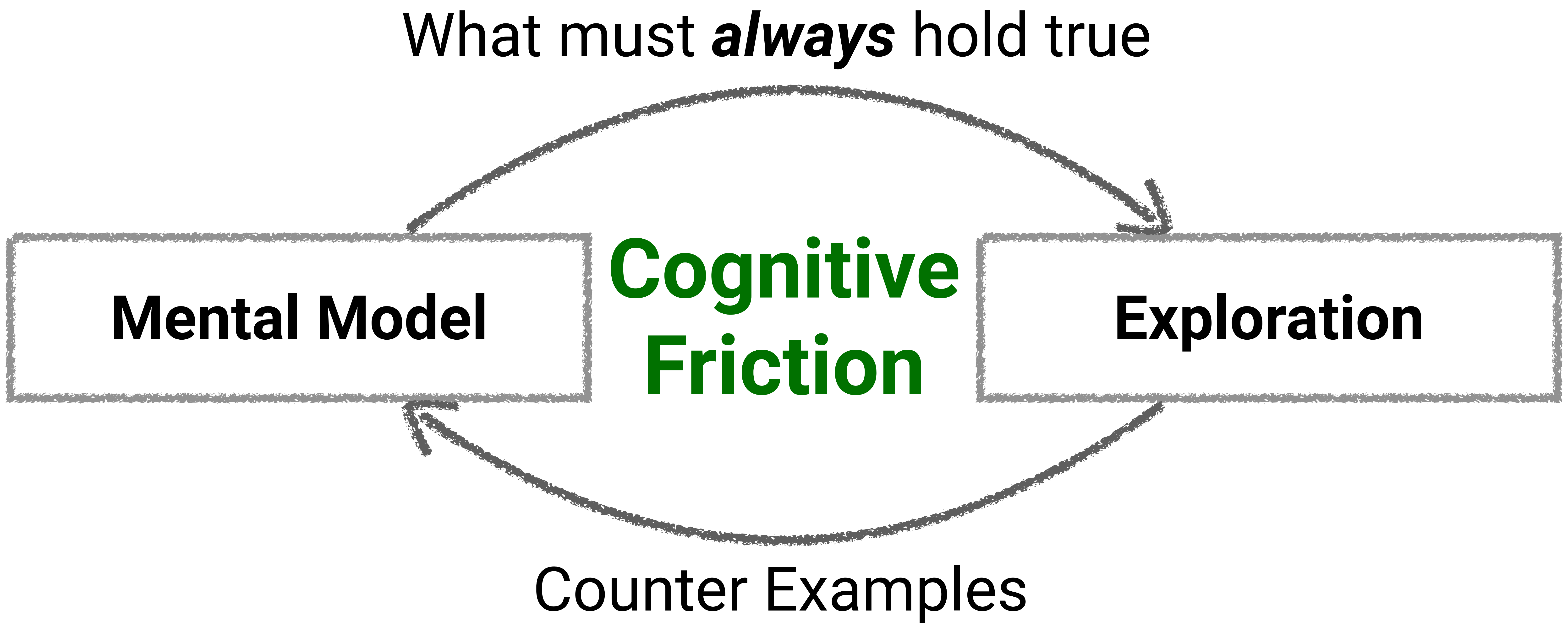
What must ***always*** hold true

**Mental Model** → **Exploration**

# Counter Examples are Transformative



# Counter Examples are Transformative



# Everything's Changed

# Passive Overseeing is Not Enough

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- LLMs code faster than we can read
  - Quality is not guaranteed
  - Just being “in the loop” doesn’t work

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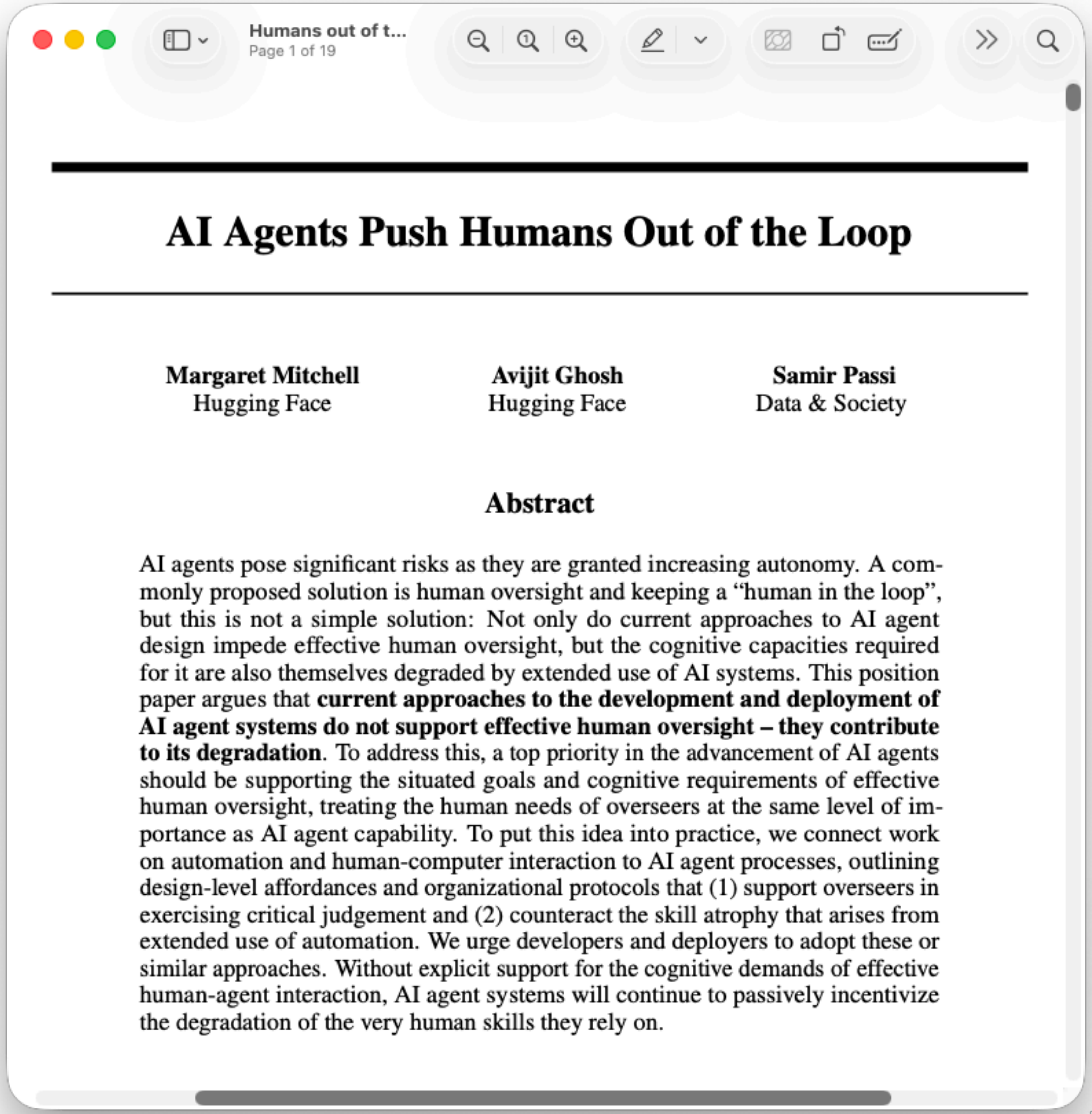



# AI Agents Push Humans Out of the Loop

**Margaret Mitchell**
Hugging Face

**Avijit Ghosh**
Hugging Face

**Samir Passi**
Data & Society

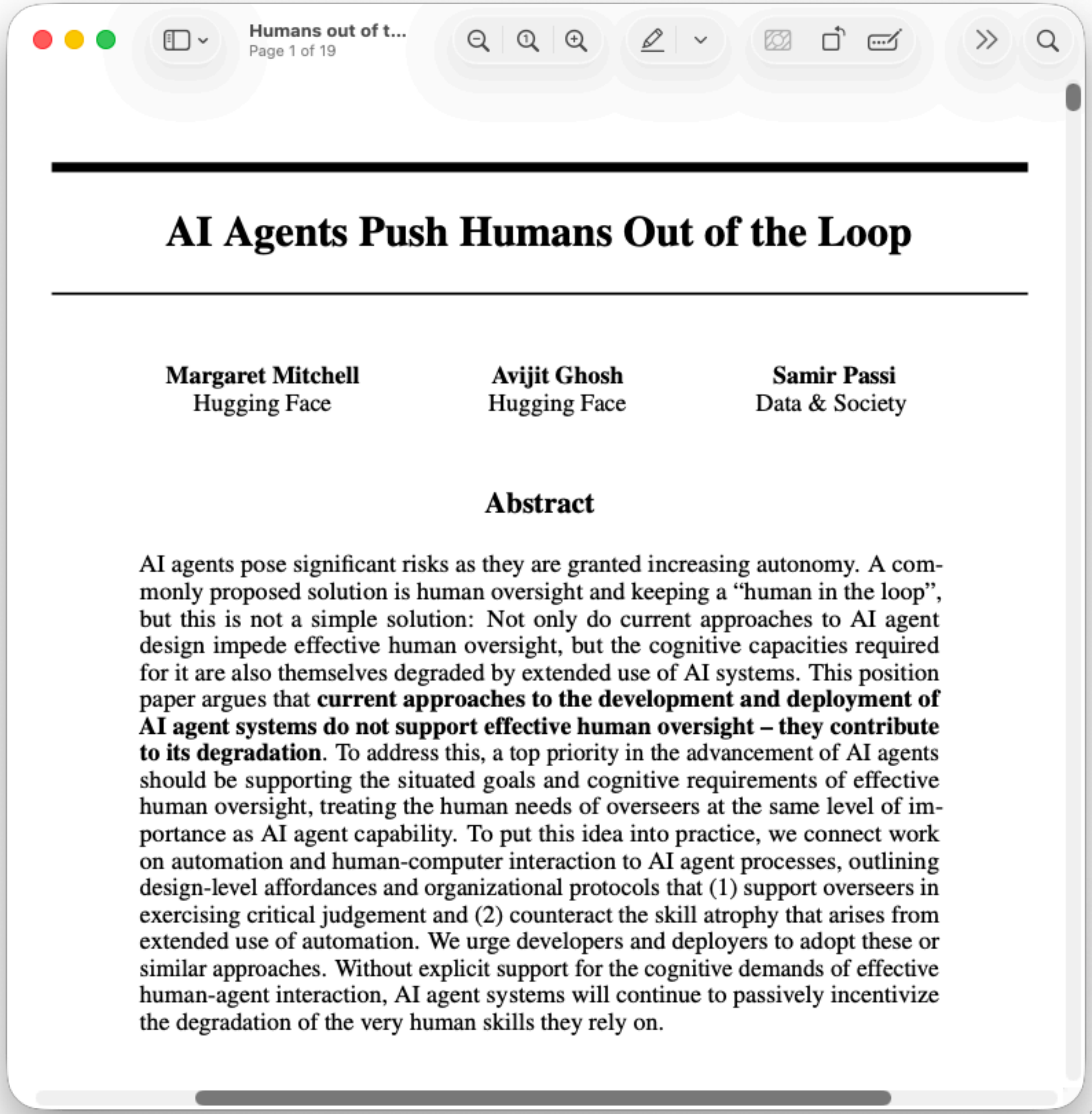
**Abstract**

AI agents pose significant risks as they are granted increasing autonomy. A commonly proposed solution is human oversight and keeping a "human in the loop", but this is not a simple solution: Not only do current approaches to AI agent design impede effective human oversight, but the cognitive capacities required for it are also themselves degraded by extended use of AI systems. This position paper argues that **current approaches to the development and deployment of AI agent systems do not support effective human oversight – they contribute to its degradation**. To address this, a top priority in the advancement of AI agents should be supporting the situated goals and cognitive requirements of effective human oversight, treating the human needs of overseers at the same level of importance as AI agent capability. To put this idea into practice, we connect work on automation and human-computer interaction to AI agent processes, outlining design-level affordances and organizational protocols that (1) support overseers in exercising critical judgement and (2) counteract the skill atrophy that arises from extended use of automation. We urge developers and deployers to adopt these or similar approaches. Without explicit support for the cognitive demands of effective human-agent interaction, AI agent systems will continue to passively incentivize the degradation of the very human skills they rely on.

Source: [https://arxiv.org/abs/2608.23642](https://arxiv.org/abs/2608.23642) (Aug 2026)

# Passive Overseeing is Not Enough

- LLMs code faster than we can read
  - Quality is not guaranteed
  - Just being “in the loop” doesn’t work
- Strategic cognitive friction
  - Embed human comprehension into software engineering

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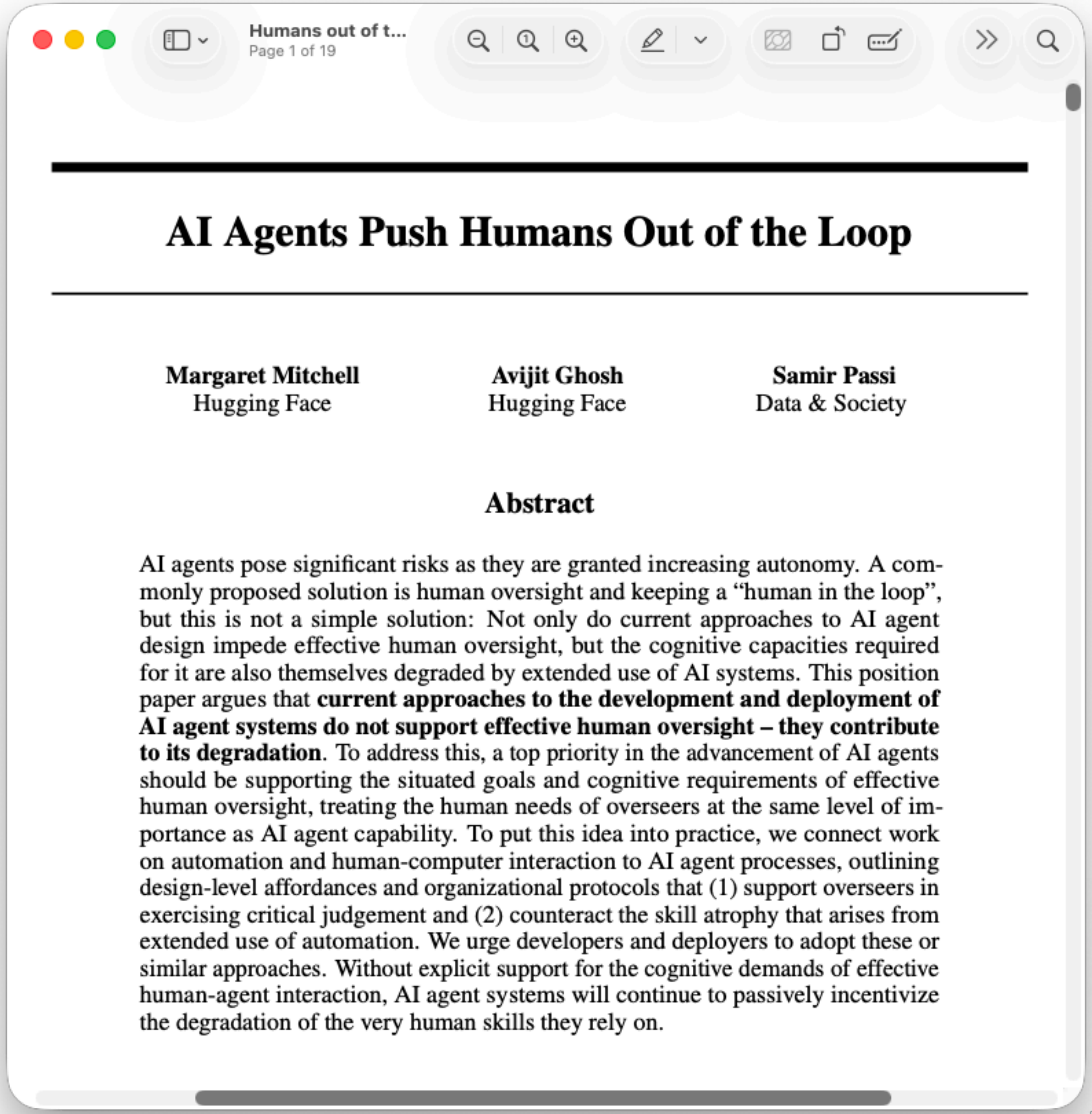
# Passive Overseeing is Not Enough

- LLMs code faster than we can read
  - Quality is not guaranteed
  - Just being “in the loop” doesn’t work
- Strategic cognitive friction
  - Embed human comprehension into software engineering
- Exploratory testing brings cognitive friction back

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# Division of Cognitive Labour

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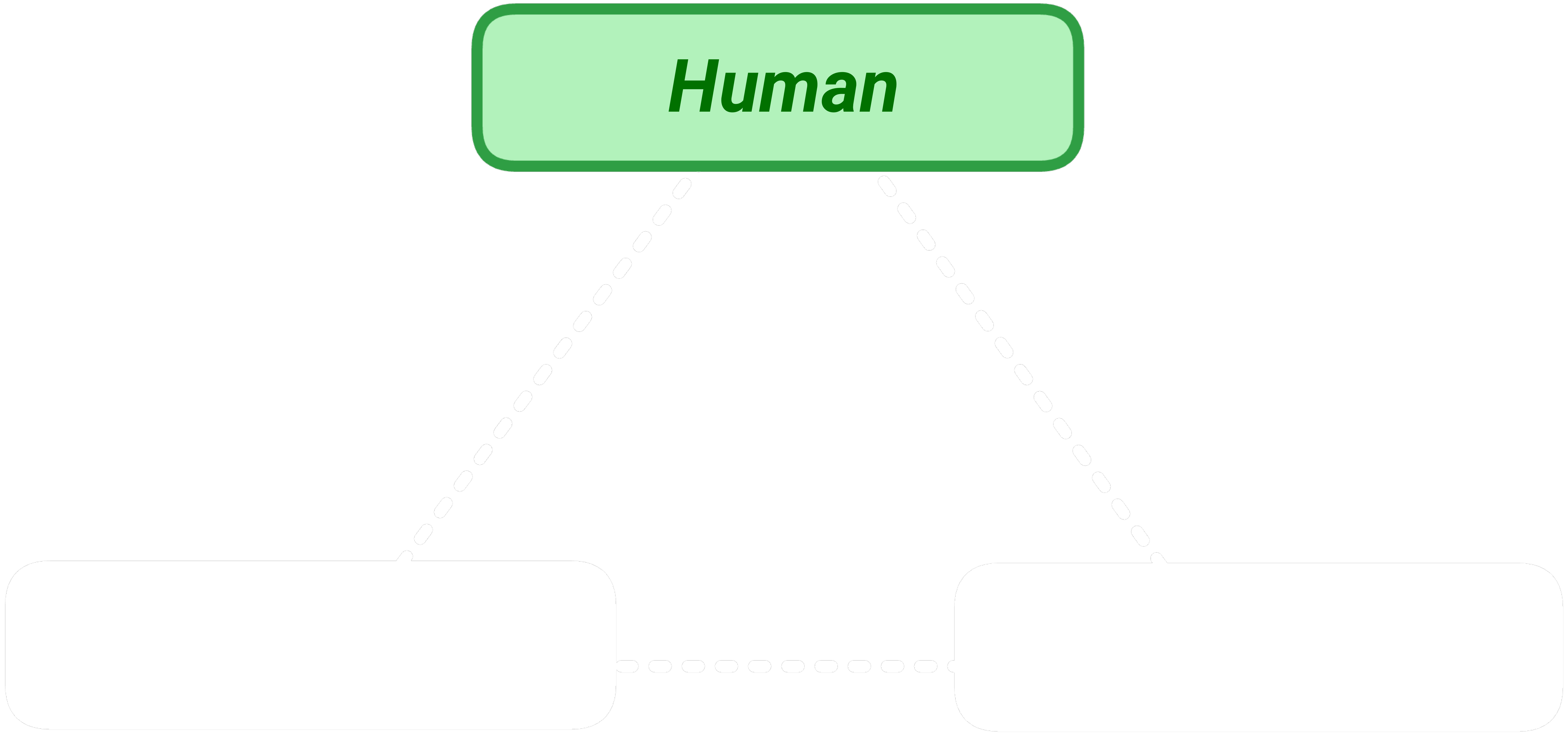

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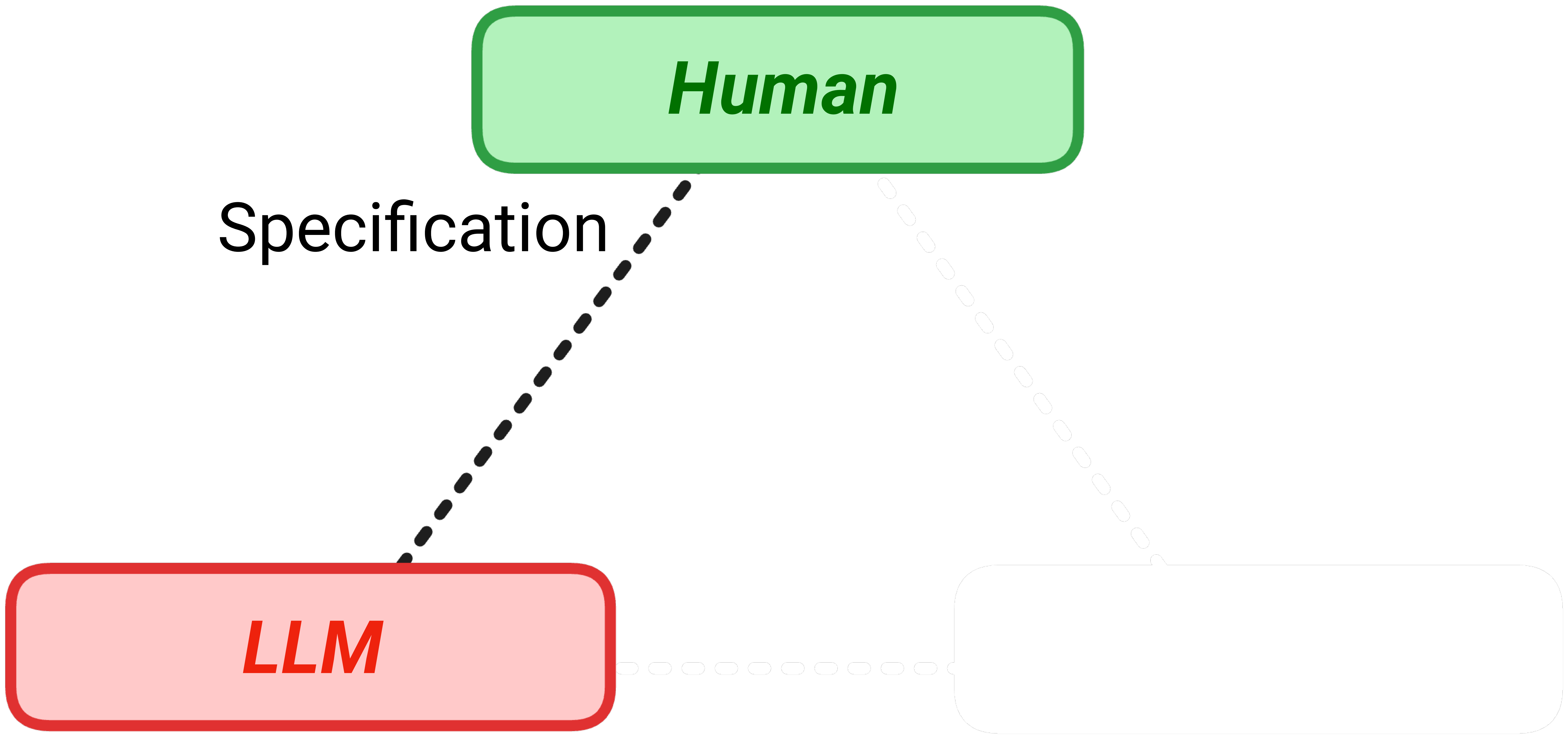

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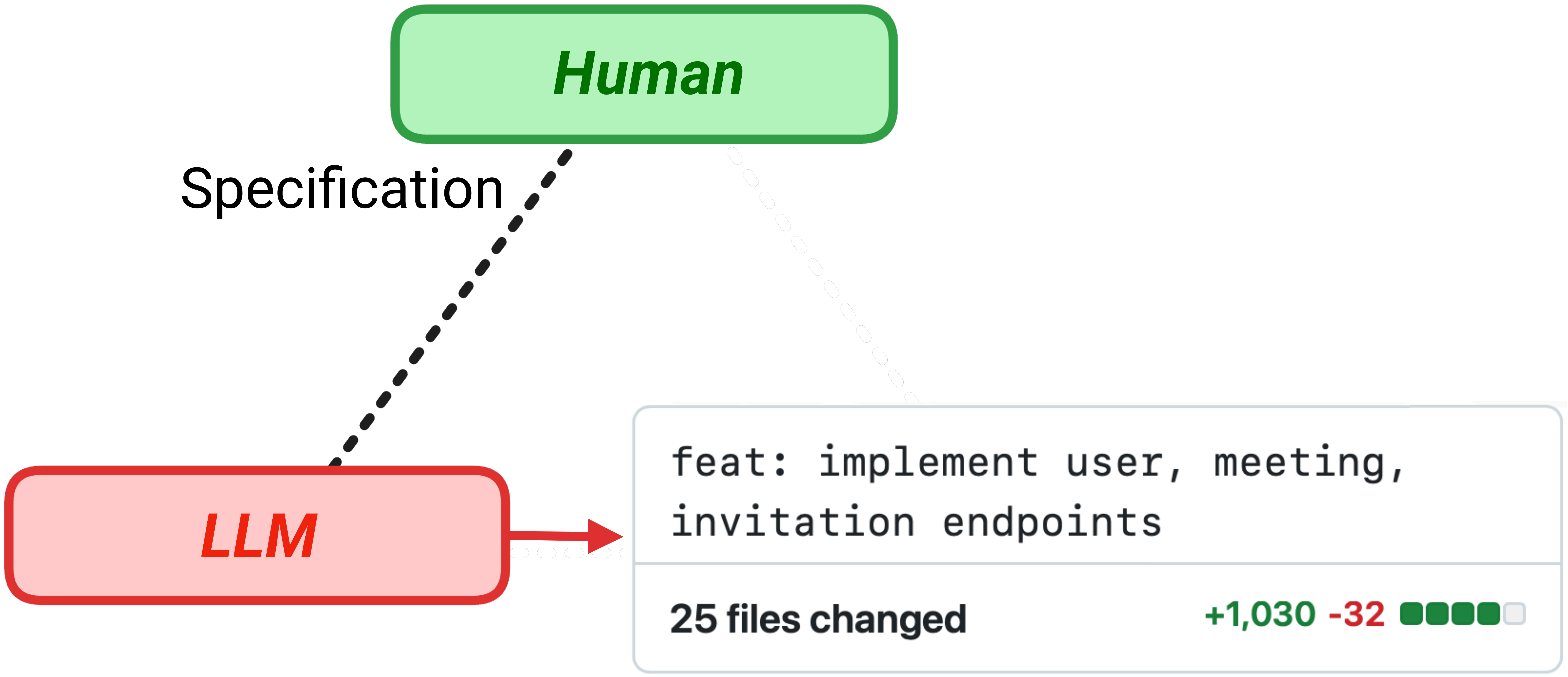

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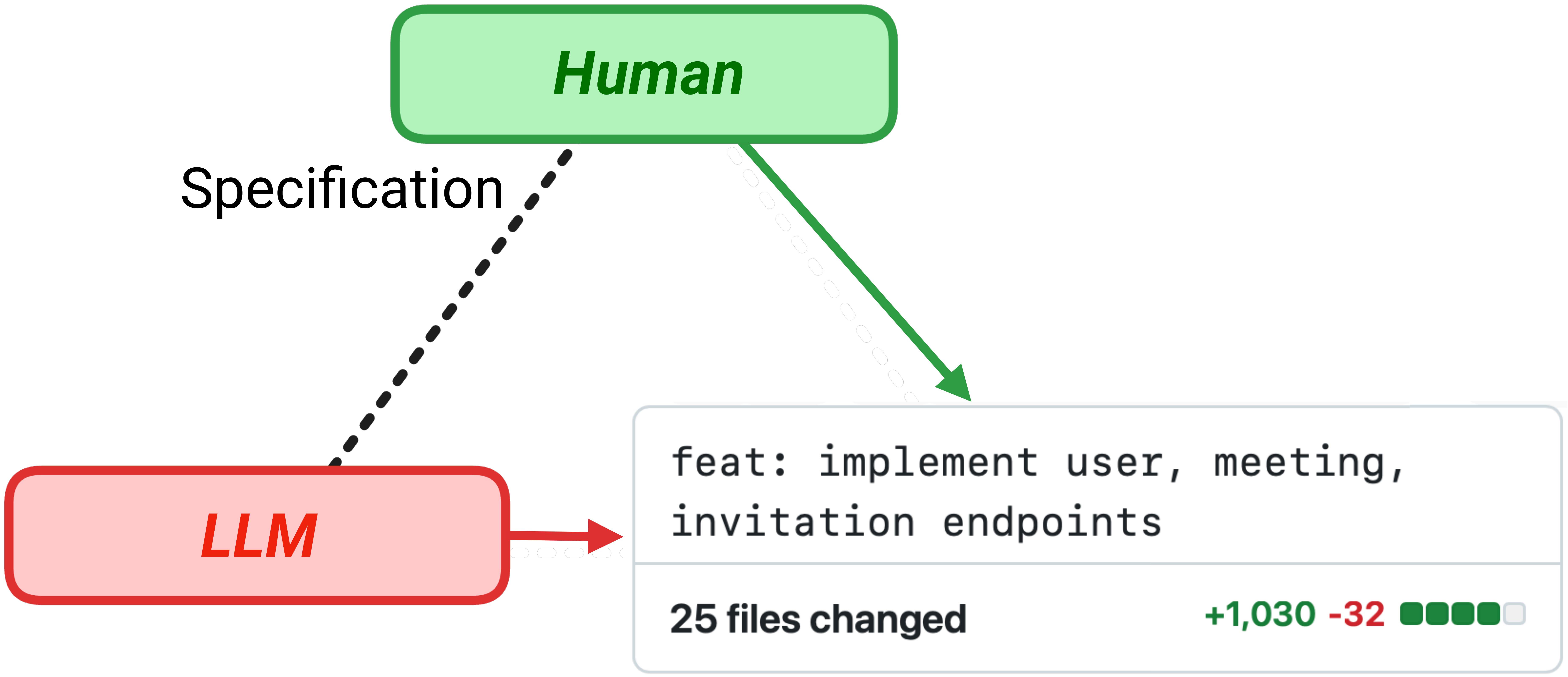

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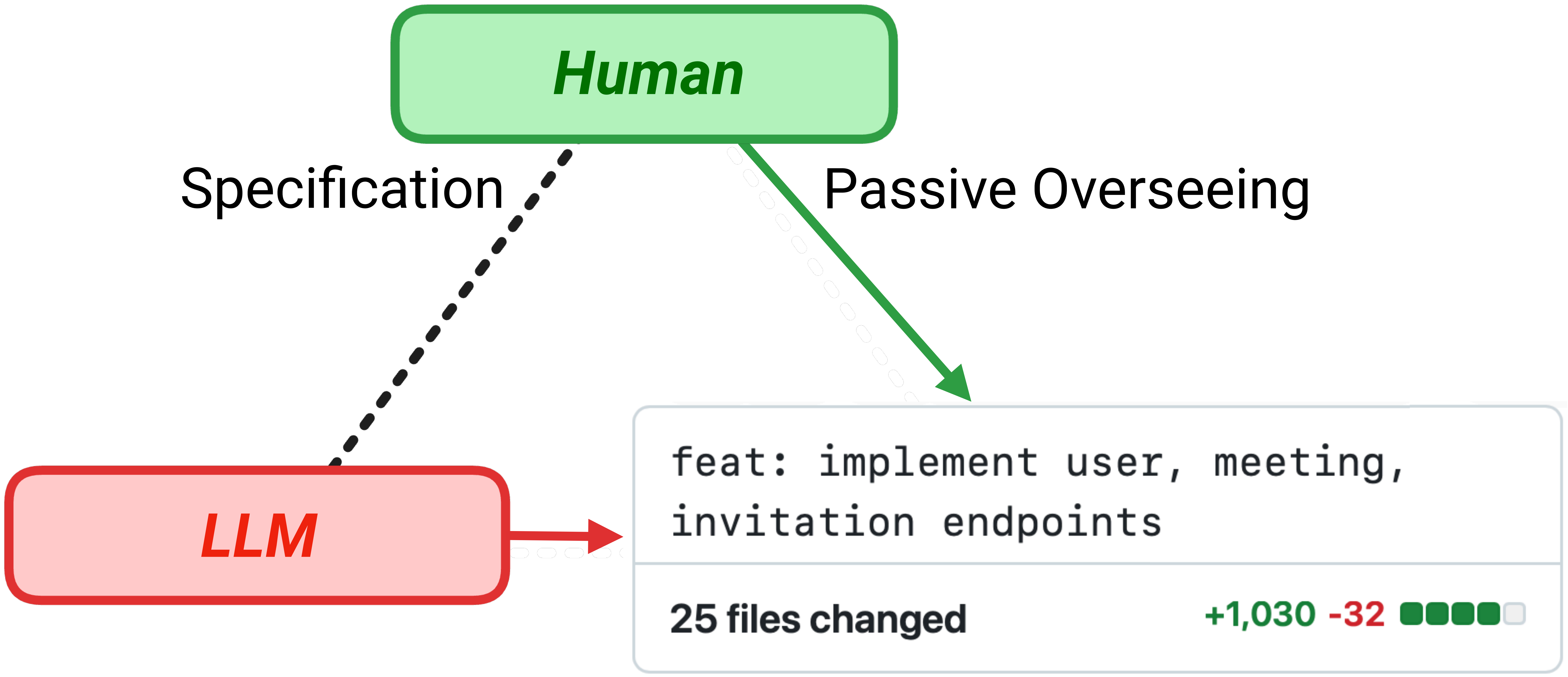

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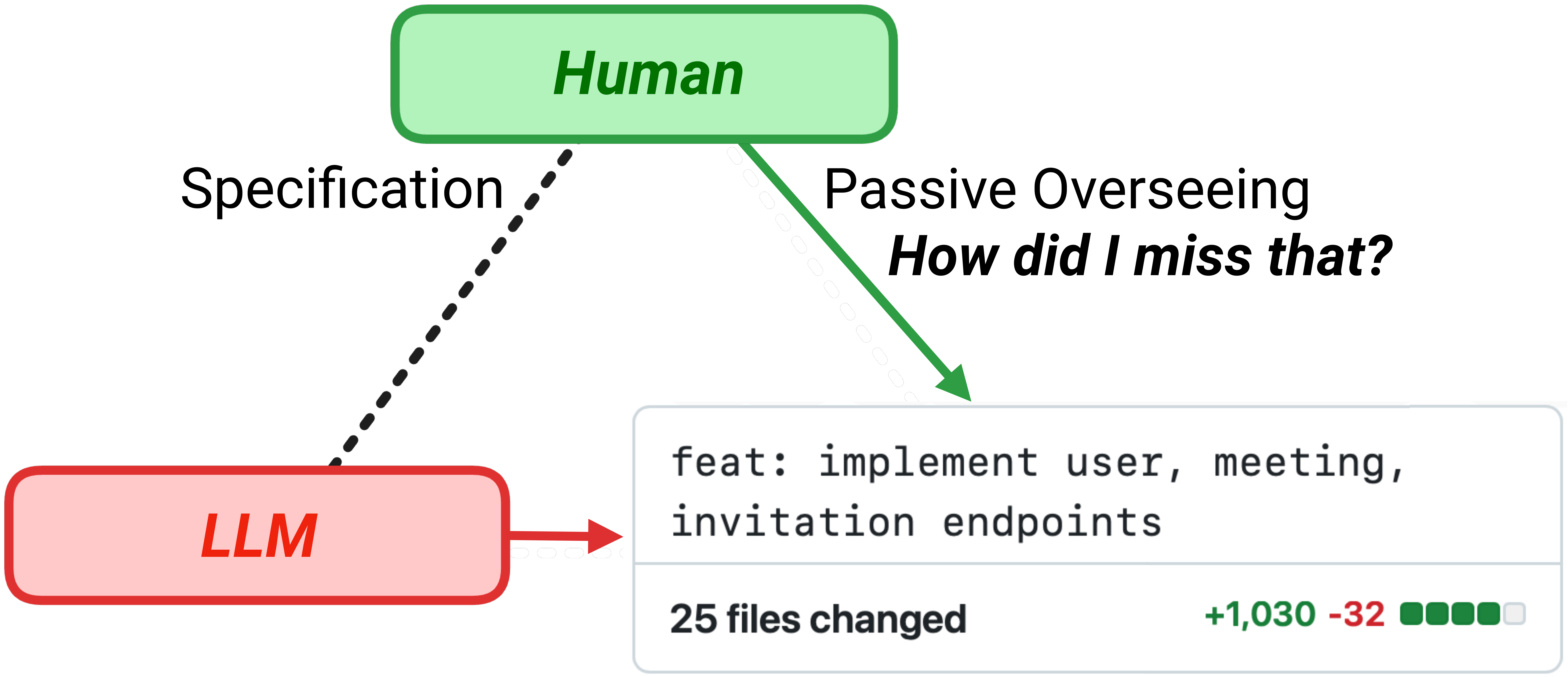

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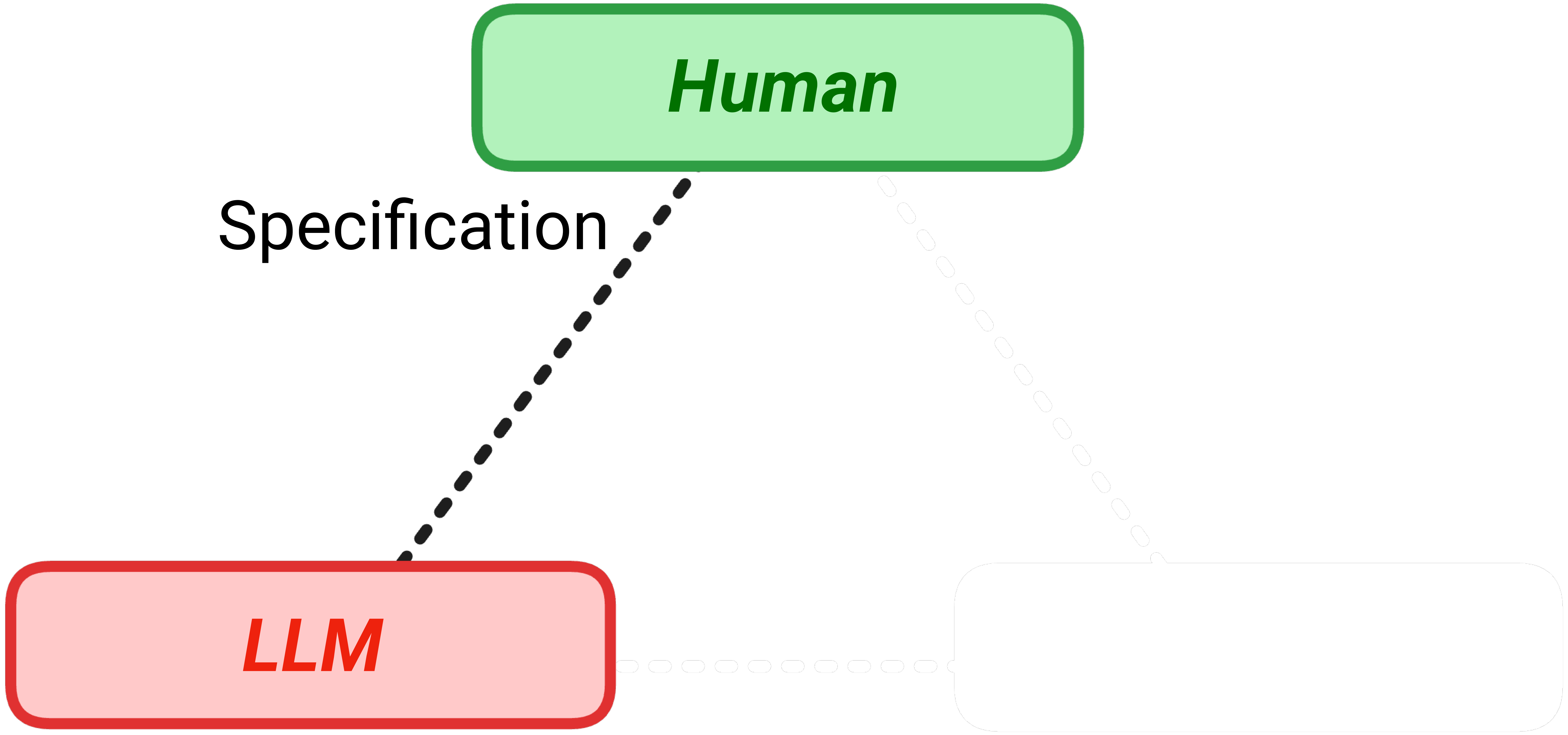
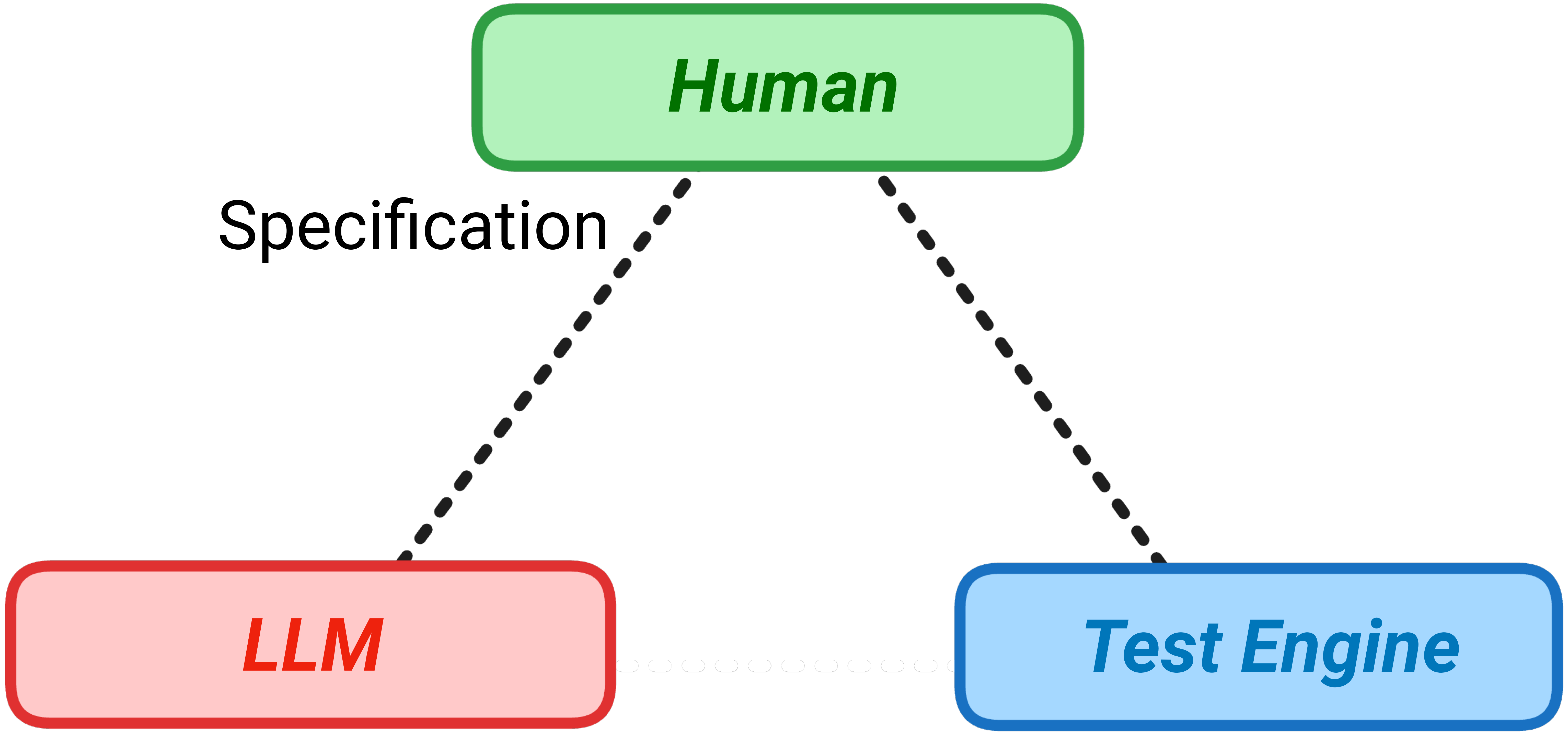


# Division of Cognitive Labour



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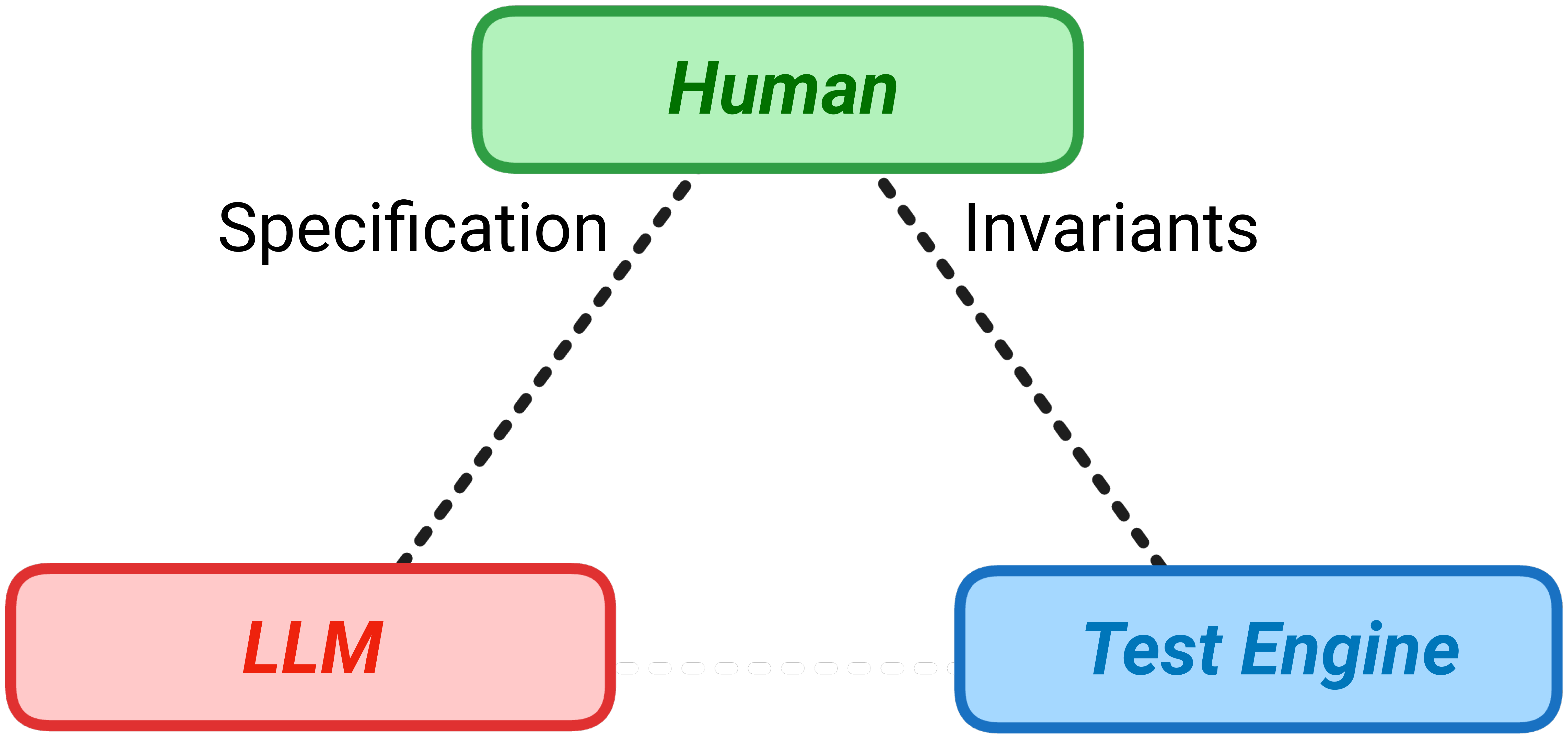

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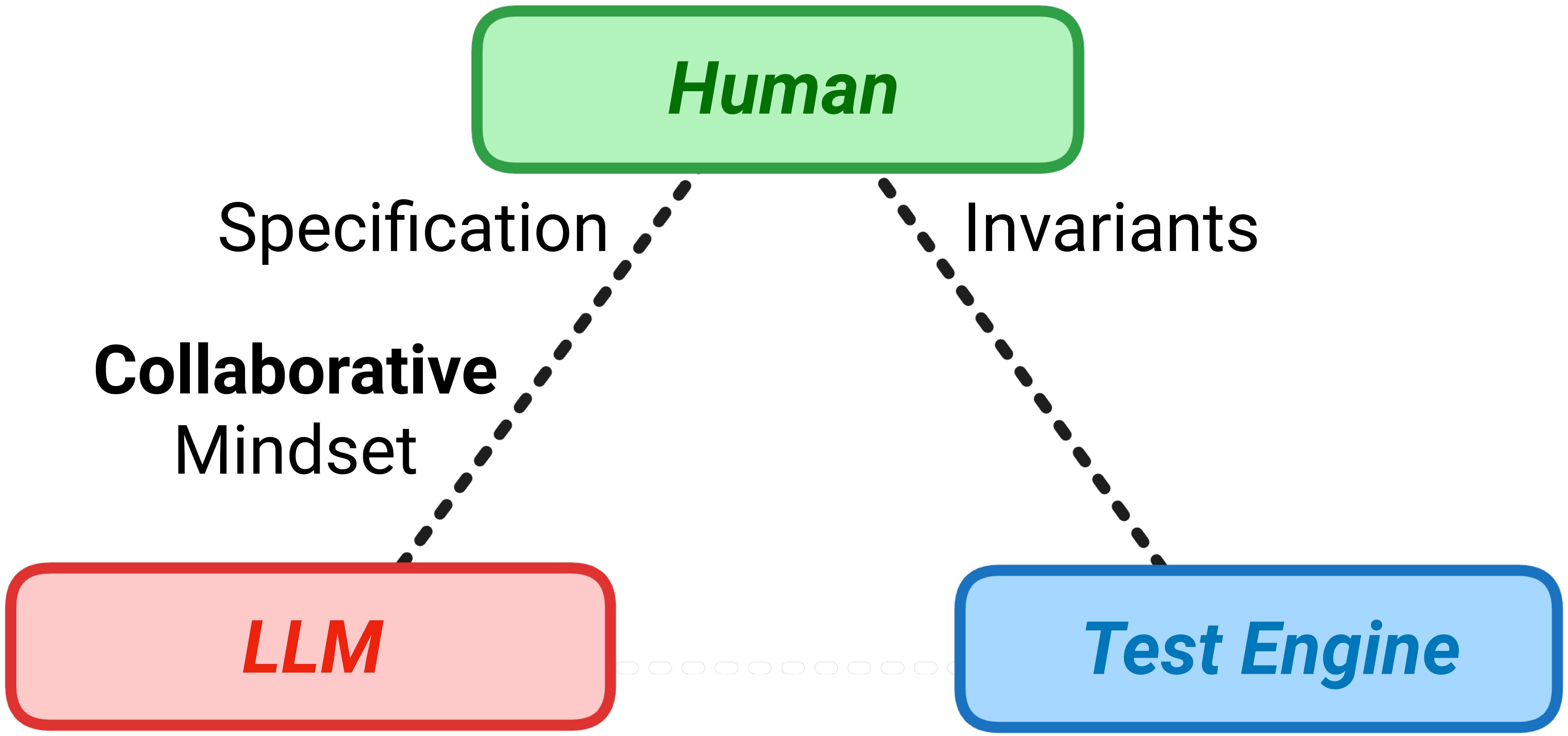

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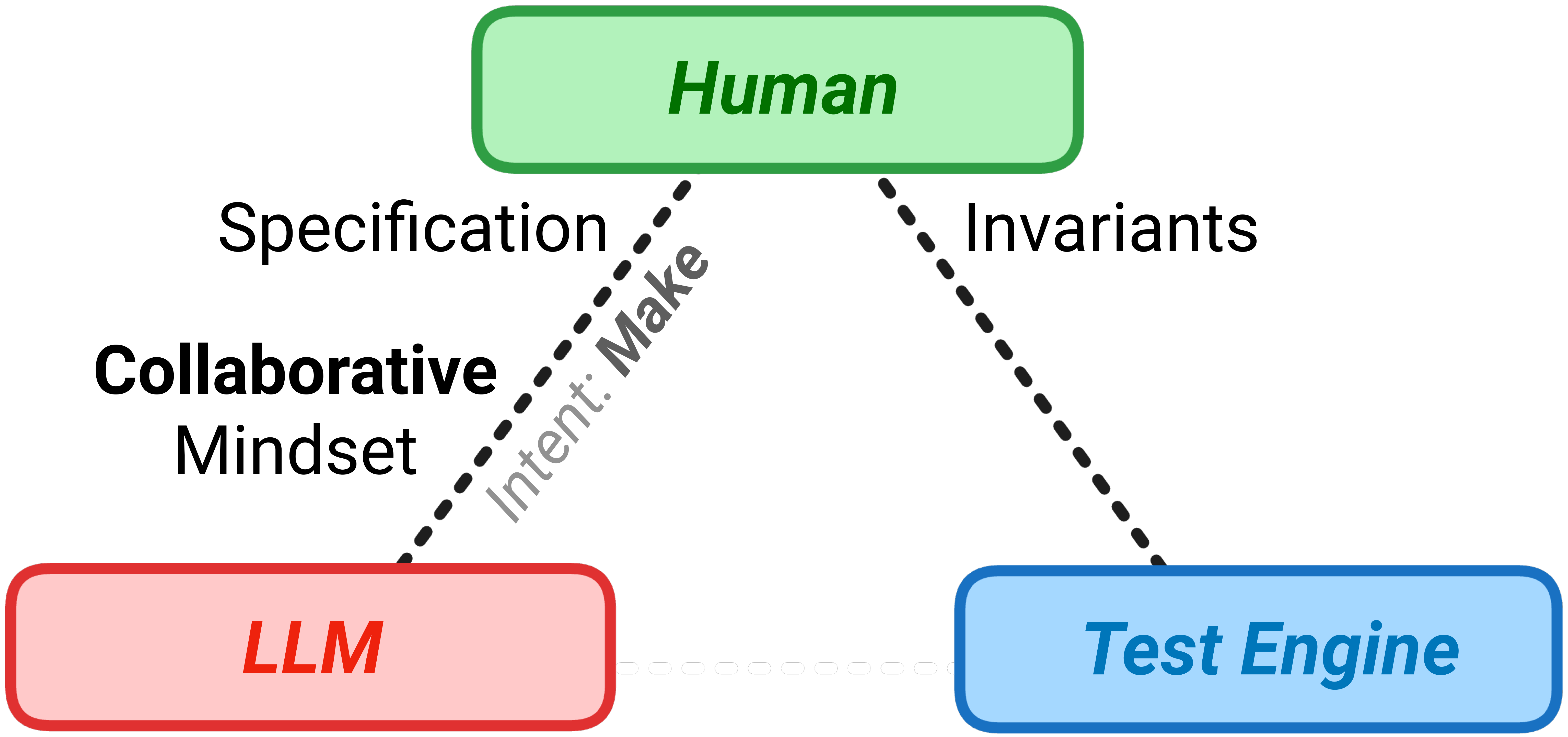

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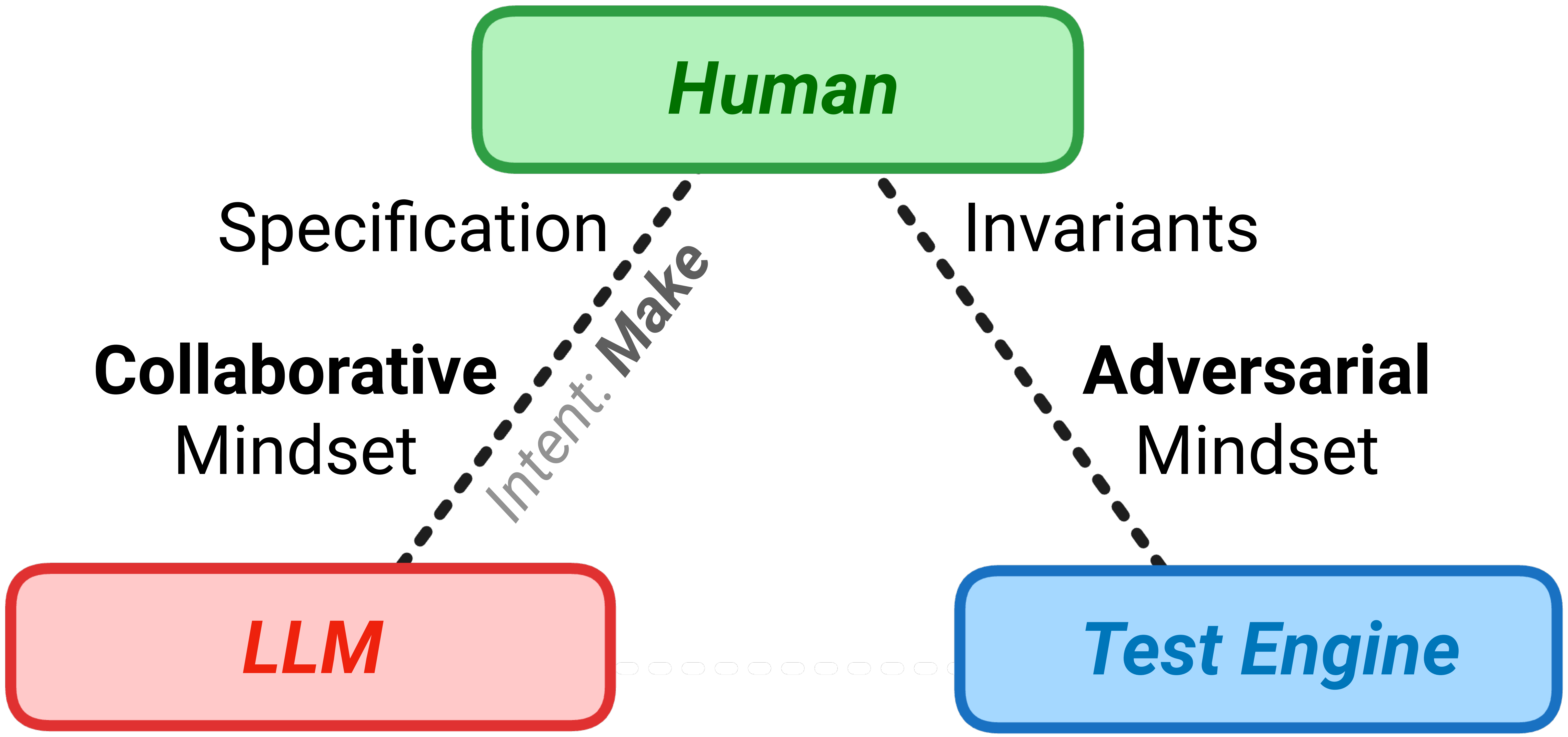

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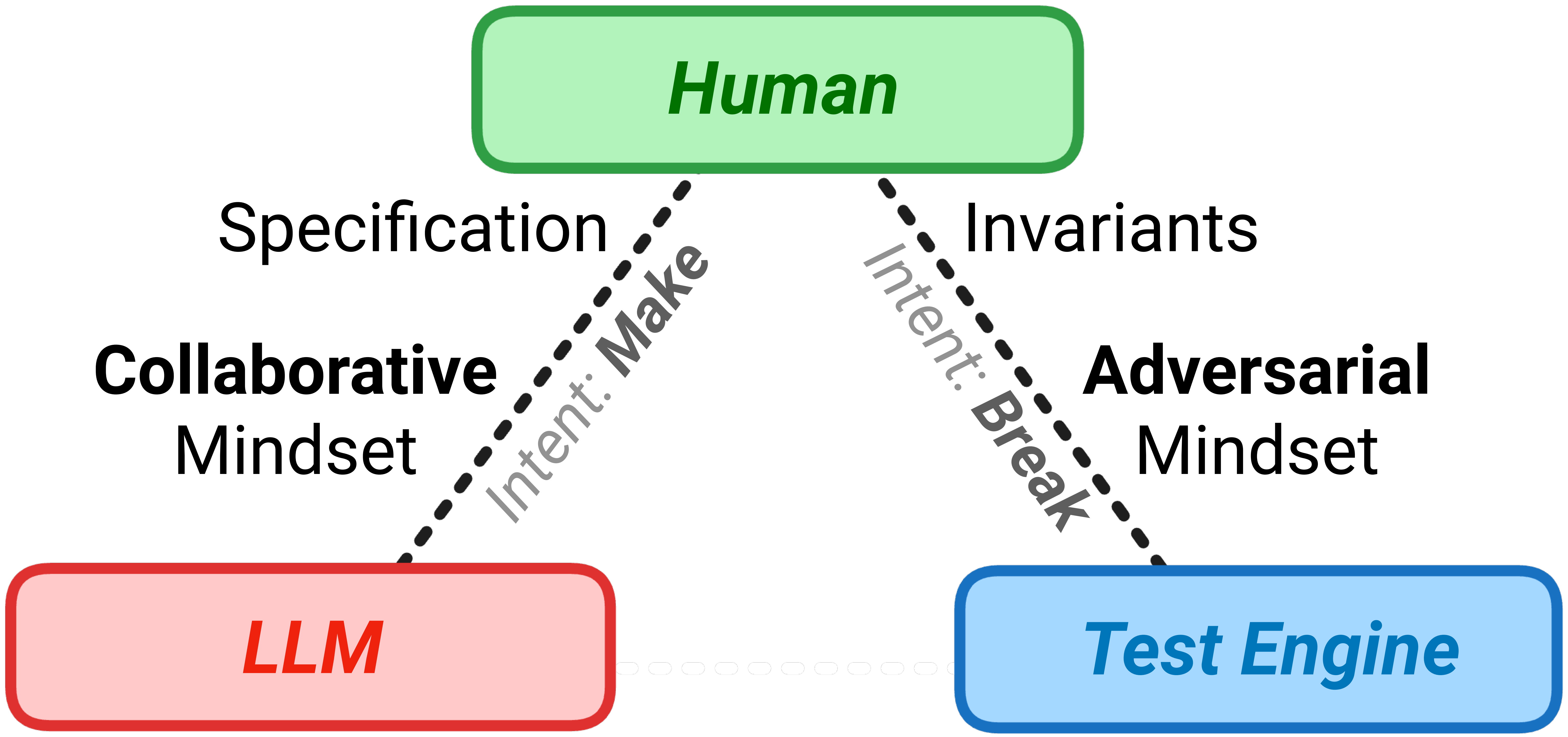

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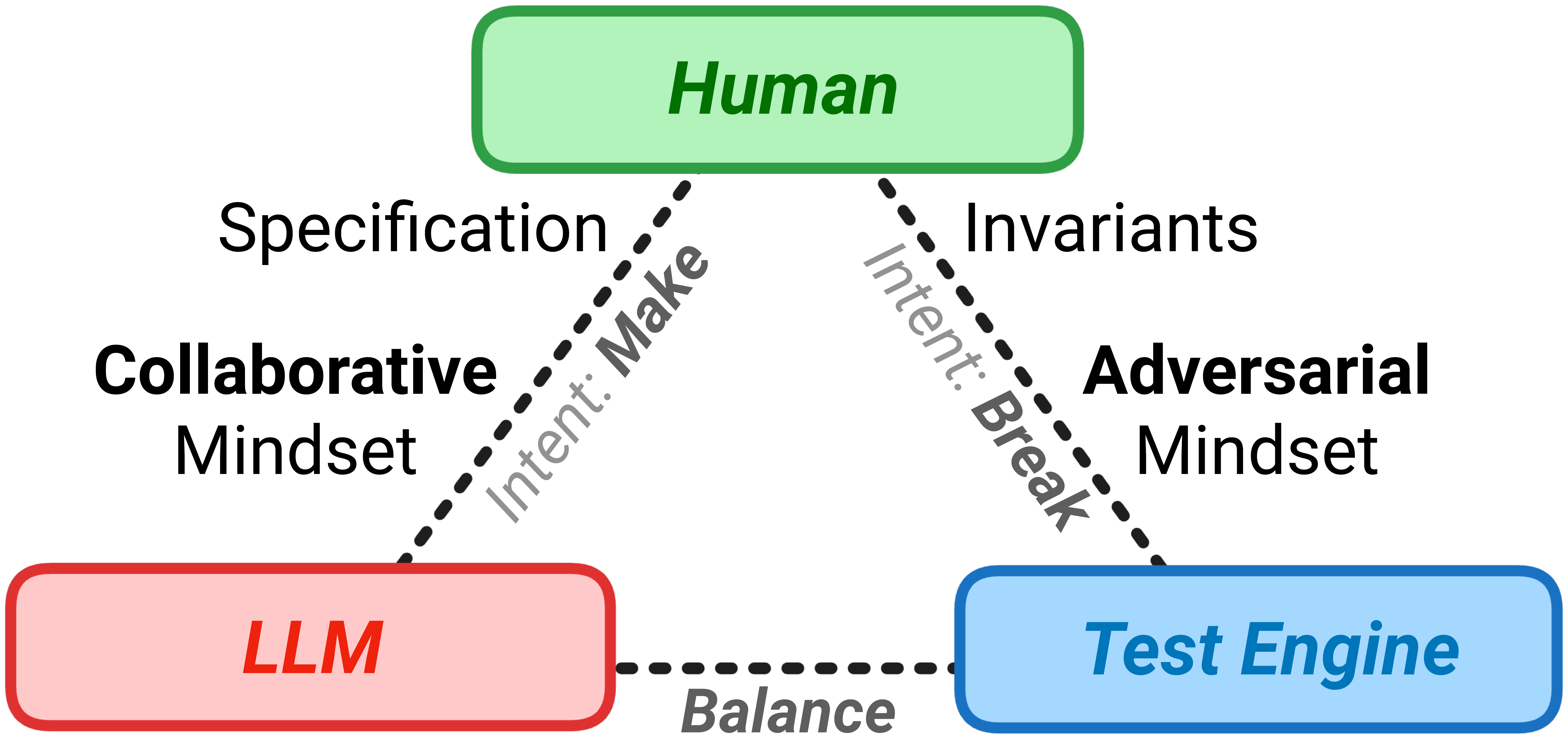
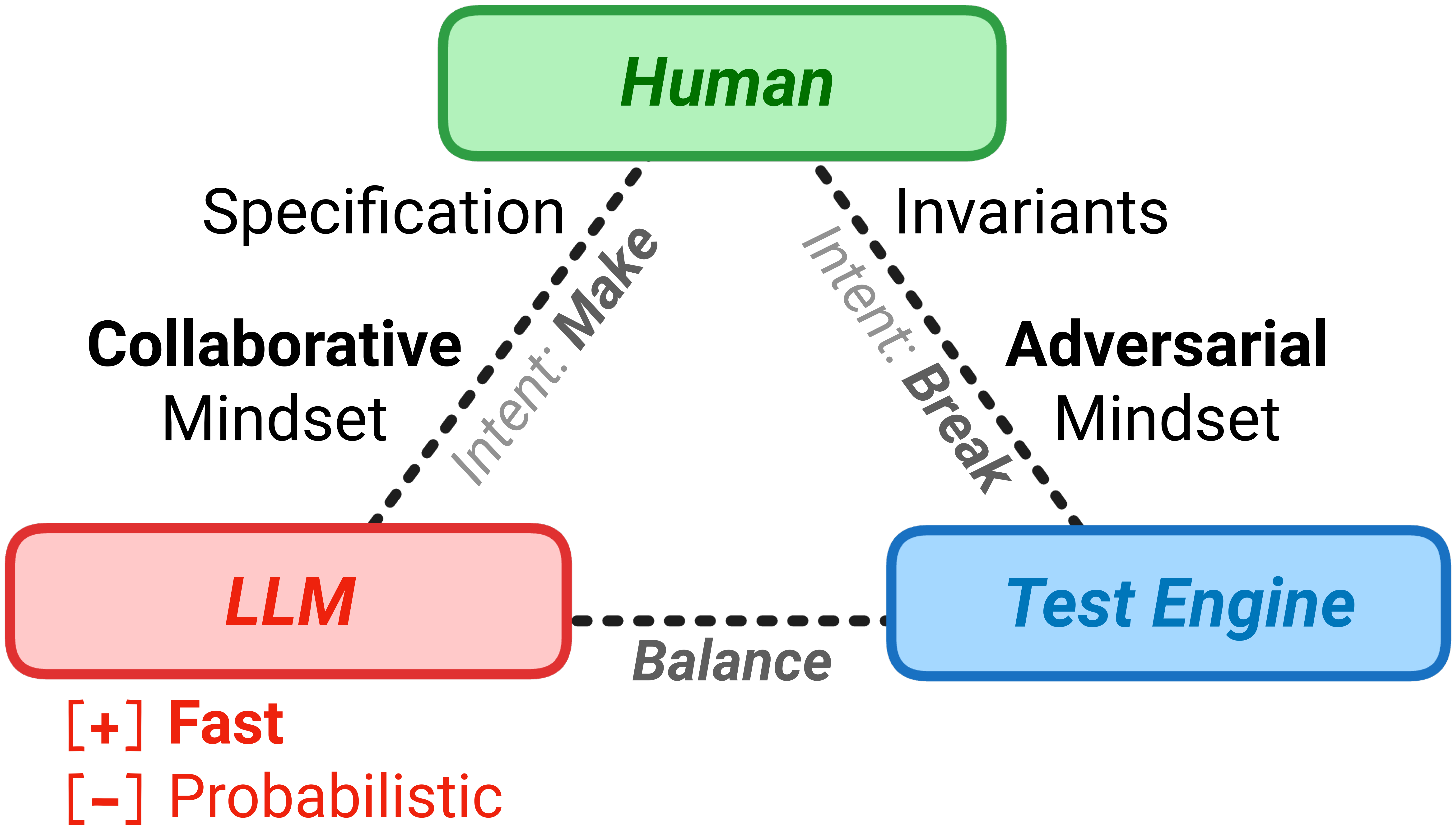
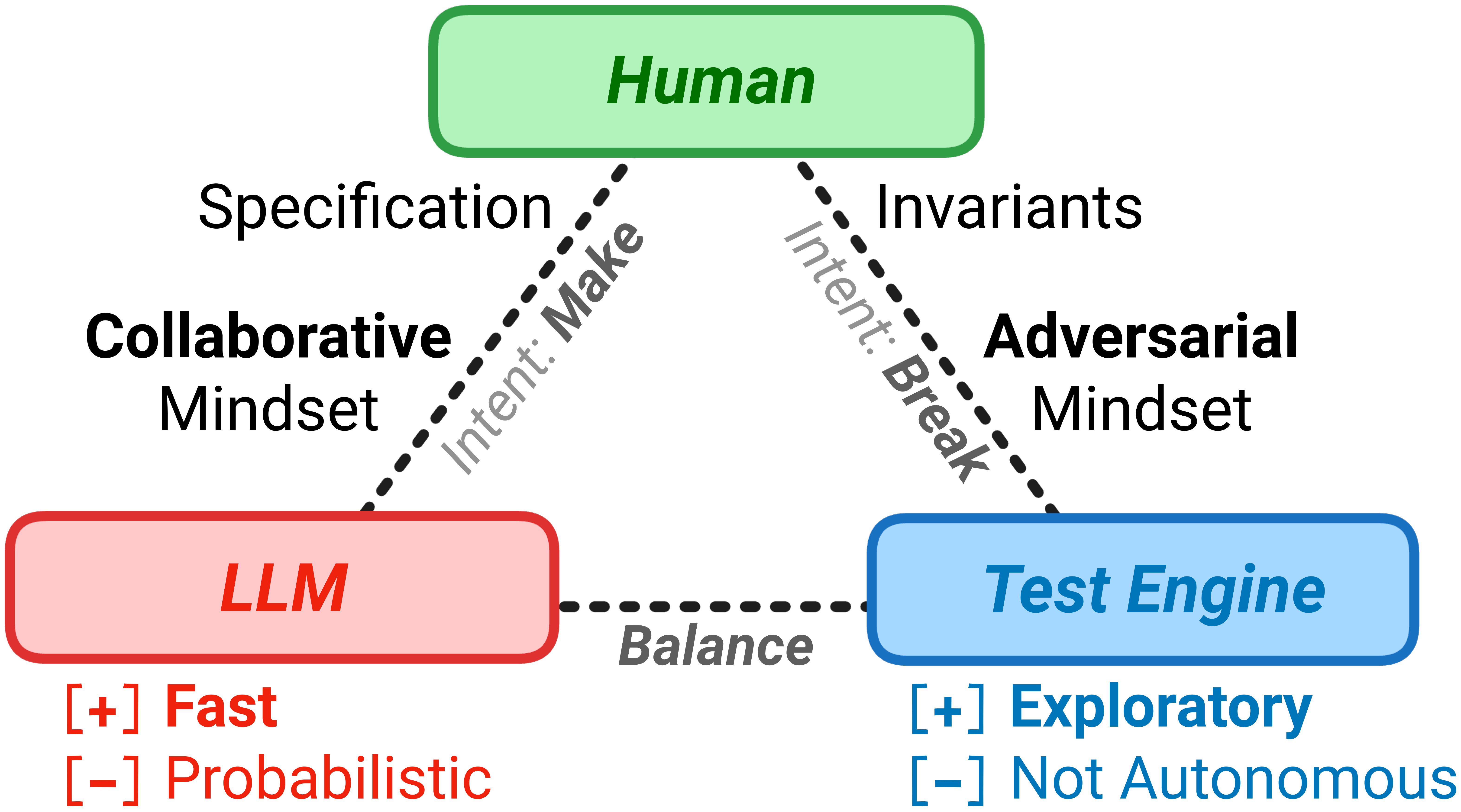


# Division of Cognitive Labour



# Division of Cognitive Labour



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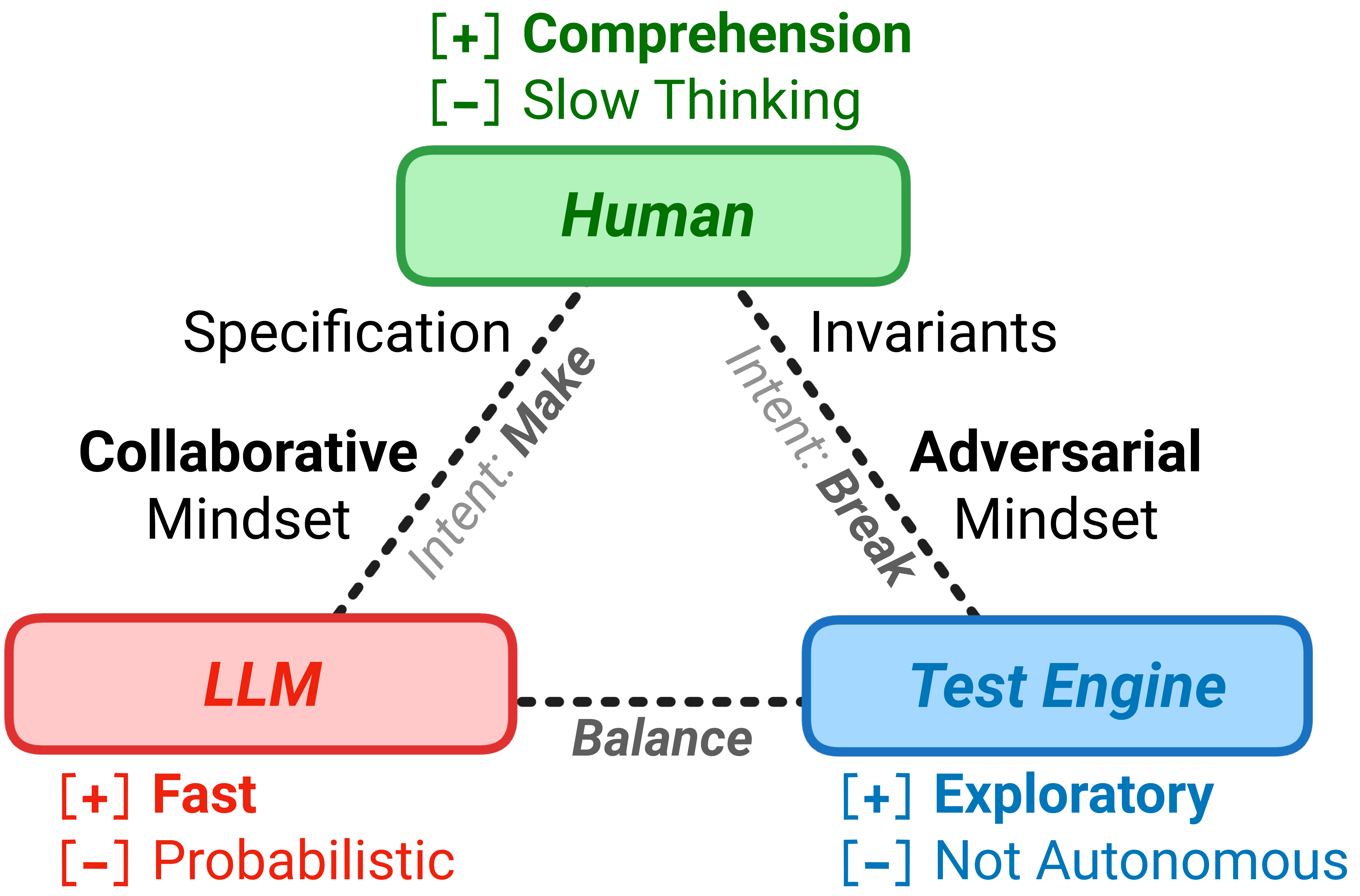

# Thank You!

- mourjo.me
- github.com/mourjo/prompt-meetings
- mourjo.me/slides/2026-09-11-signals.pdf

linkedin.com/in/mourjo